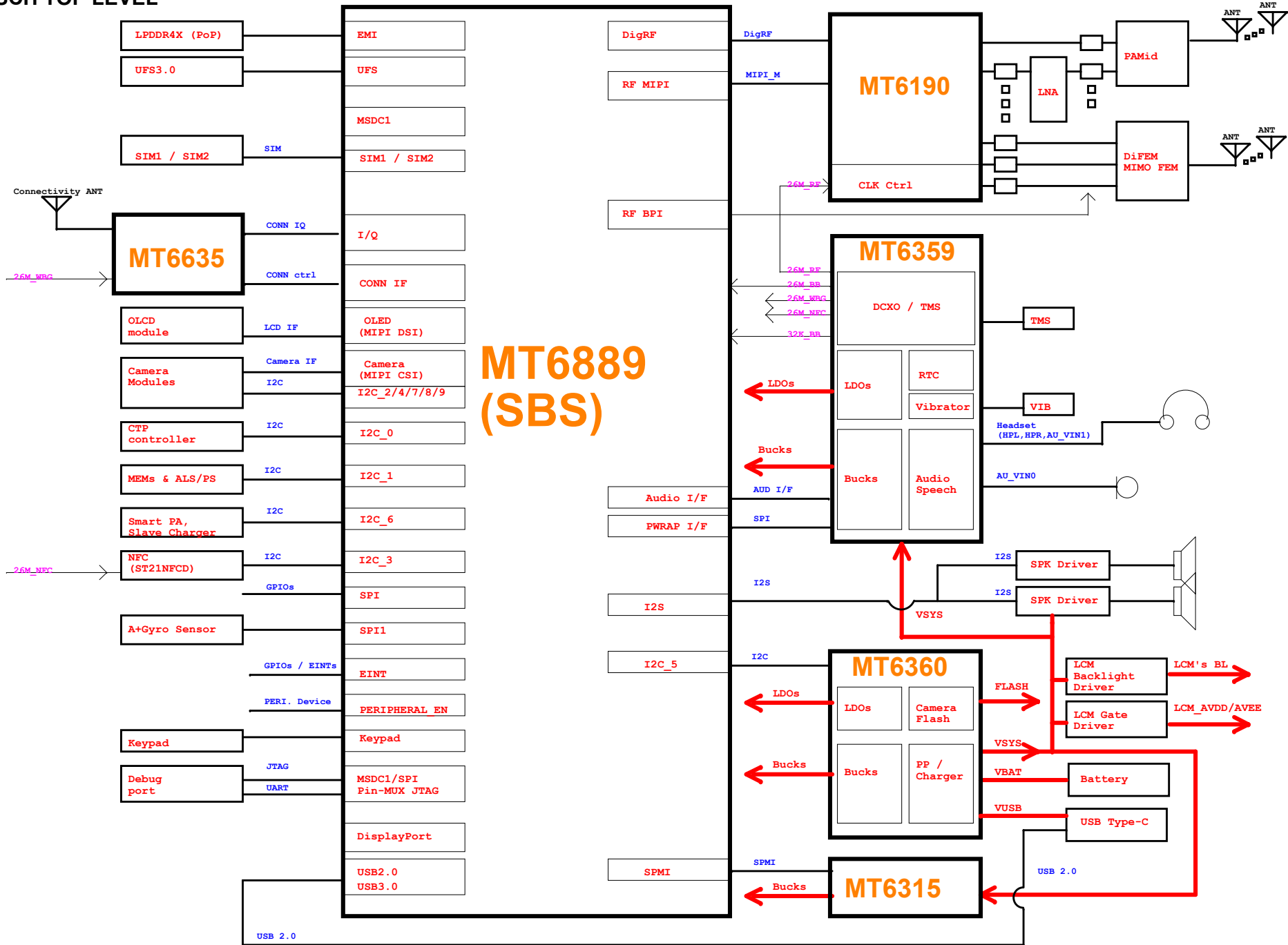
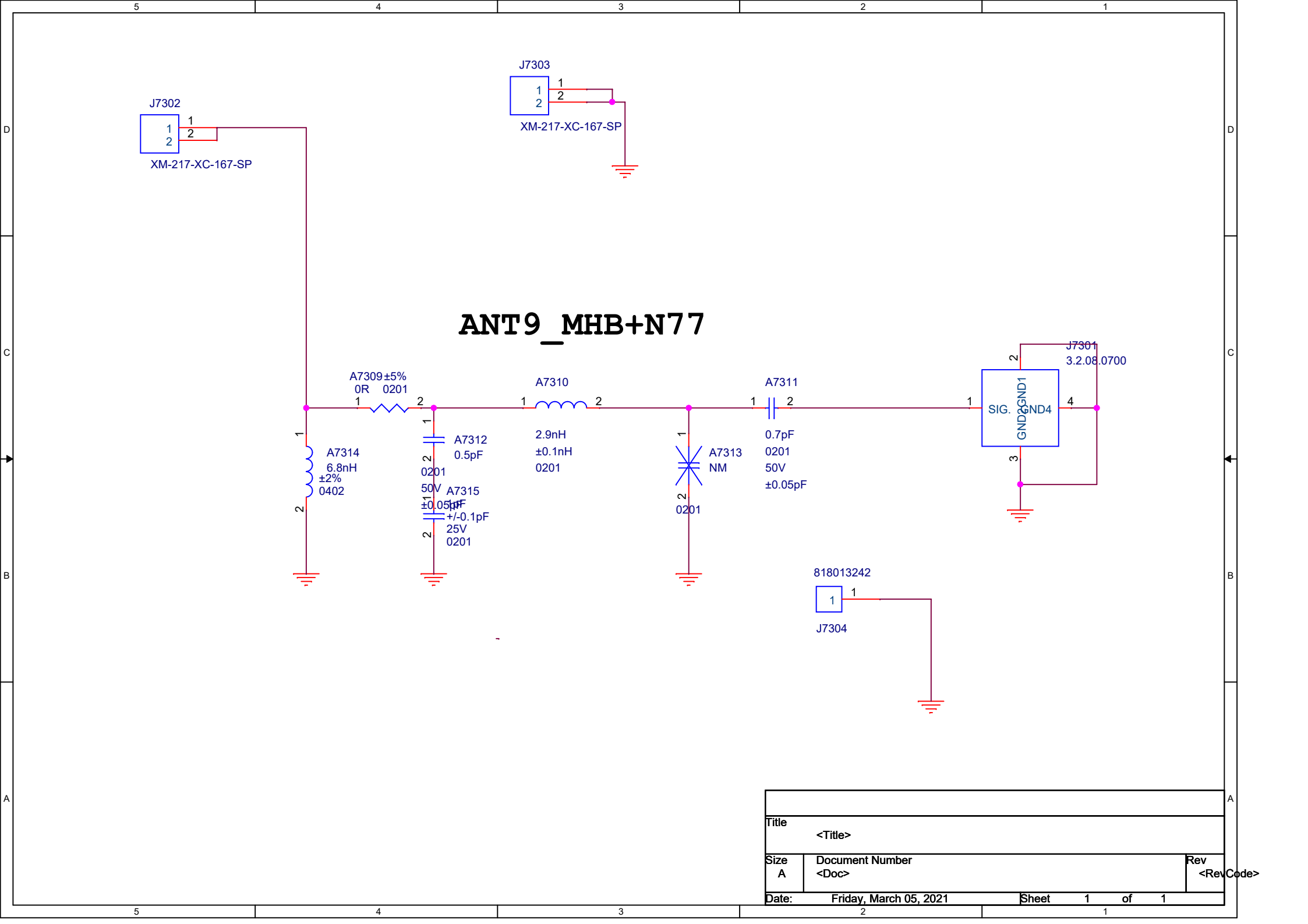


Project : MT6889, LPDDR4X (SBS)
REF_SCH TOP LEVEL





ANT9_MHB+N77

J7302
XM-217-XC-167-SP

J7303
XM-217-XC-167-SP

A7314
6.8nH
±2%
0402

A7309±5%
0R 0201

A7312
0.5pF
50V 0201

A7315
±0.05pF
25V 0201

A7310
2.9nH
±0.1nH
0201

A7313
NM
50V
0201

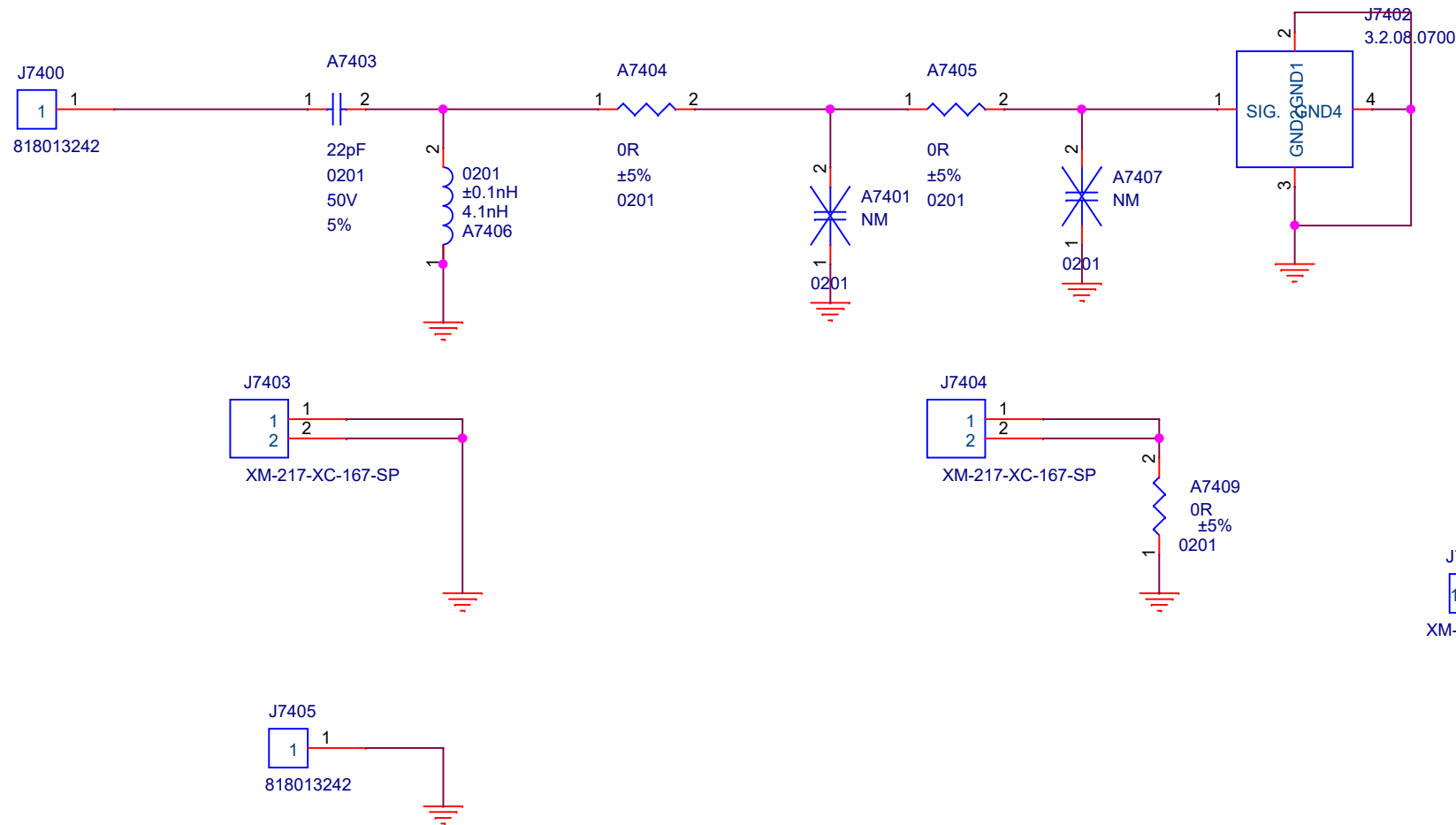
A7311
0.7pF
50V
0201
±0.05pF

J7301
3.2.08.0700

SIG.
GND1
GND2
GND4

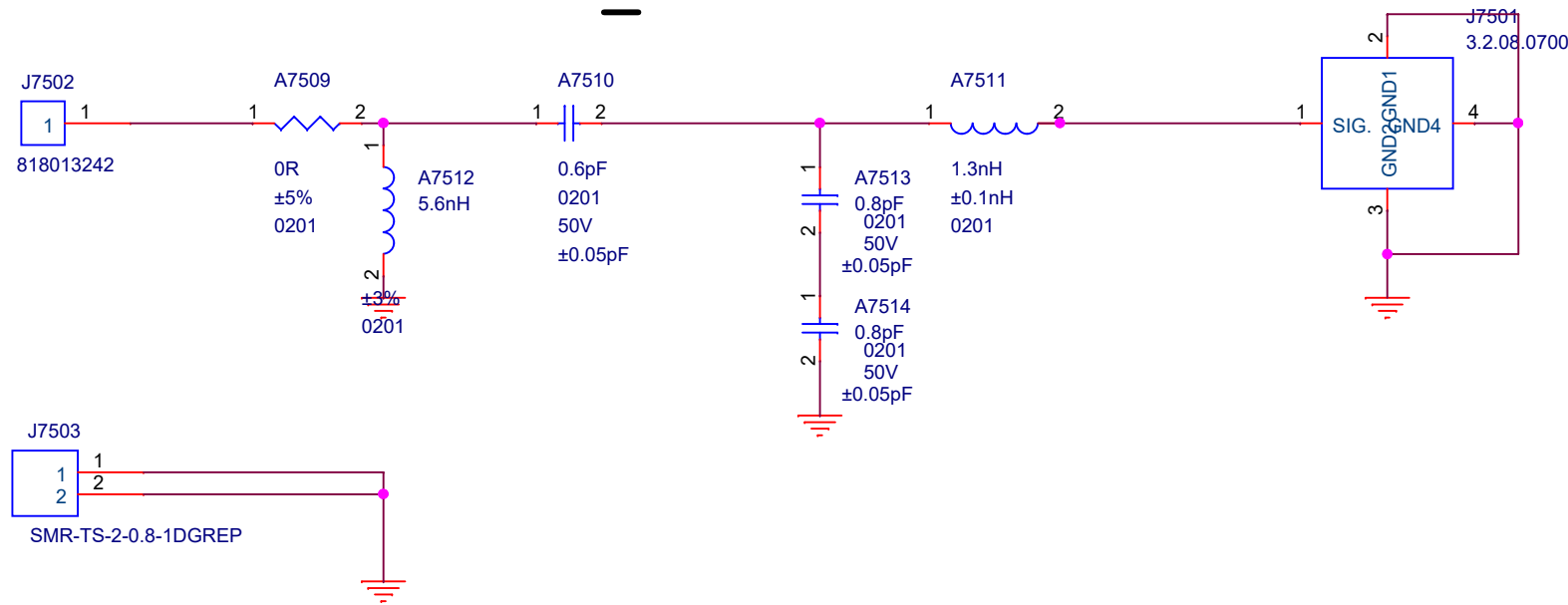
818013242
J7304

ANT1_LB+N41



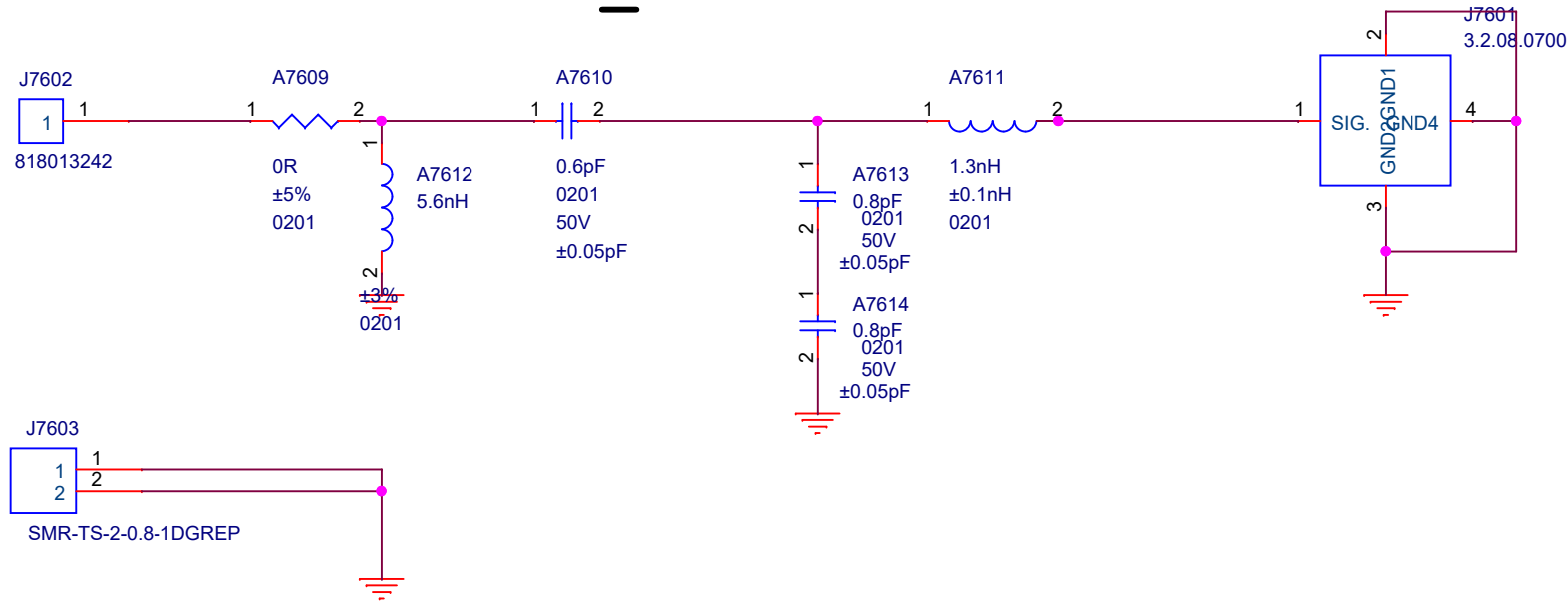
				A	
Title <Title>					
Size A	Document Number <Doc>				Rev <RevCode>
Date: Friday, March 05, 2021					Sheet 1 of 1

ANT10_WiFi2.4G+5G

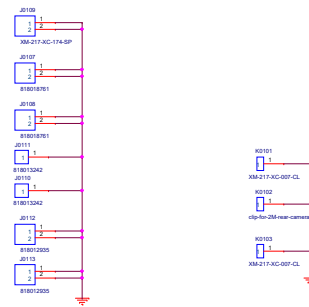


Title			
<Title>			
Size	Document Number		Rev
A	<Doc>		<RevCode>
Date:	Friday, March 05, 2021		Sheet 1 of 1

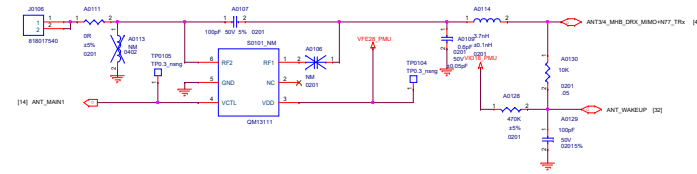
ANT10_WiFi2.4G+5G



Title			
<Title>			
Size	Document Number		Rev
A	<Doc>		<RevCode>
Date:	Friday, March 05, 2021		Sheet 1 of 1

[illegible]

The schematic shows the internal wiring of the ADXL345 module. It starts with a USB Type-A connector (J1) providing power to a 5V regulator (U1) which outputs +5V to the ATtiny85 microcontroller (U2). The ATtiny85 is configured as a serial-to-parallel interface, receiving SCL and DATA signals from the ADXL345 via pins A5 and A4 respectively. The ADXL345 is powered by +5V and GND. Other components include a push button (SW1) connected to pin D0, and two LEDs (H1, H2) connected to pins D1 and D2 through current-limiting resistors. Various other components like capacitors (C1-C6) and resistors (R1-R7) are used for timing and signal conditioning.

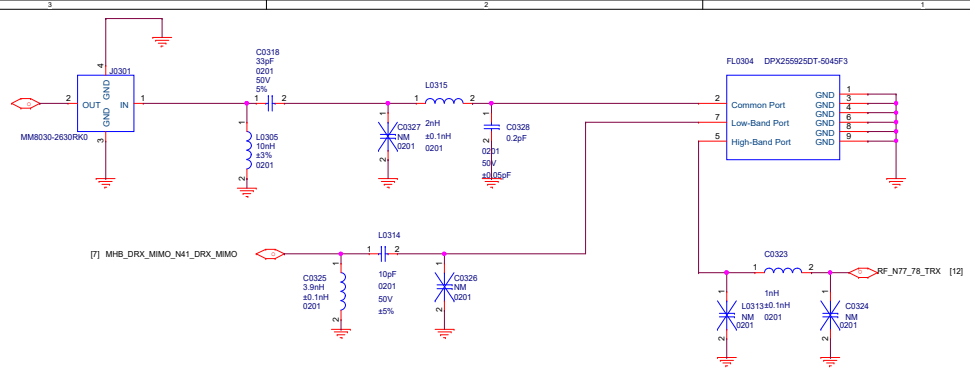


File		
GTA MAIN		
Size	Document Number	
A1	<Doc>	
Date:	Friday, March 05, 2021	Sheet 2 of 73

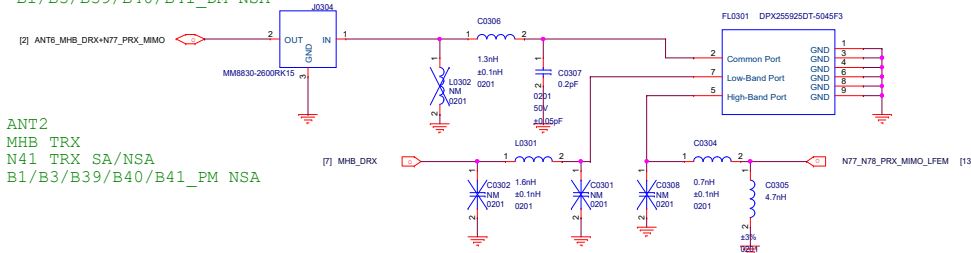
[illegible]

Title		
G7A MAIN		
Size	Document Number	
A2	<Doc>	
Date:	Friday, March 05, 2021	Sheet

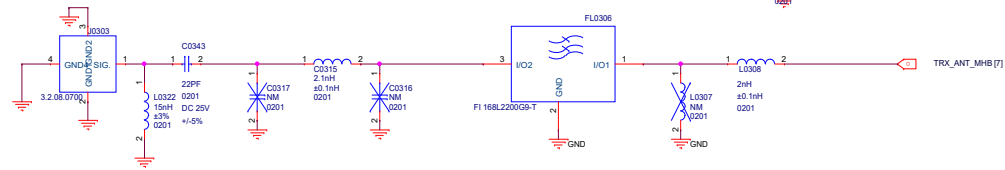
ANT3&4
N77 TRX
B1/B3/B39/B40/B41 DM
N41 DM SA/NSA
B1/B3/B39/B40/B41_DRX NSA



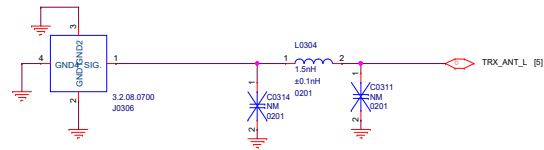
ANT6
MHB DRX+N77 PM
N41 DRX SA/NSA
B1/B3/B39/B40/B41_DM NSA



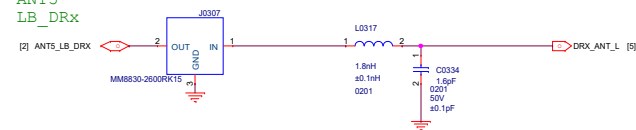
ANT2
MHB TRX
N41 TRX SA/NSA
B1/B3/B39/B40/B41_PM NSA



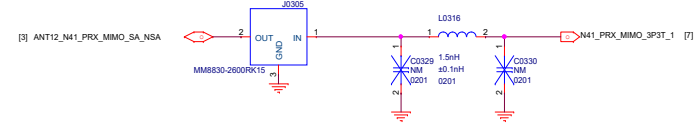
ANT1
LB_TRX



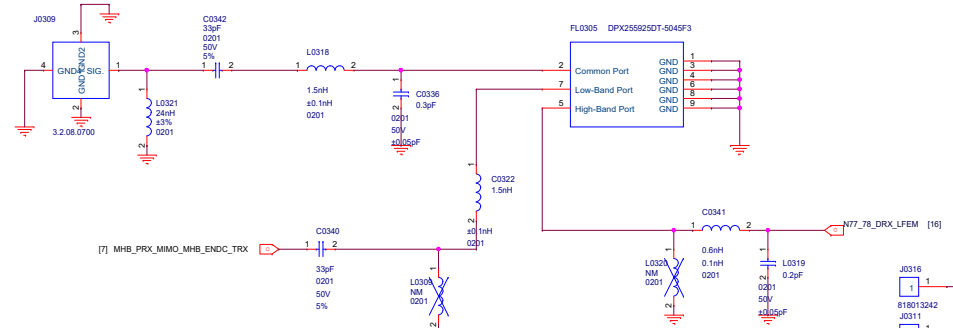
ANT5
LB_DRX



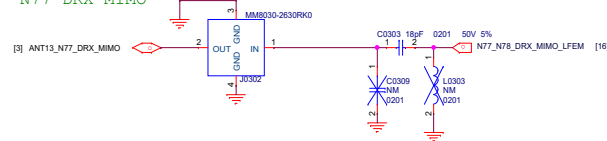
ANT12
N41 PRX MIMO NSA



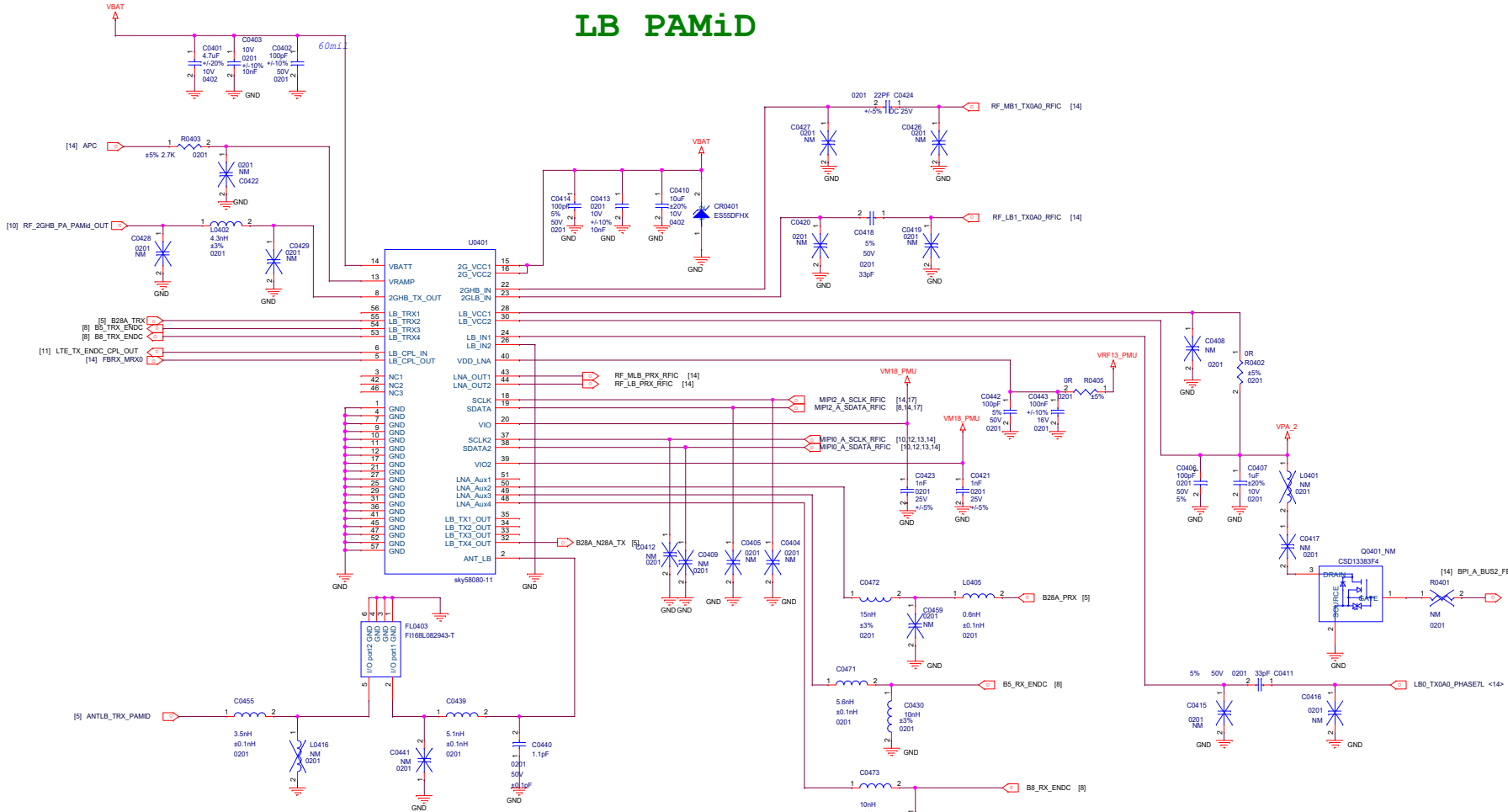
ANT9
N77 DRX
B1/B3/B39/B40/B41/N41 SA PM
B1/B3/B39/B40/B41_TRX NSA



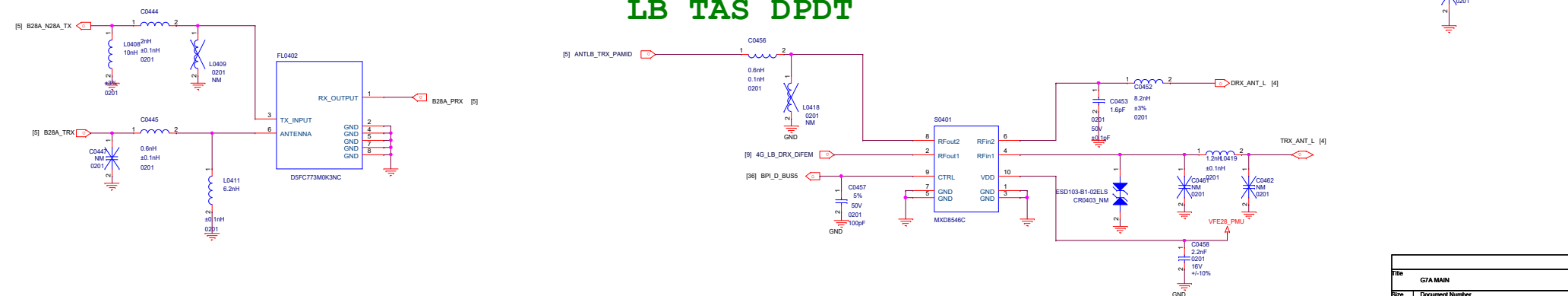
ANT13
N77 DRX MIMO



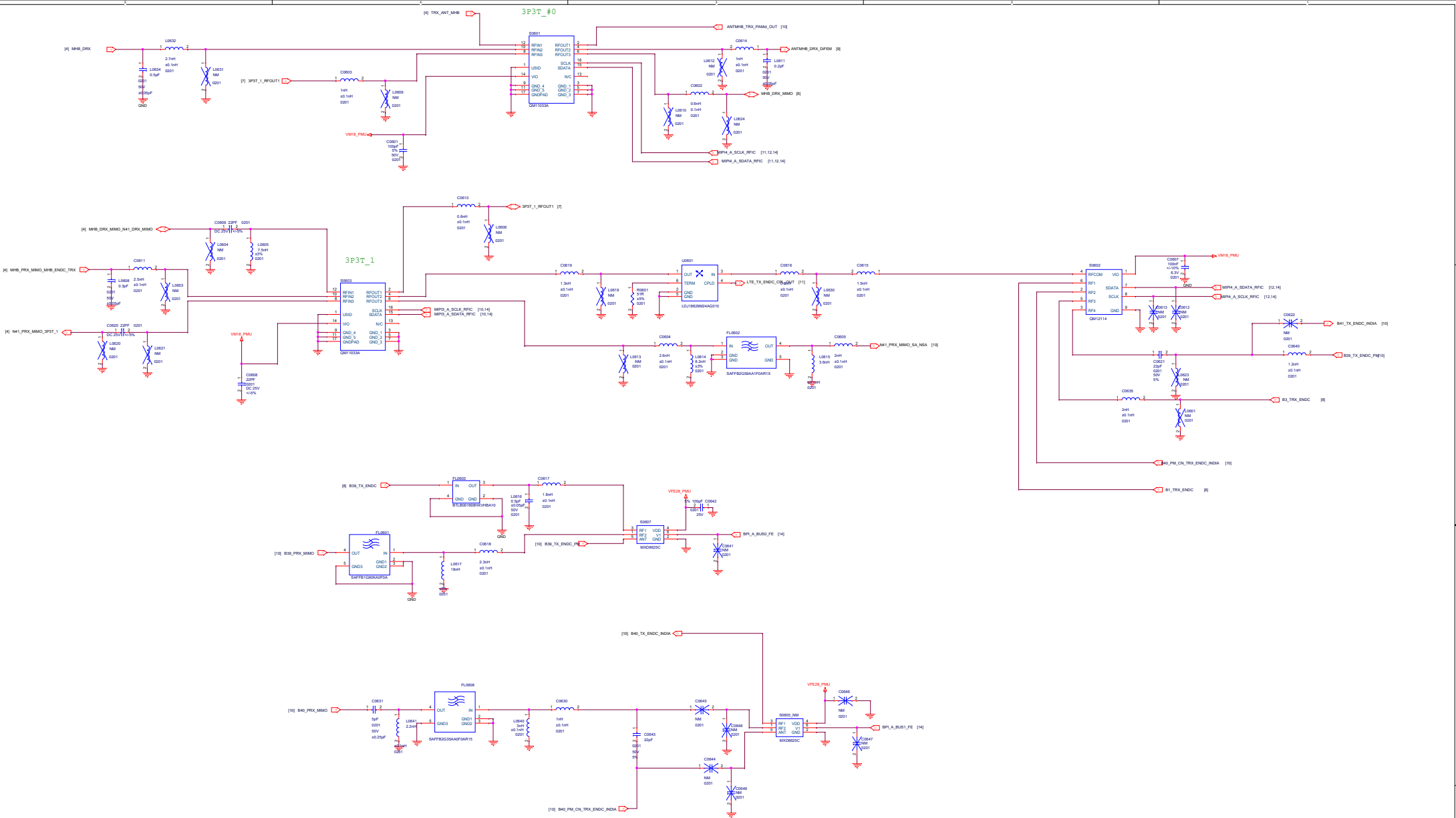
LB PAMiD

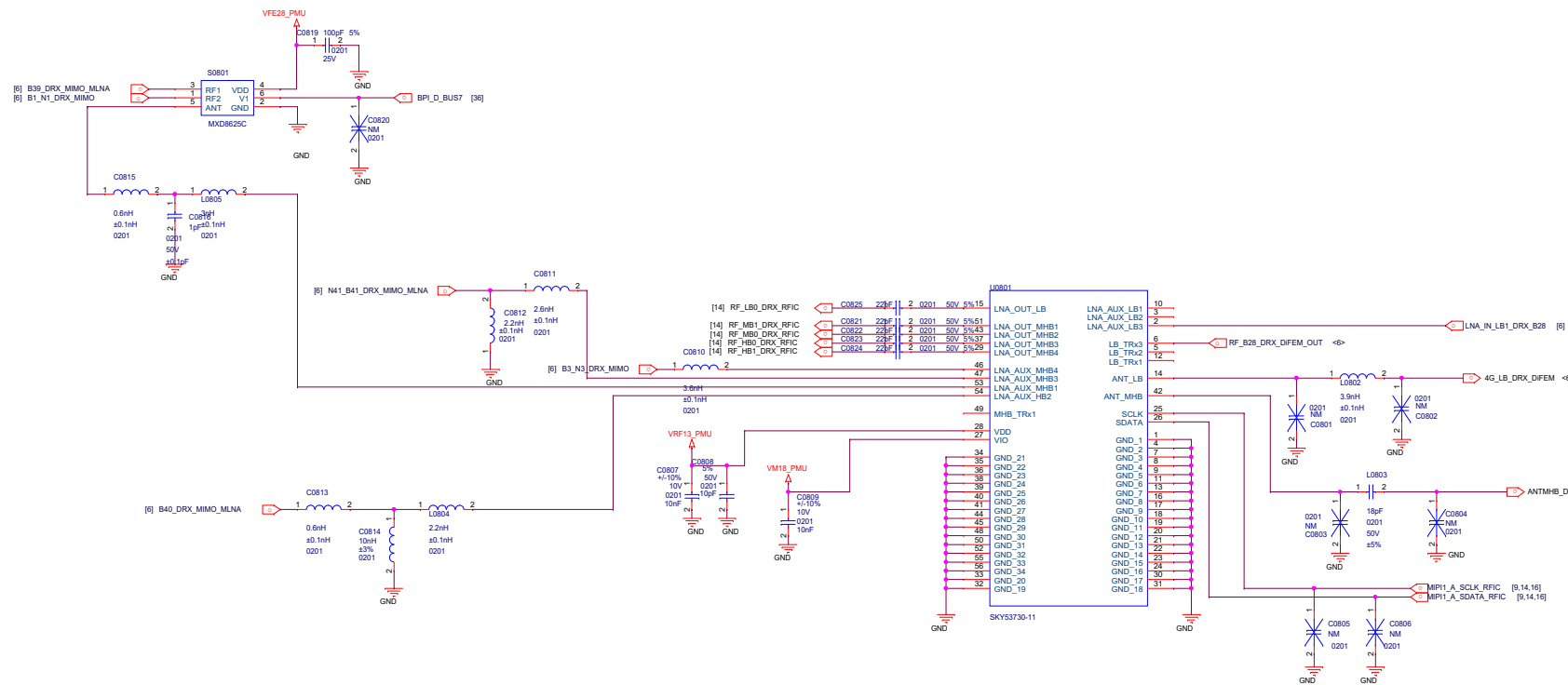


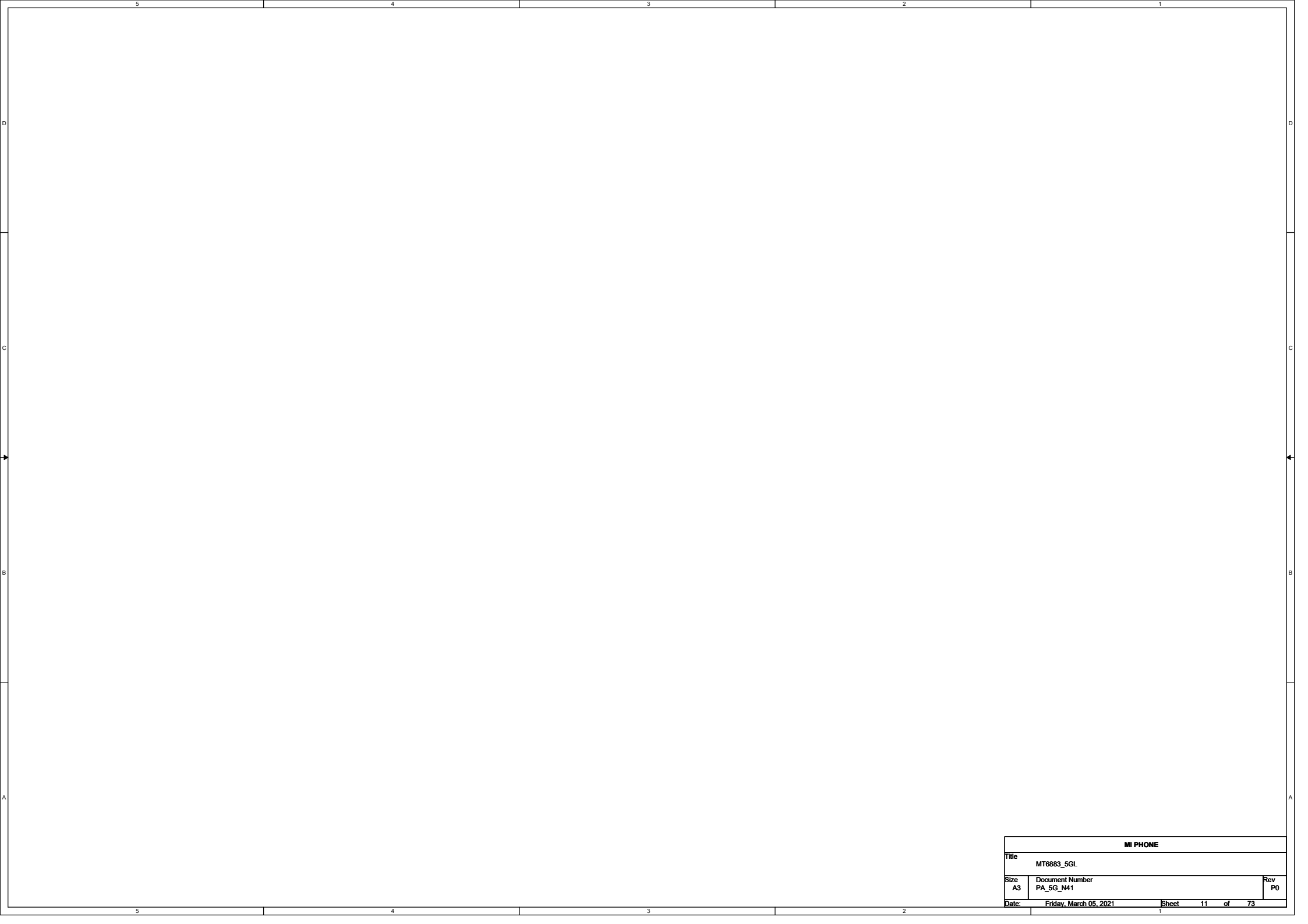
LB TAS DPDT



File	GTA MAIN		
Size	Document Number		
A2	<Doc>		
Date:	Friday, March 05, 2021	Sheet	

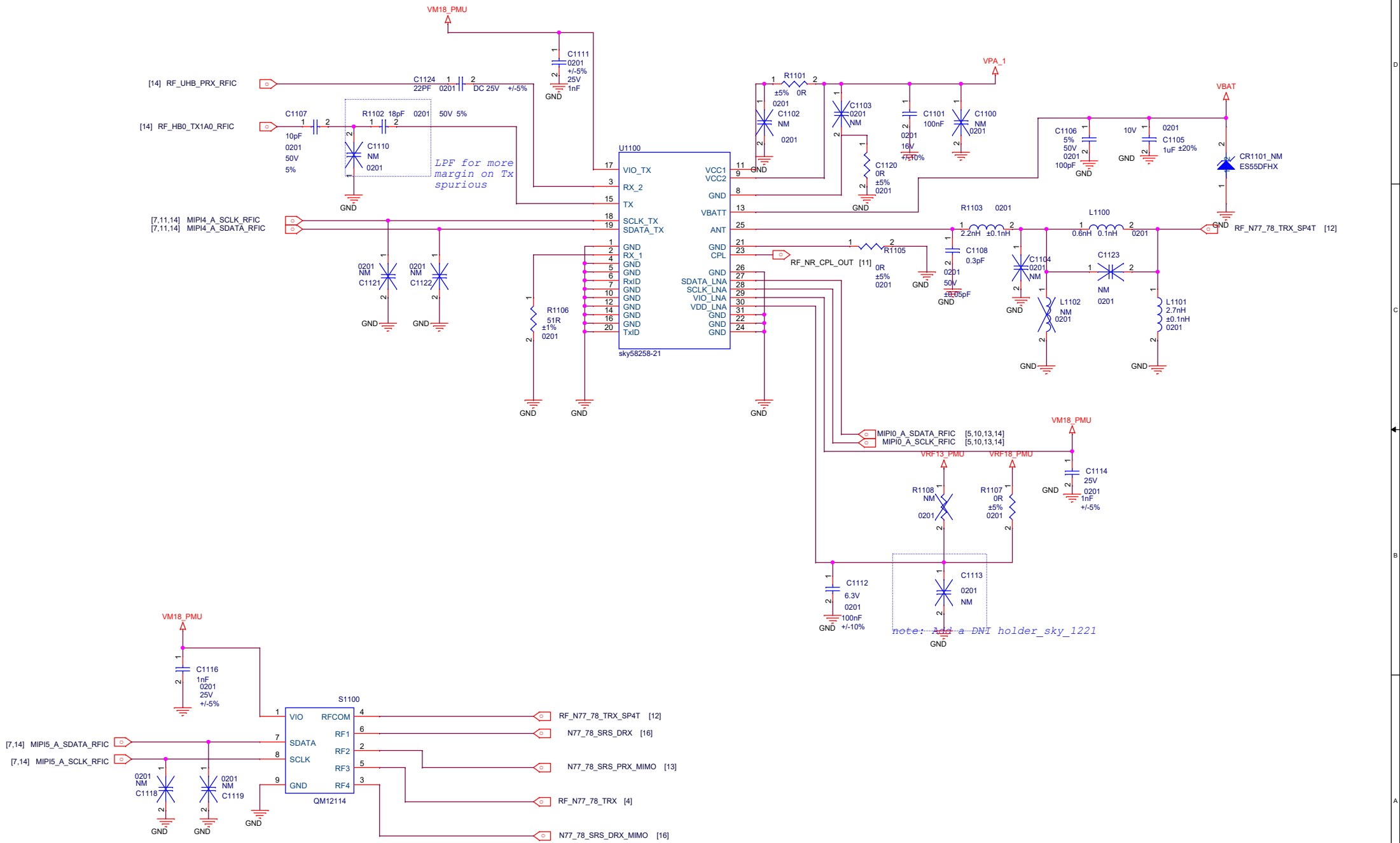






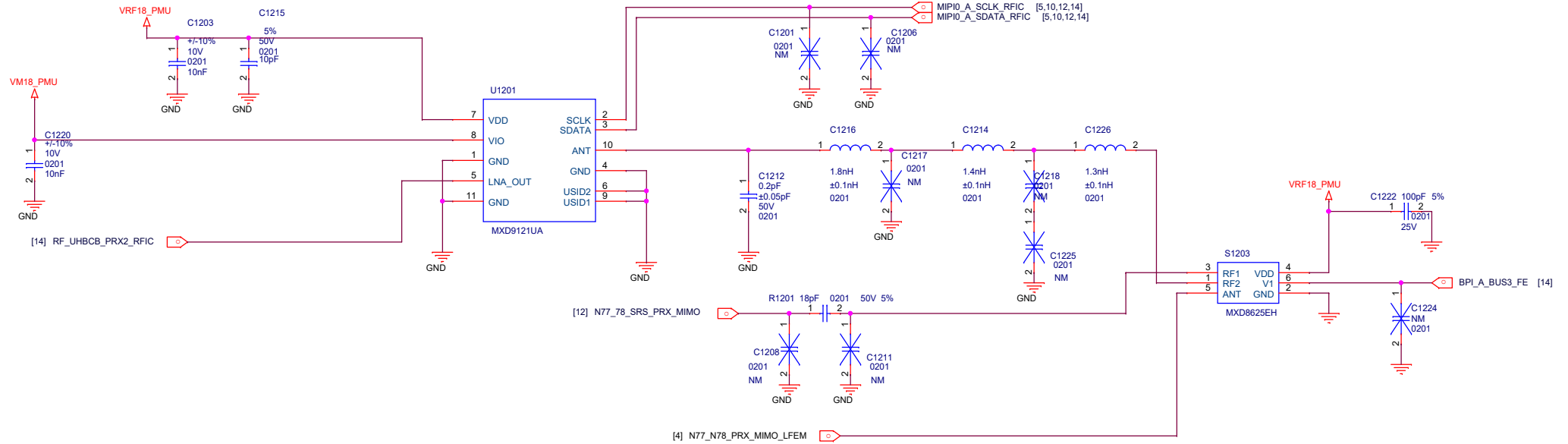
MI PHONE			
Title		MT6883_5GL	
Size	Document Number		Rev
A3	PA_5G_N41		P0
Date:	Friday, March 05, 2021		Sheet 11 of 73

NR N77/N78/N79 PAMID 0(Main)
TX1A0

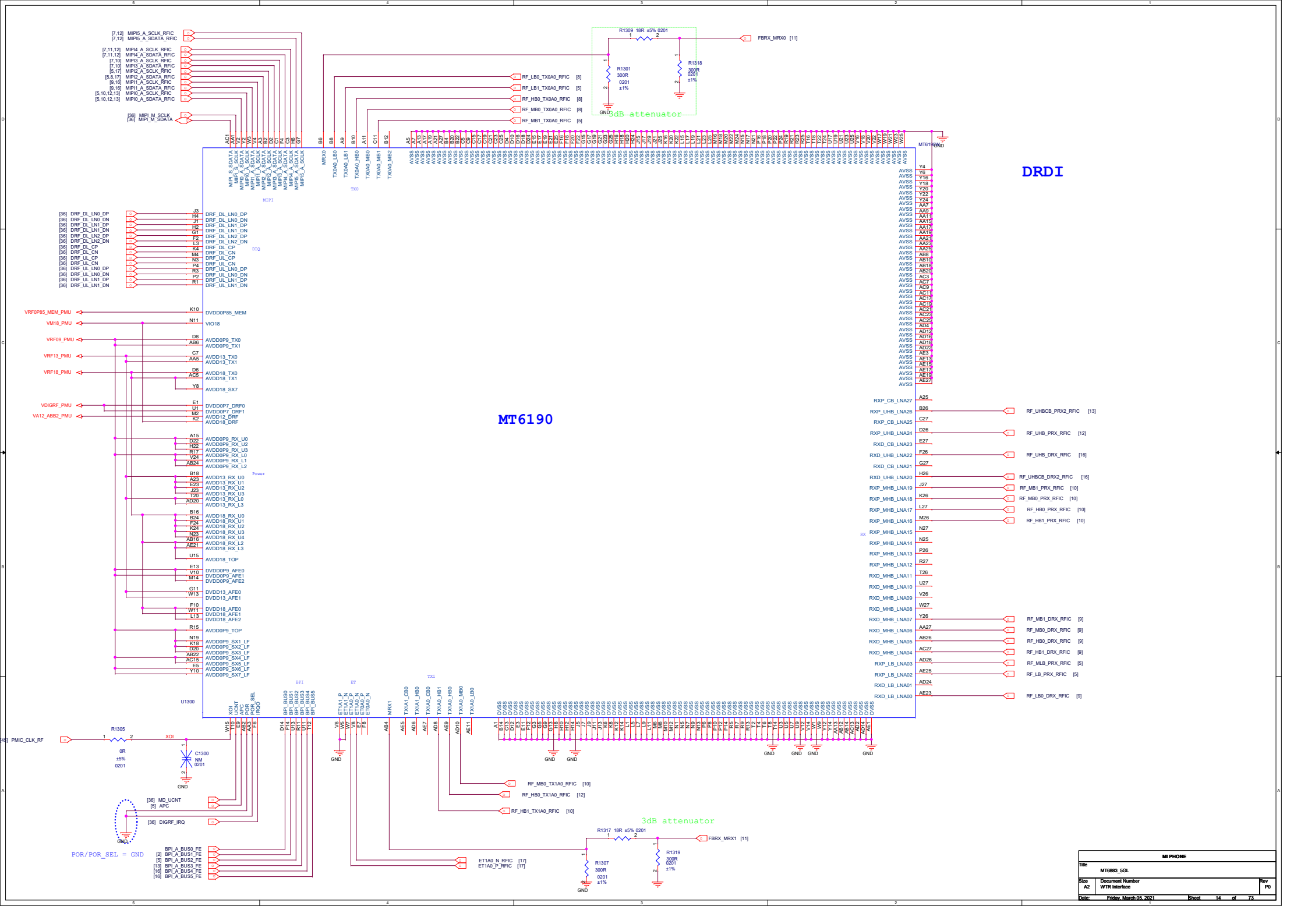


MI PHONE		
Title		
MT6883_5GL		
Size A3		
Document Number		Rev
PA_5G_TX0_N78_N79		P0
Date: Friday, March 05, 2021		
Sheet 12 of 73		

NR N77/N78/N79 PRx-MIMO LFEM



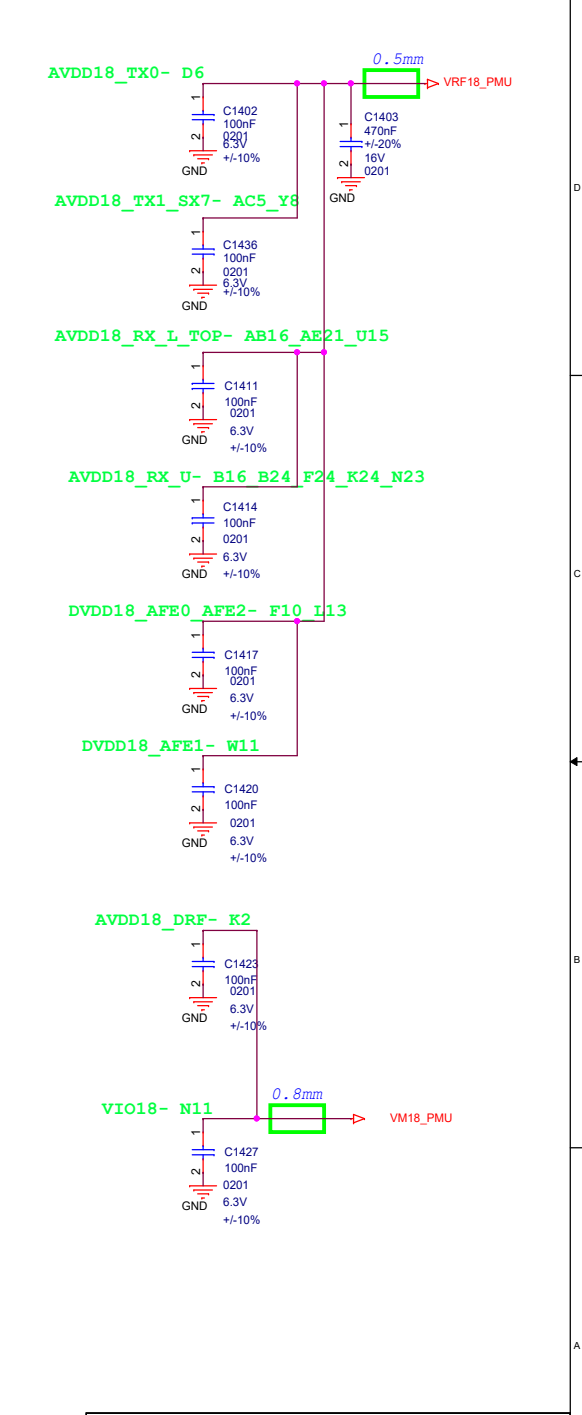
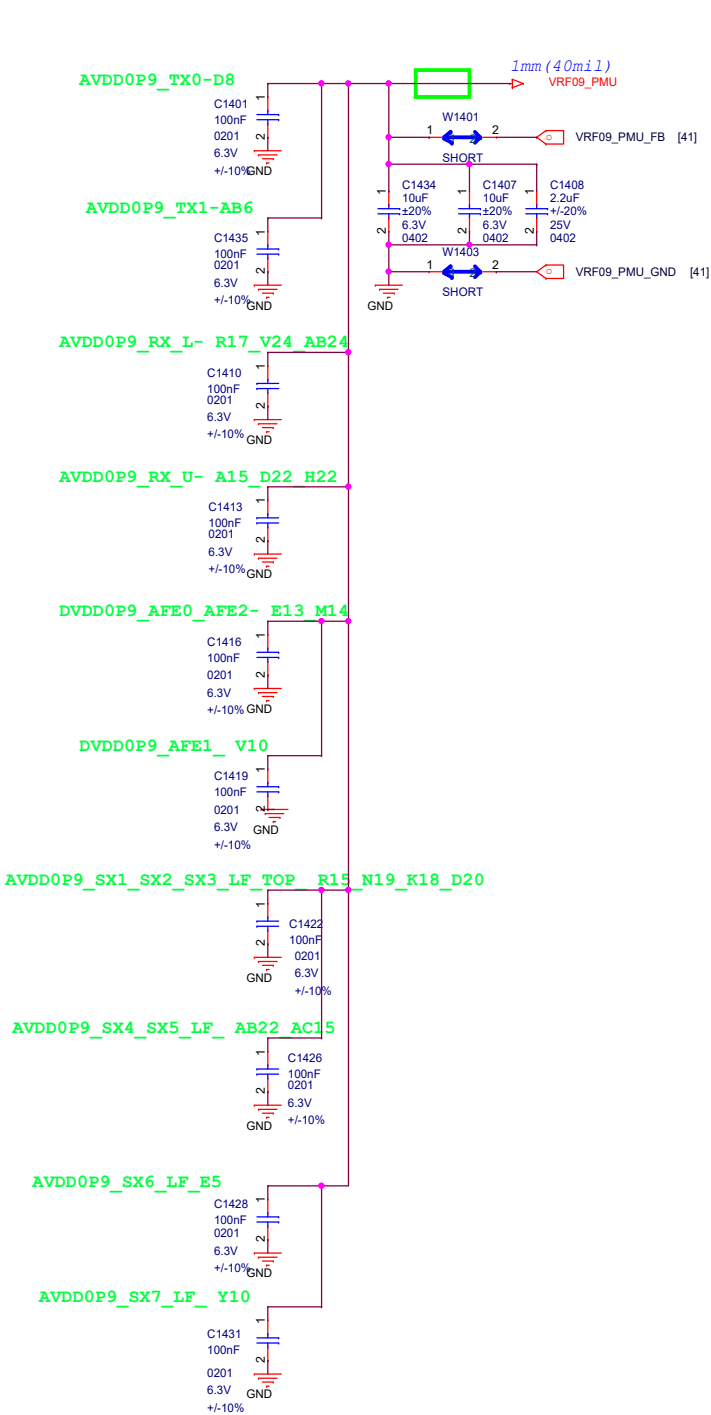
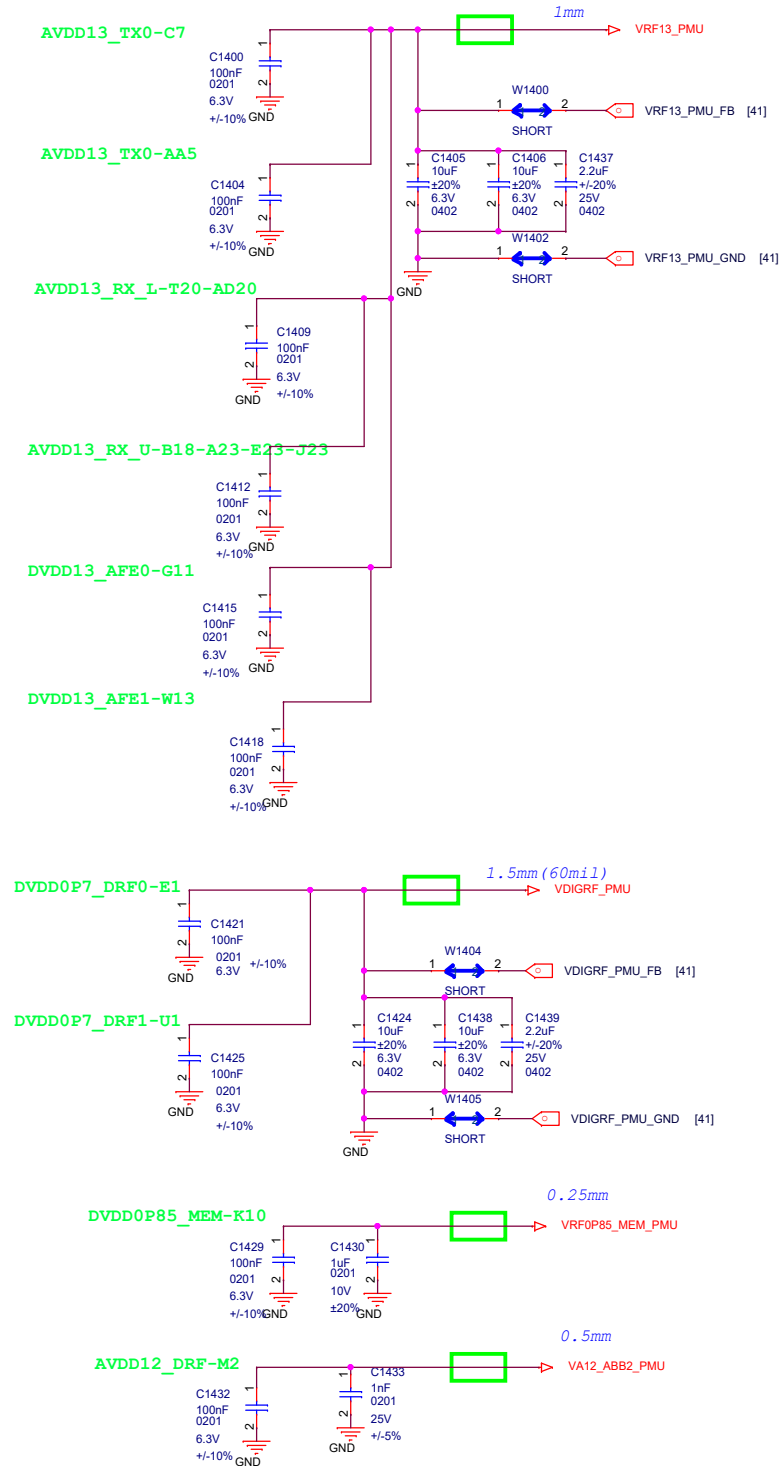
MI PHONE		
Title	MT6883_5GL	
Size	Document Number	Rev
A3		
Date:	Friday, March 05, 2021	Sheet 1 of 13



DRDI

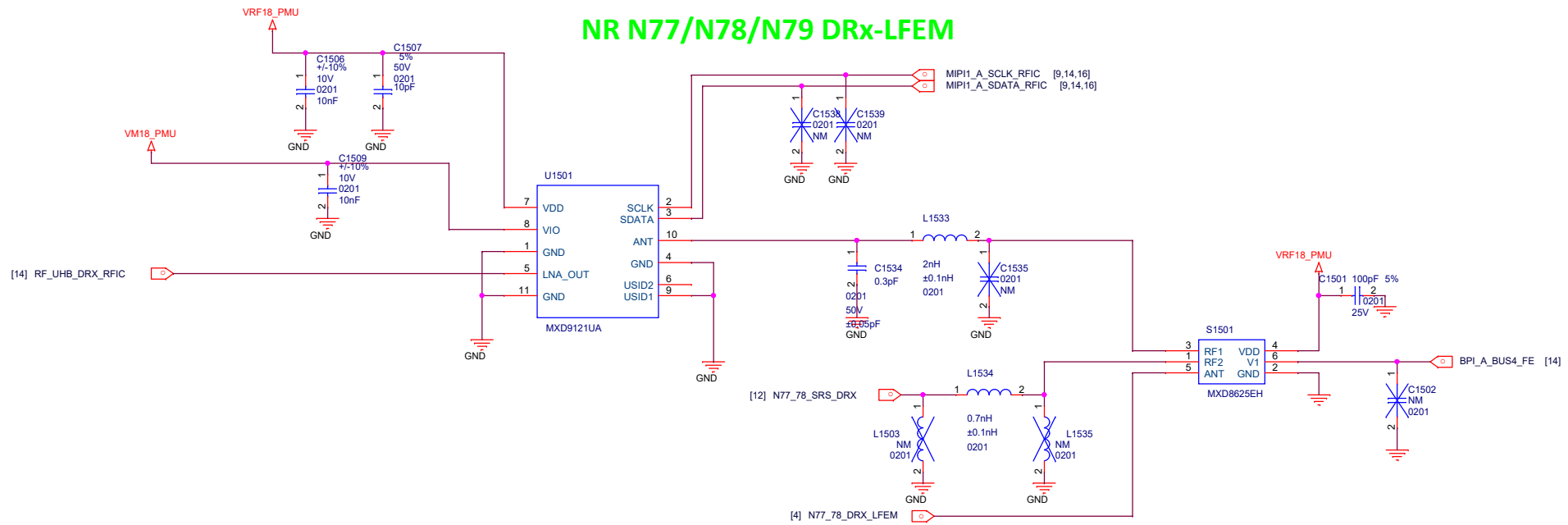
MT6190

MT PHONE			
Rev 1.0			
Size	Document Number	Rev	P0
A2	WTR Interface		
Date: Friday, March 05, 2021			
Sheet 14 of 73			

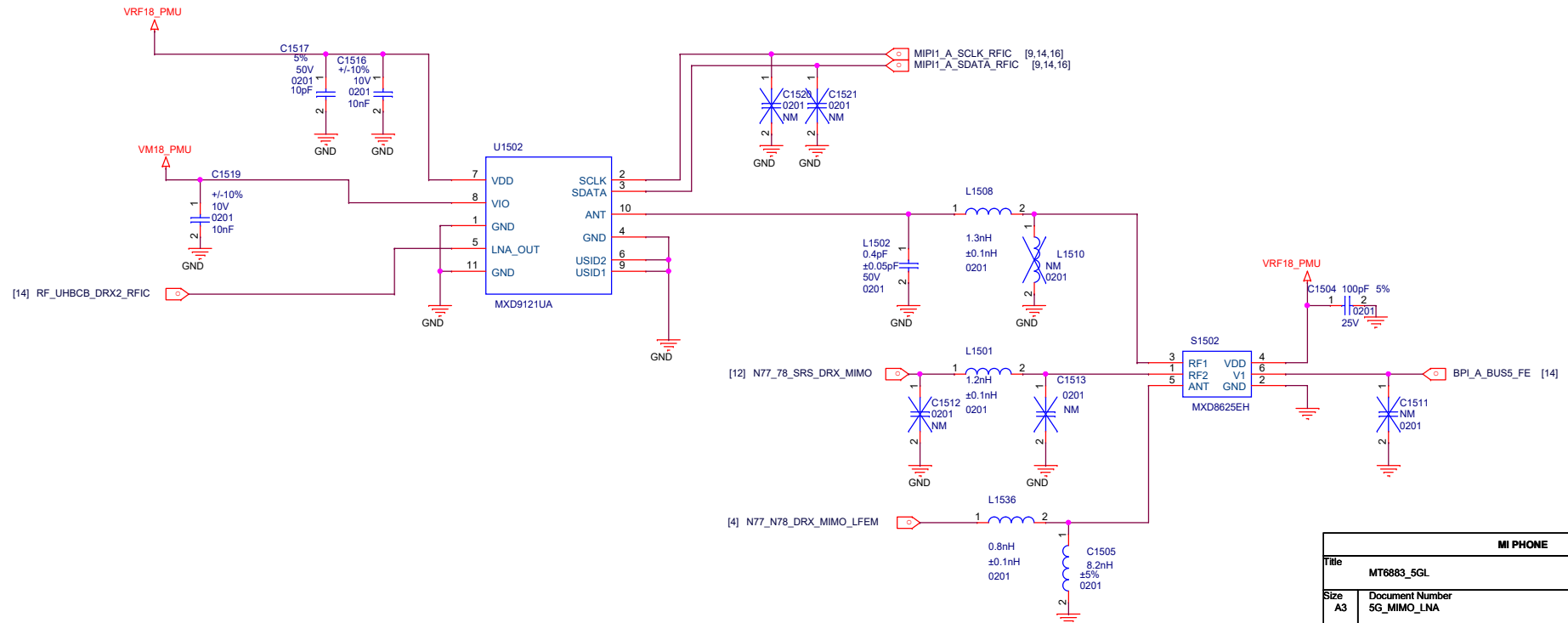


MI PHONE			
Title	MT6883_5GL		
Size A3	Document Number	Rev P0	
Date:	Friday, March 05, 2021	Sheet	15 of 73

NR N77/N78/N79 DRx-LFEM



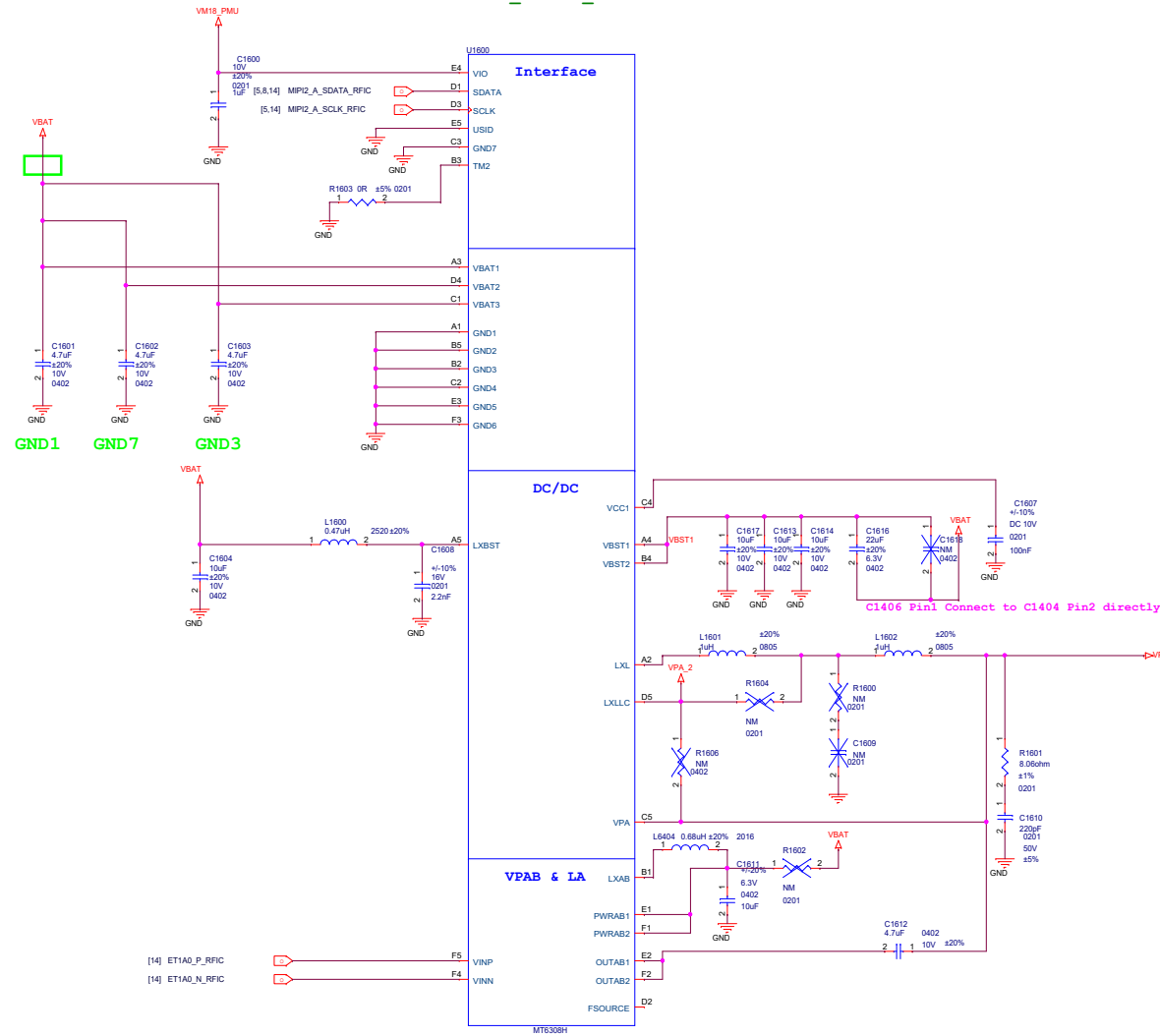
NR N77/N78/N79 DRx_MIMO-LFEM



MI PHONE			
Title	MT6883_5GL		
Size A3	Document Number	Rev P0	
Date:	Friday, March 05, 2021	Sheet	16 of 73

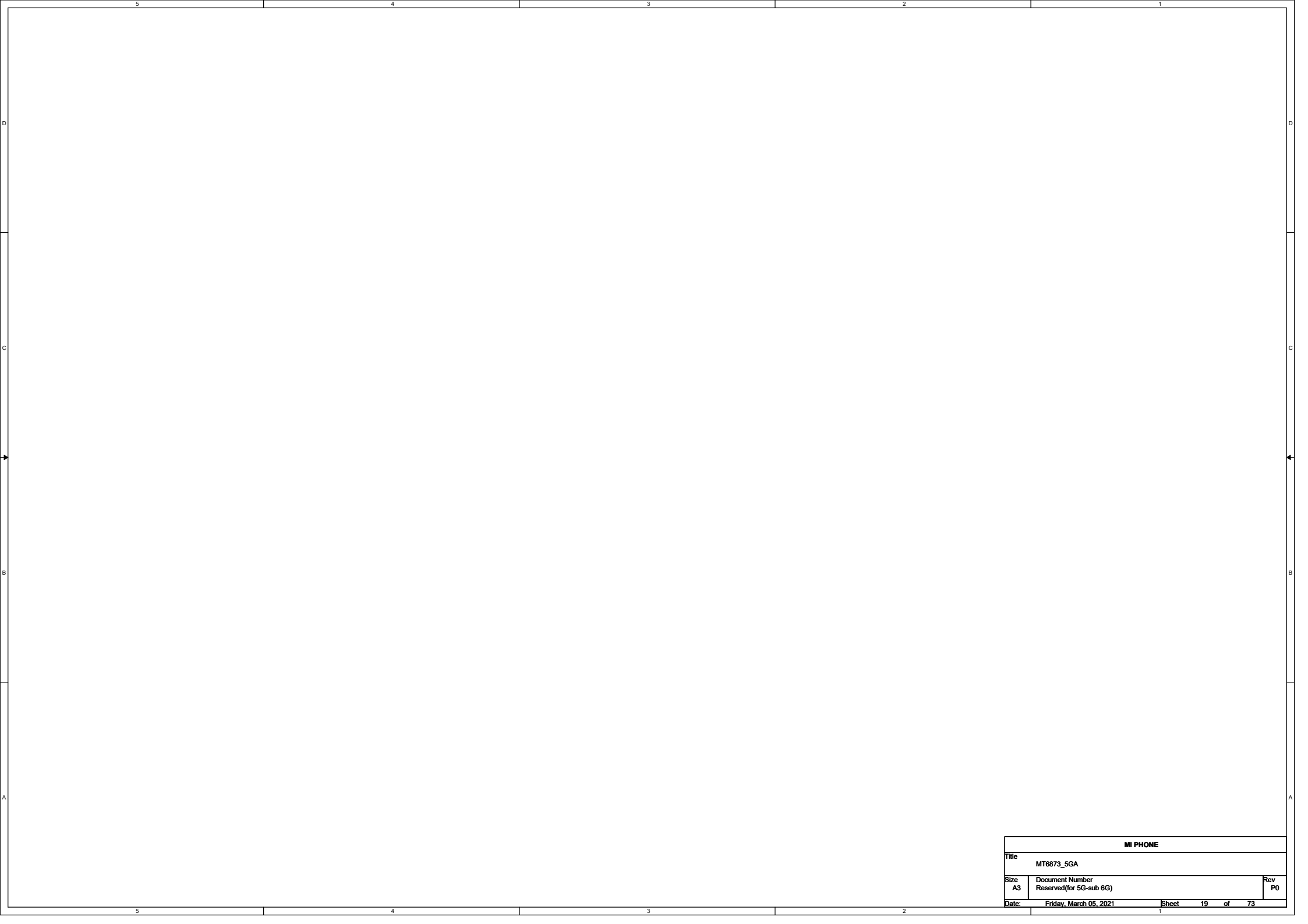
NR_PAMid_n41_SatPA_ET1_NR1/ET Ref design

L+MH_PAMId_ET

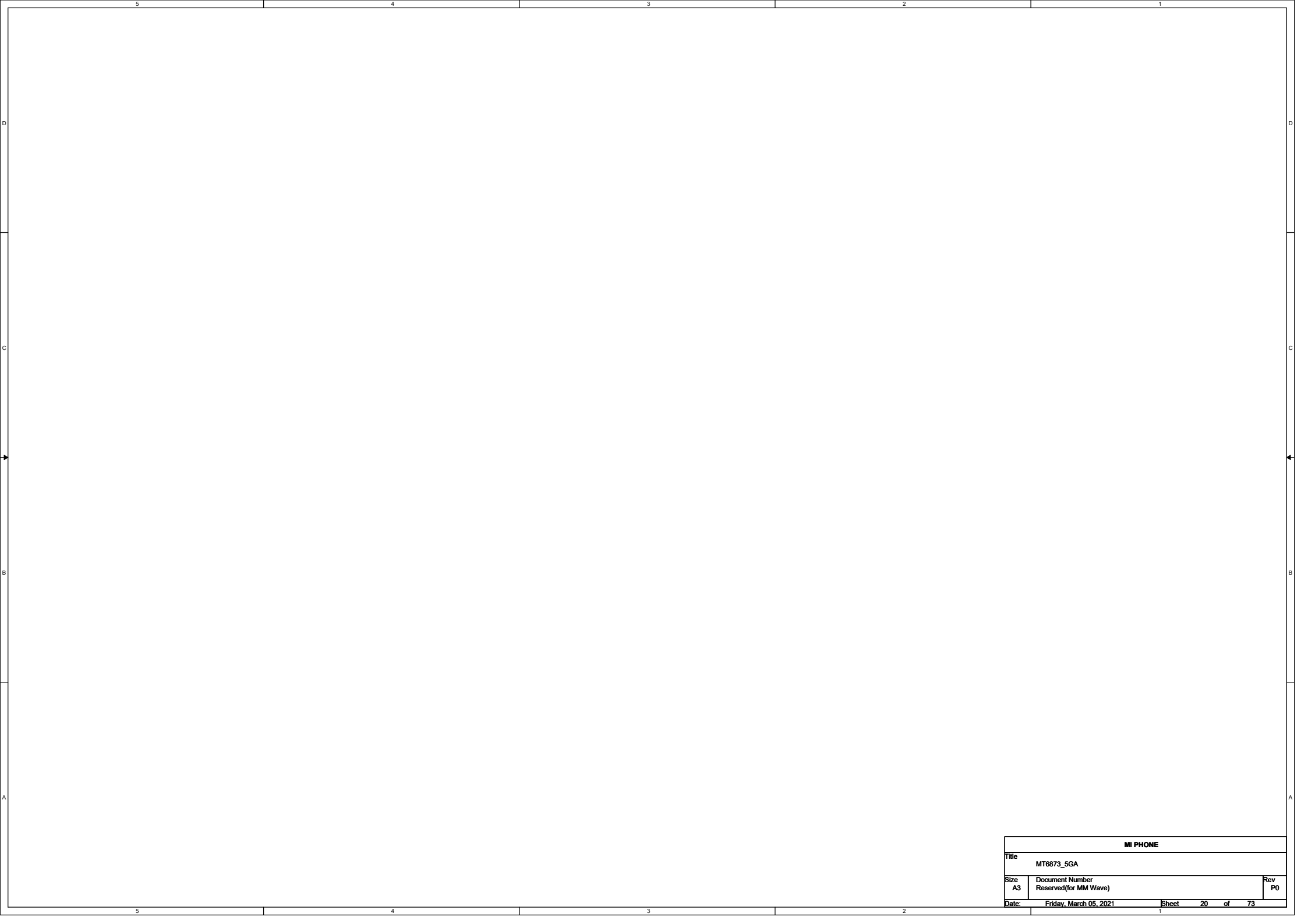


NR_PAMid + n41 PAD /APT Ref Design

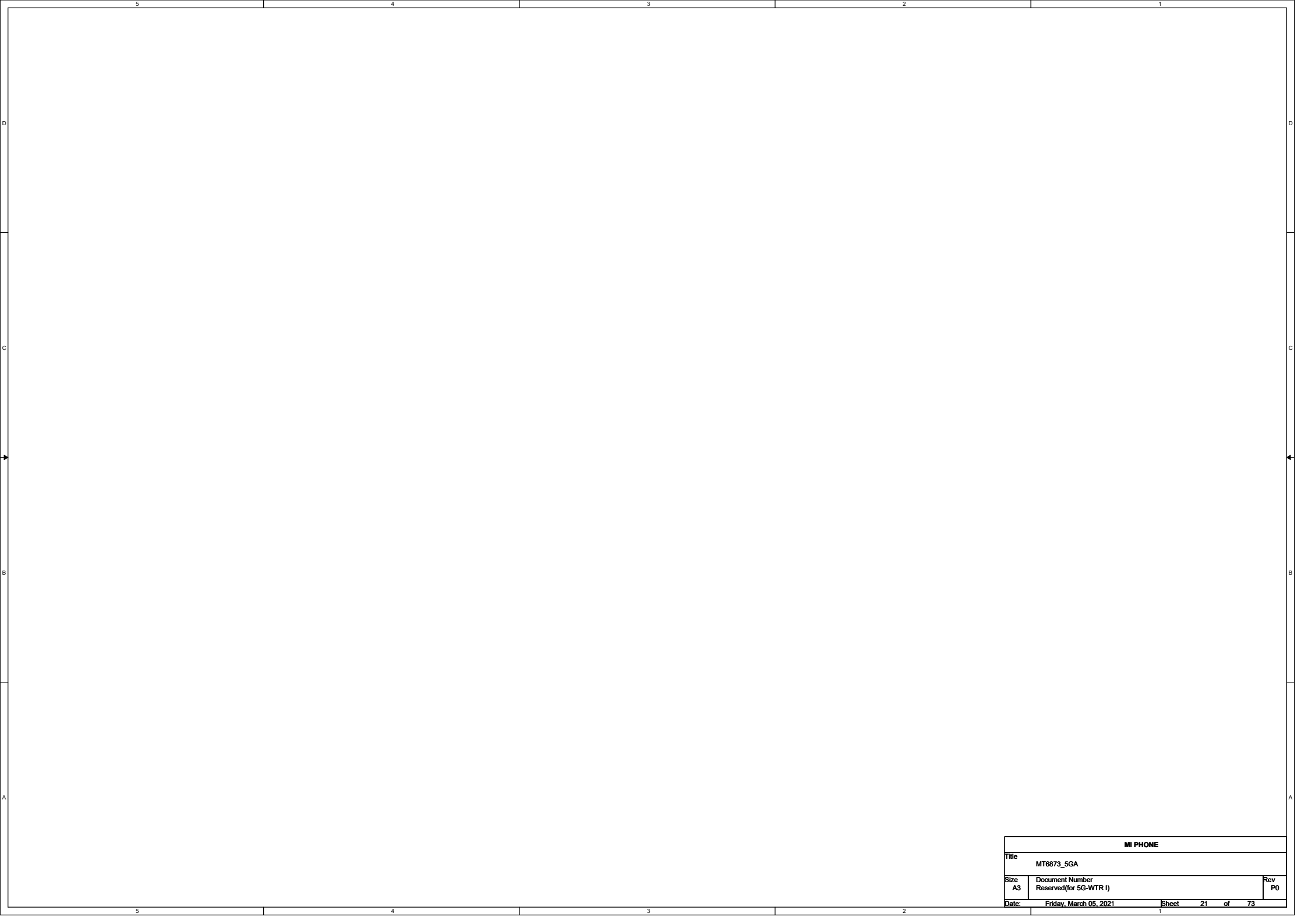
M PHONE			
Title			
MT0889_SGL			
Size	Document Number		Rev
A2	RF_SG_DCDG		P0
Date:	Friday, March 05, 2021	Sheet	18 of 73



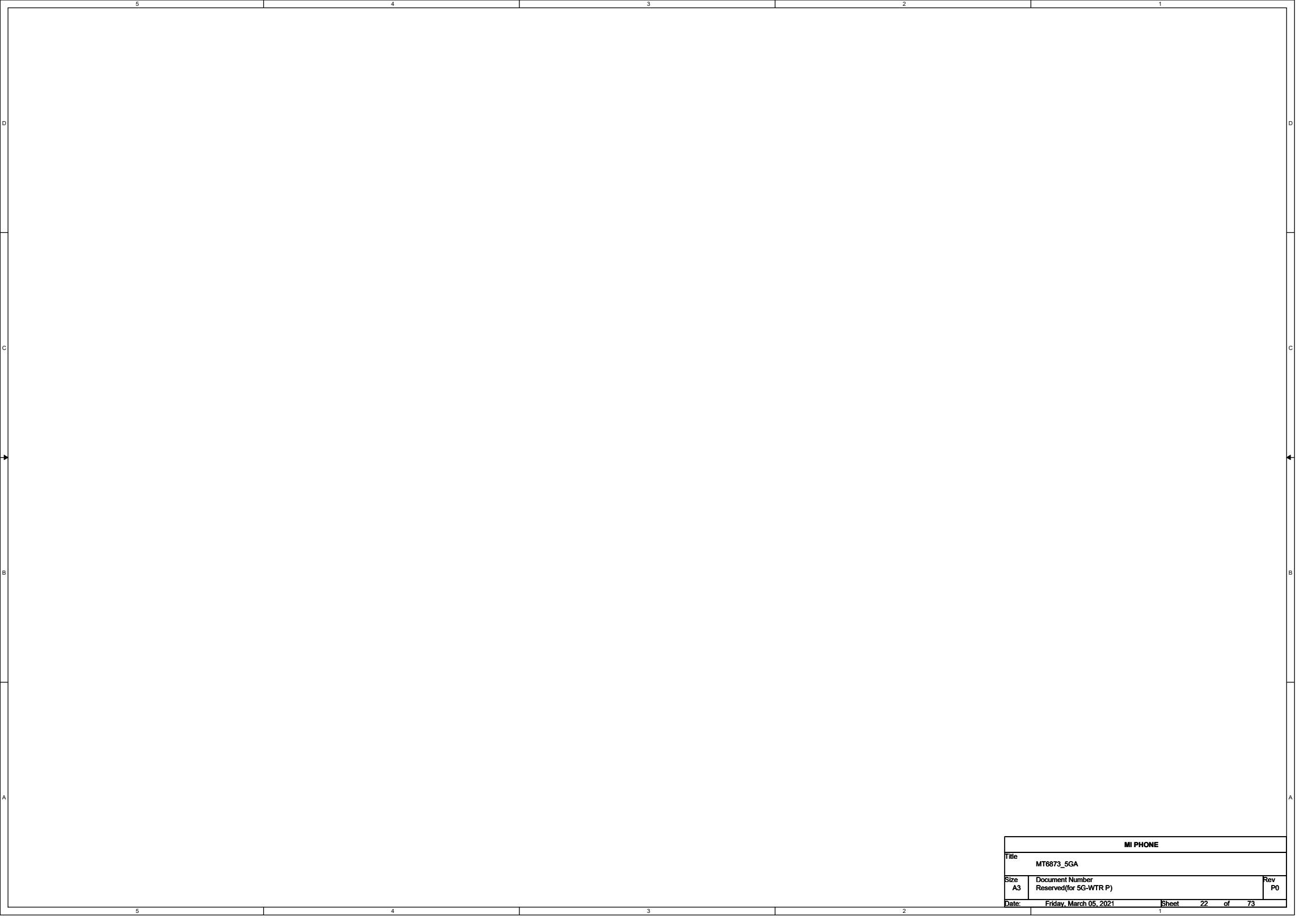
MI PHONE			
Title		MT6873_5GA	
Size	Document Number Reserved(for 5G-sub 6G)		Rev
A3			P0
Date:	Friday, March 05, 2021		Sheet 19 of 73



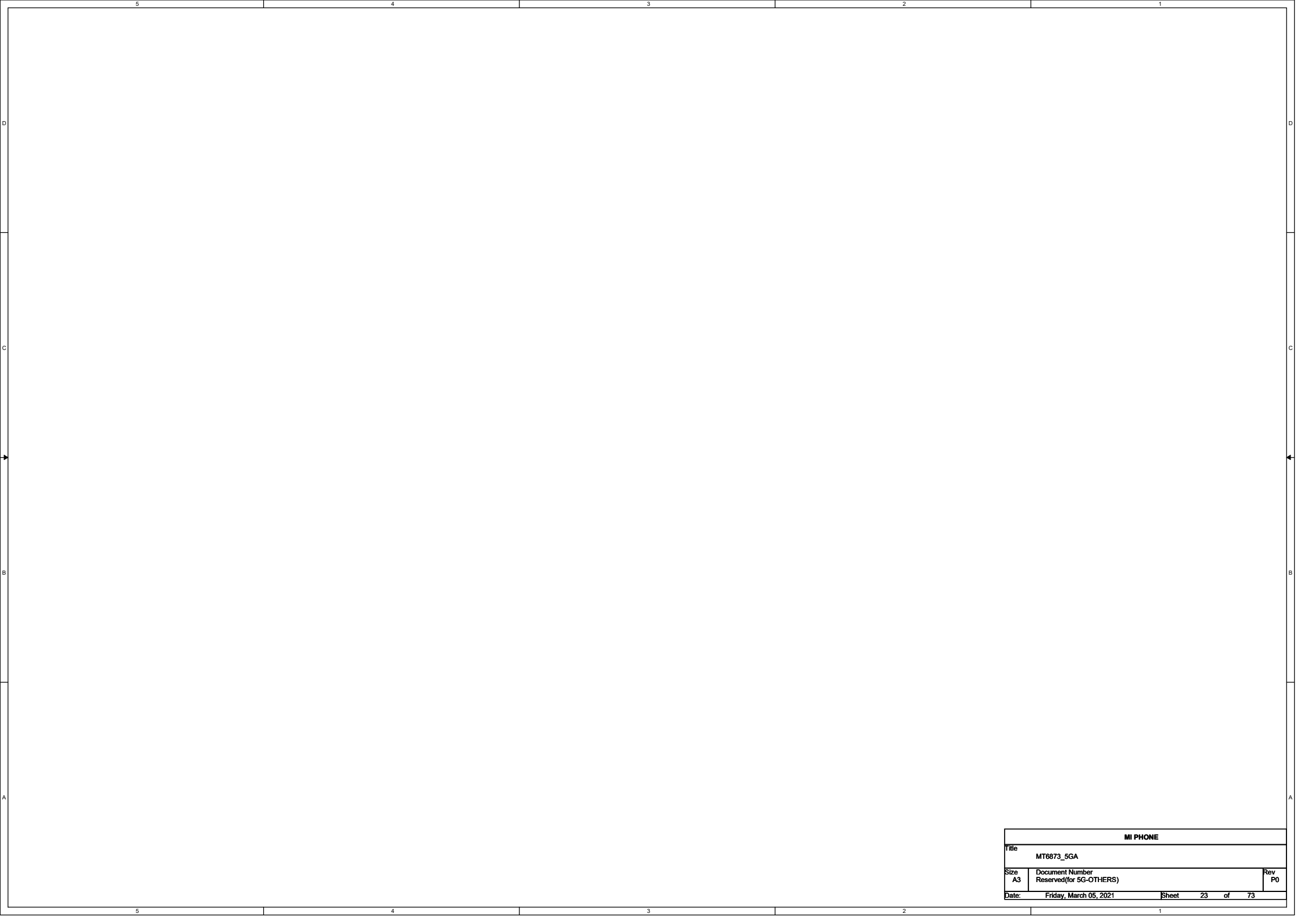
MI PHONE			
Title		MT6873_5GA	
Size	Document Number	Rev	
A3	Reserved(for MM Wave)	P0	
Date:	Friday, March 05, 2021	Sheet	20 of 73



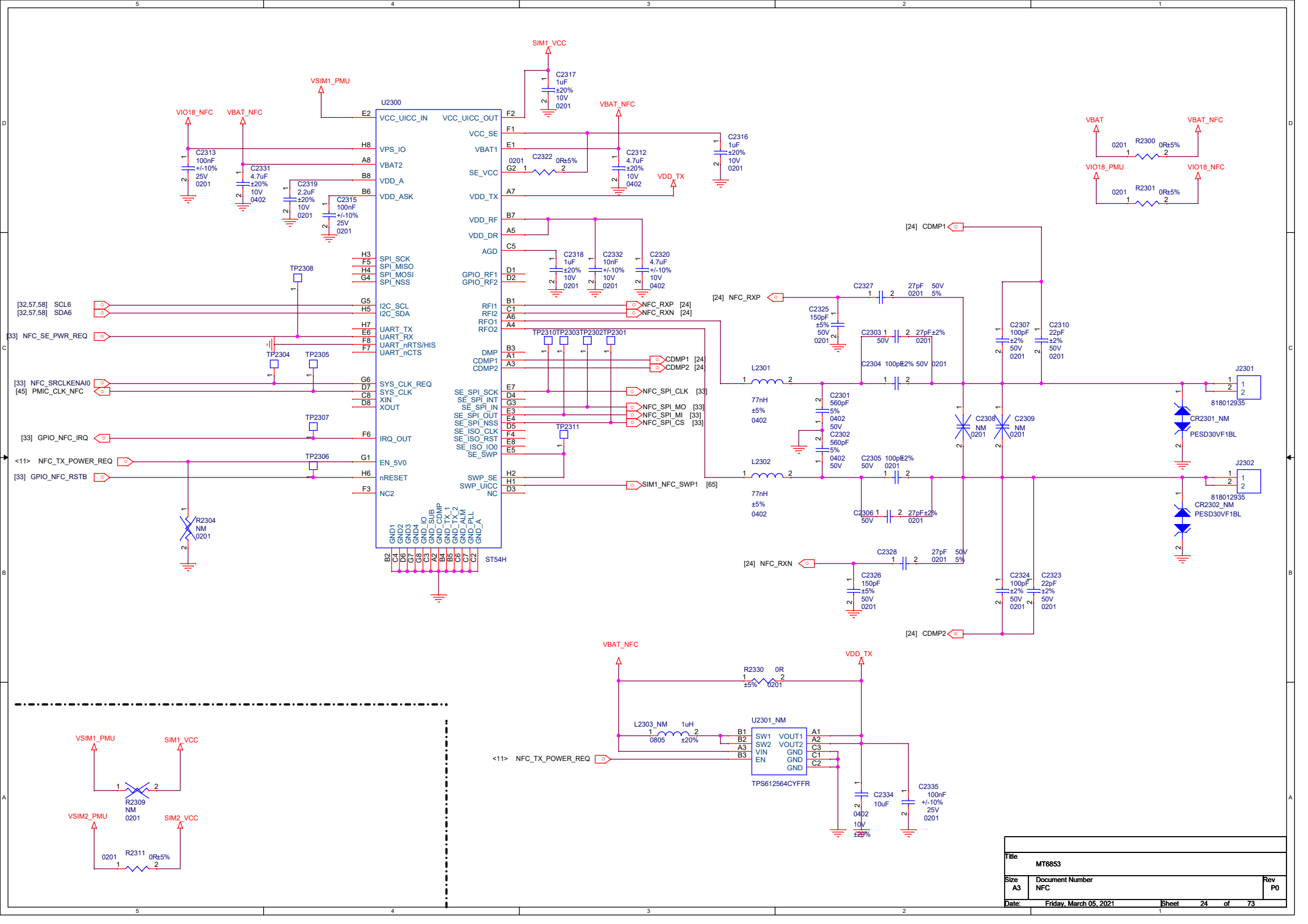
MI PHONE			
Title		MT6873_5GA	
Size	Document Number	Rev	
A3	Reserved(for 5G-WTR I)	P0	
Date:	Friday, March 05, 2021	Sheet	21 of 73

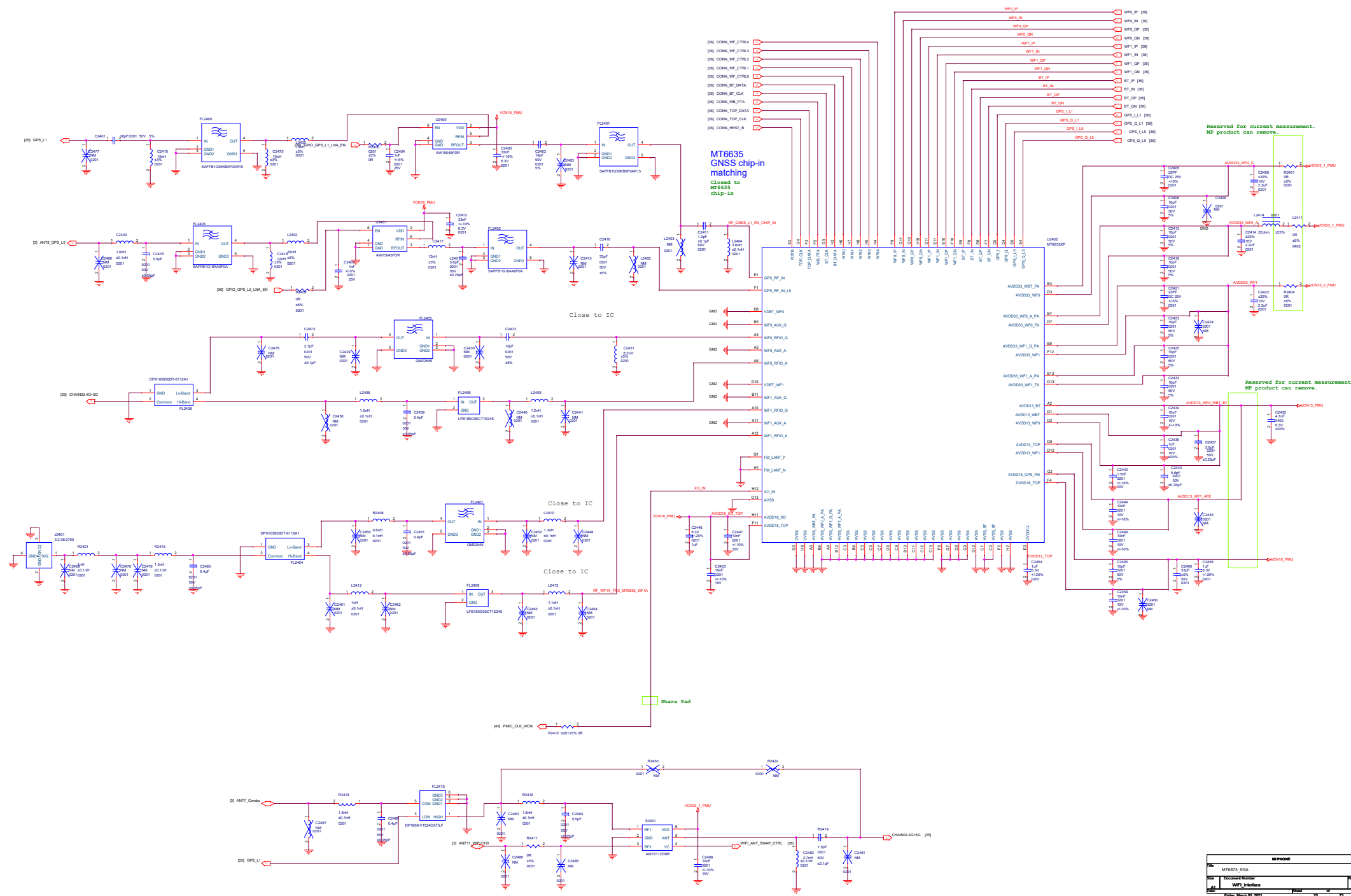


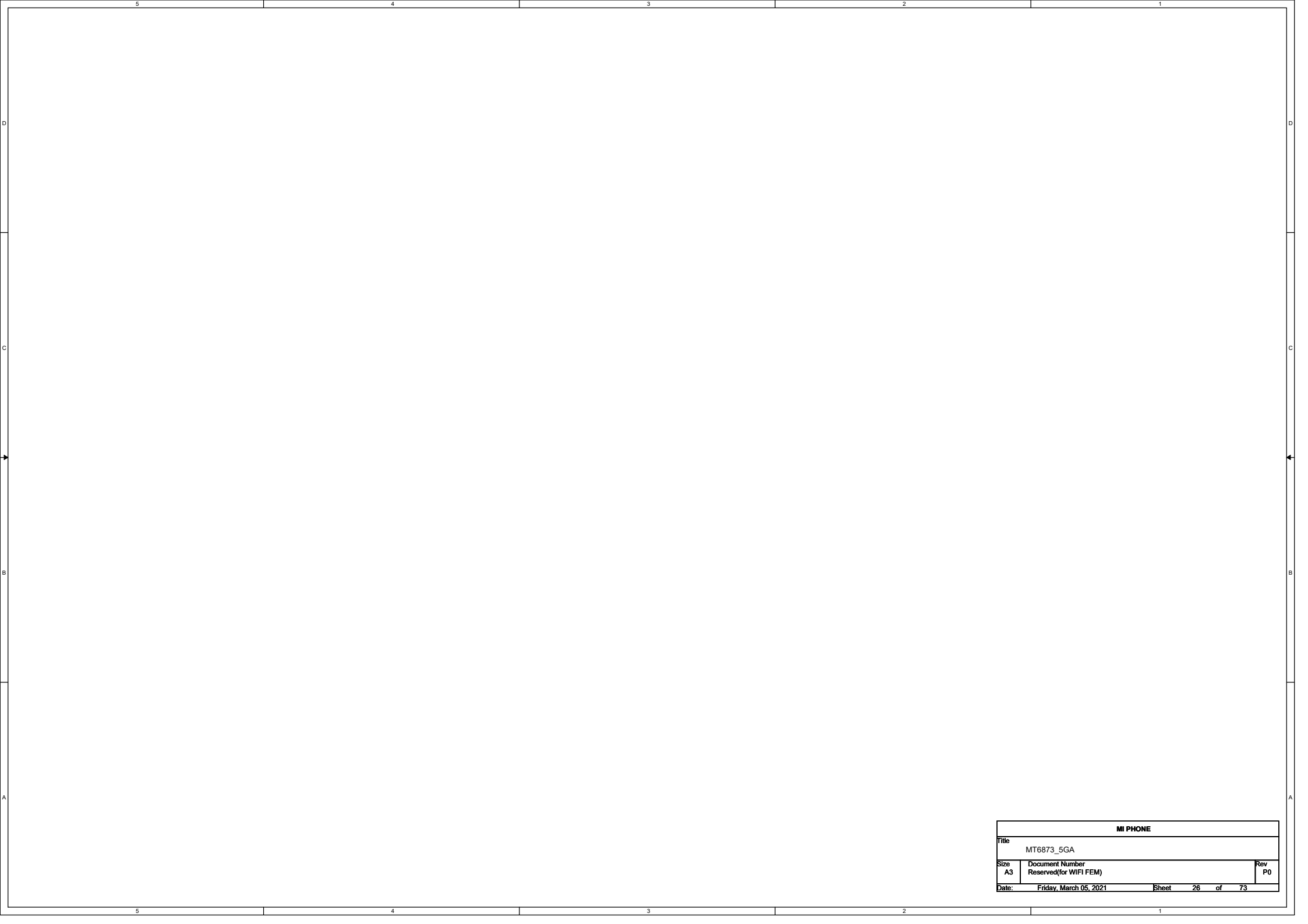
MI PHONE			
Title		MT6873_5GA	
Size	Document Number	Rev	
A3	Reserved(for 5G-WTR P)	P0	
Date:	Friday, March 05, 2021	Sheet	22 of 73



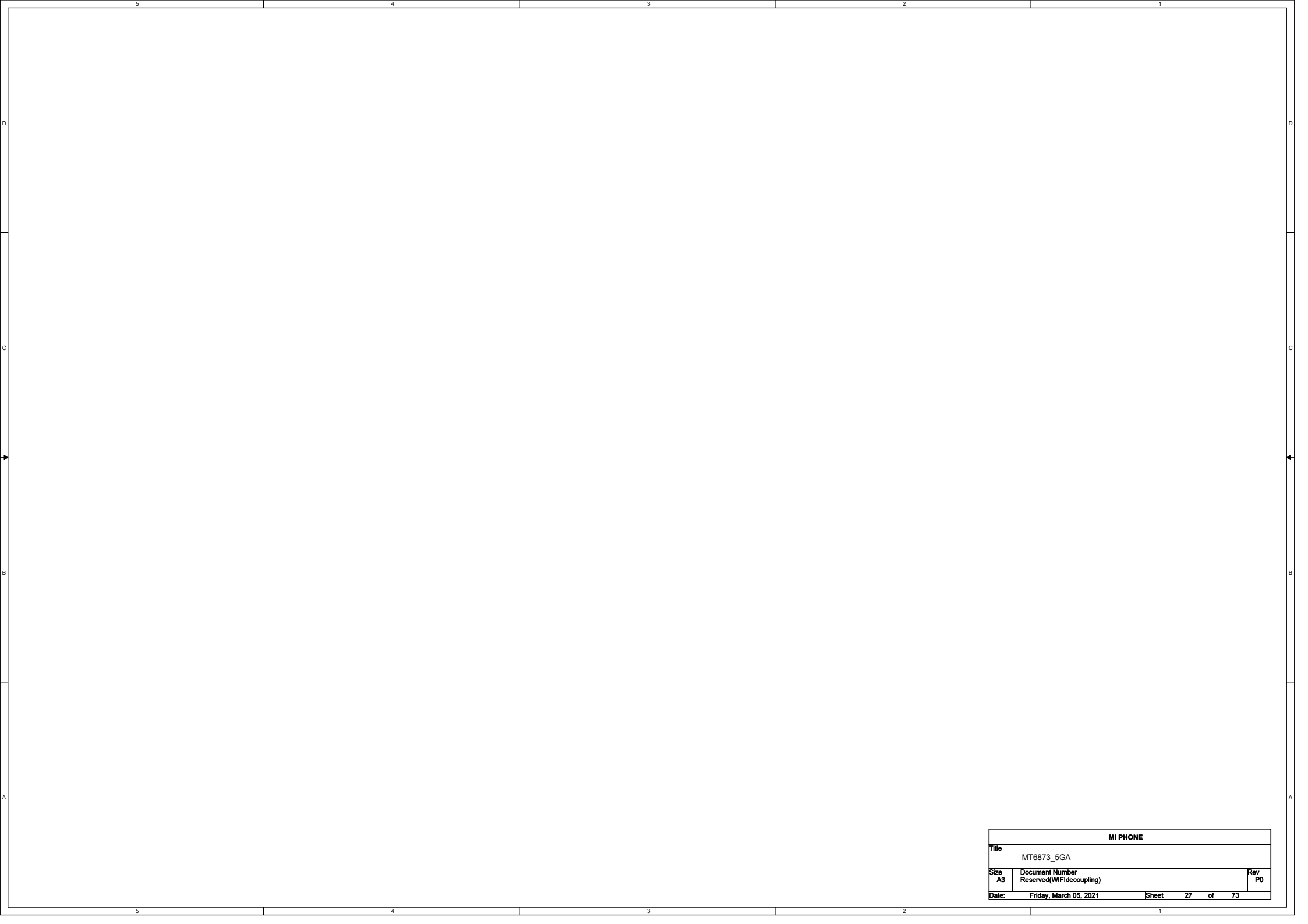
MI PHONE			
Title		MT6873_5GA	
Size	Document Number	Rev	
A3	Reserved(for 5G-OTHERS)	P0	
Date: Friday, March 05, 2021		Sheet	23 of 73



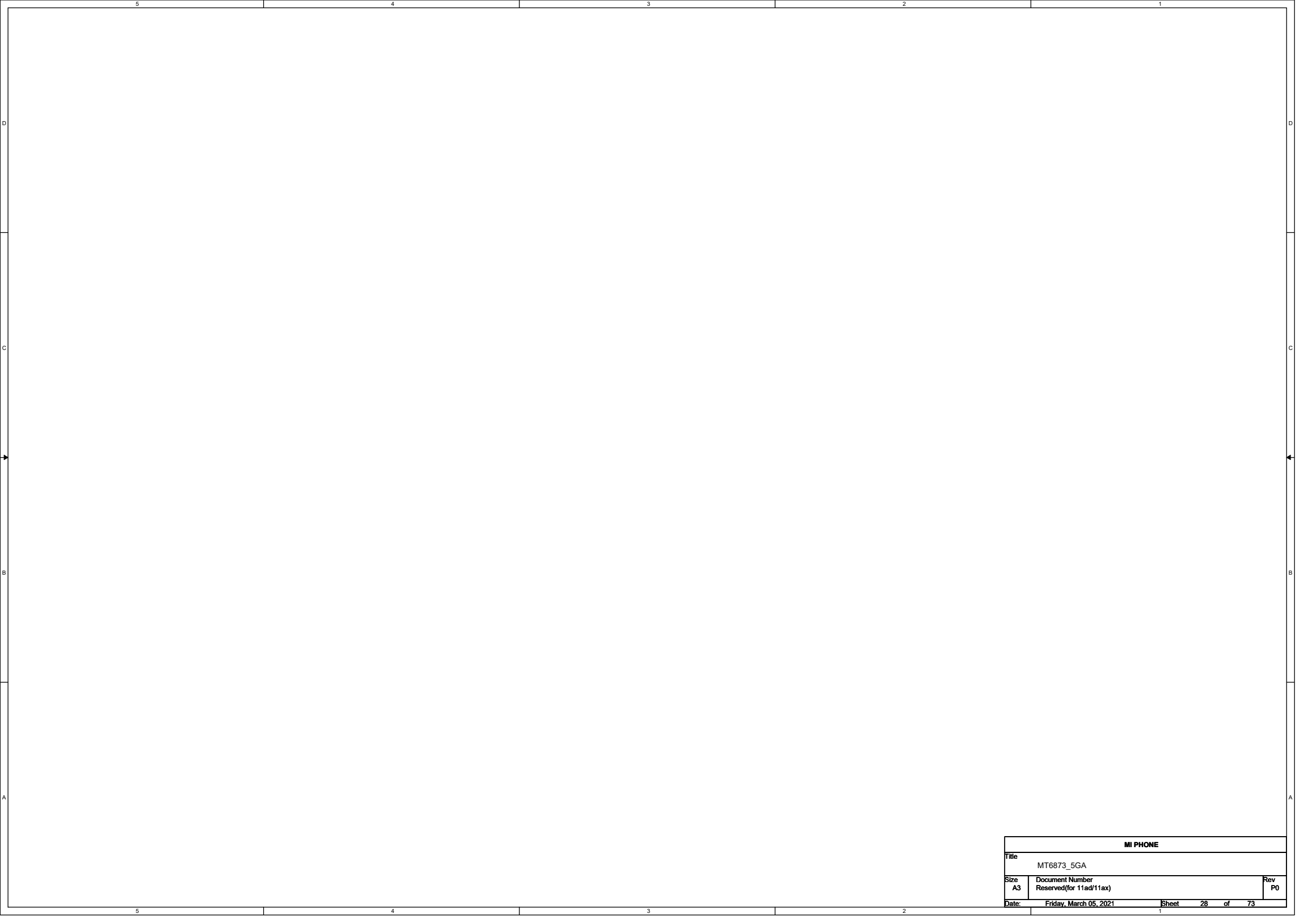




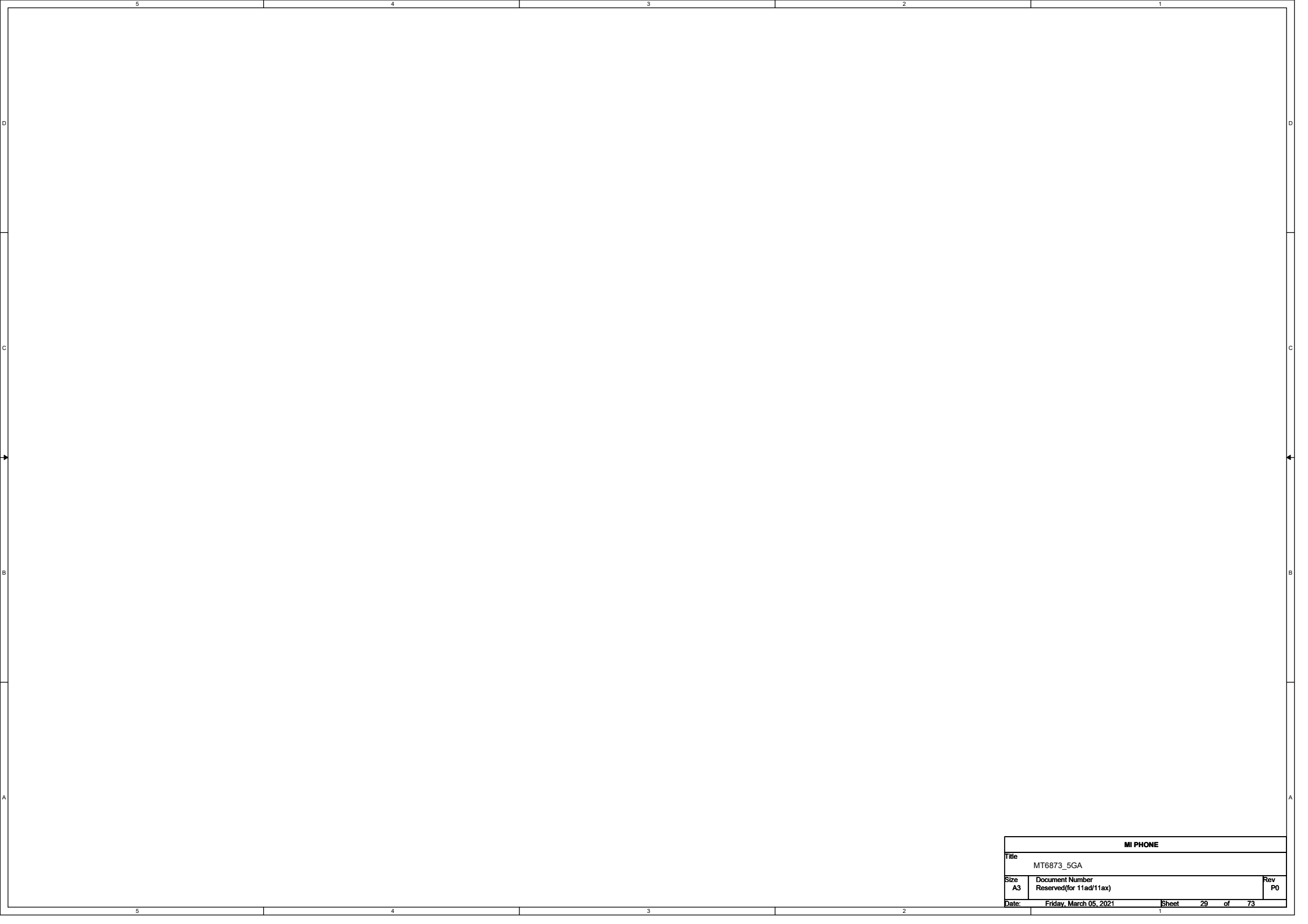
MI PHONE			
Title MT6873_5GA			
Size A3	Document Number Reserved(for WIFI FEM)		Rev P0
Date:	Friday, March 05, 2021	Sheet 26 of 73	



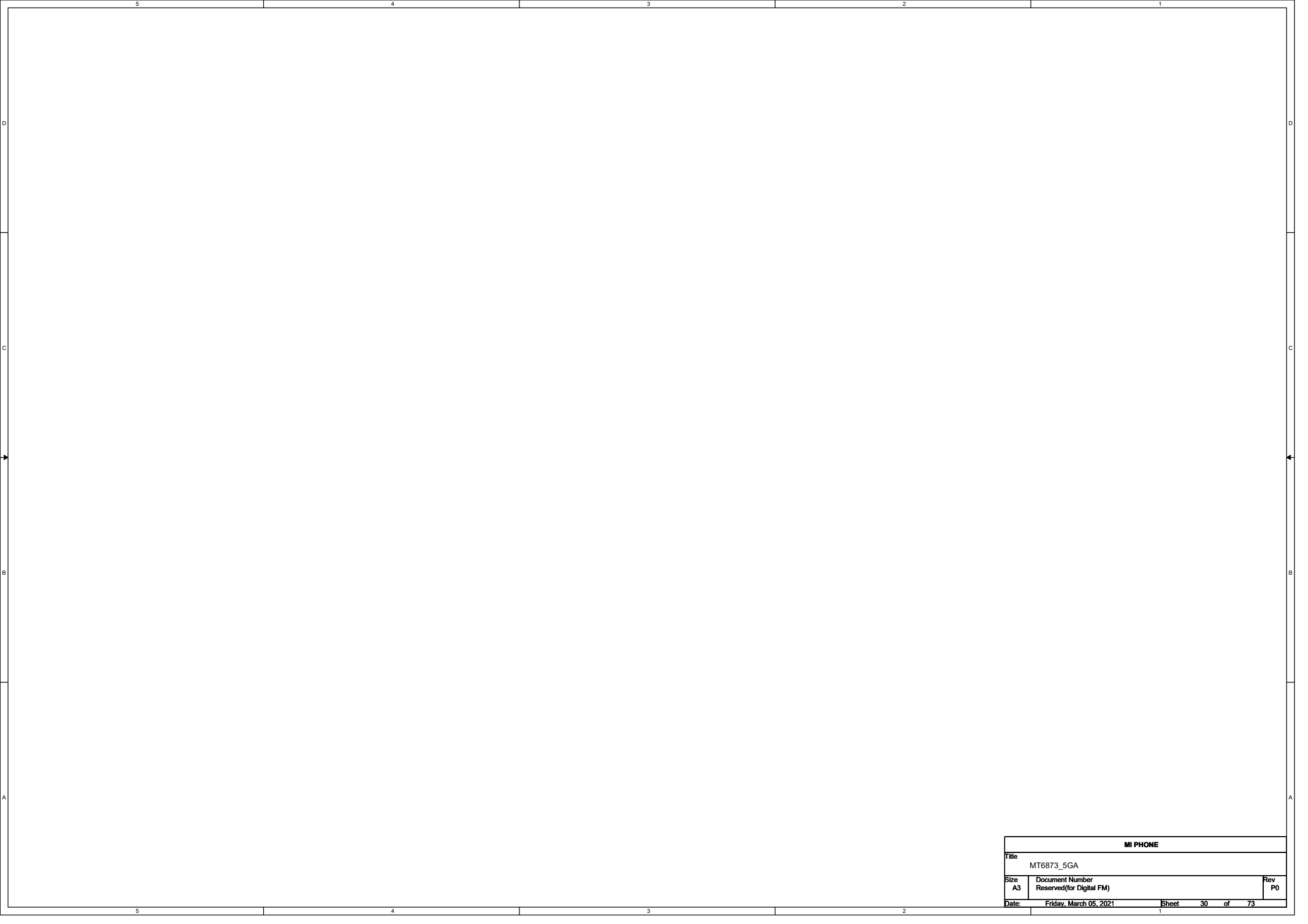
MI PHONE			
Title			
MT6873_5GA			
Size	Document Number		Rev
A3	Reserved(WIFIdecoupling)		P0
Date:	Friday, March 05, 2021	Sheet	27 of 73



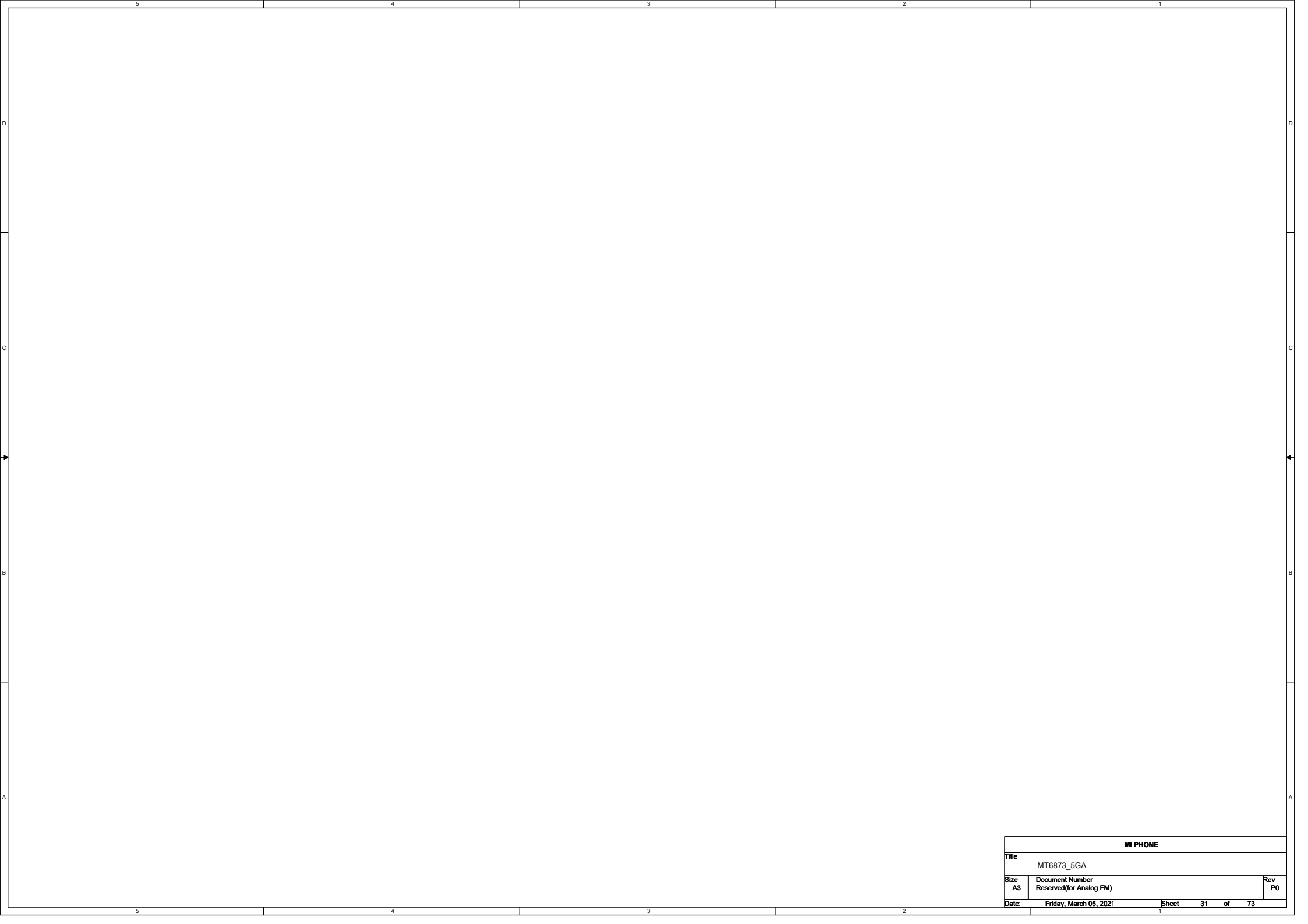
MI PHONE			
Title		MT6873_5GA	
Size	Document Number		Rev
A3	Reserved(for 11ad/11ax)		P0
Date:	Friday, March 05, 2021		Sheet 28 of 73



MI PHONE			
Title		MT6873_5GA	
Size	Document Number	Rev	
A3	Reserved(for 11ad/11ax)	P0	
Date:	Friday, March 05, 2021	Sheet	29 of 73



MI PHONE			
Title		MT6873_5GA	
Size	Document Number	Rev	
A3	Reserved(for Digital FM)	P0	
Date:	Friday, March 05, 2021	Sheet	30 of 73



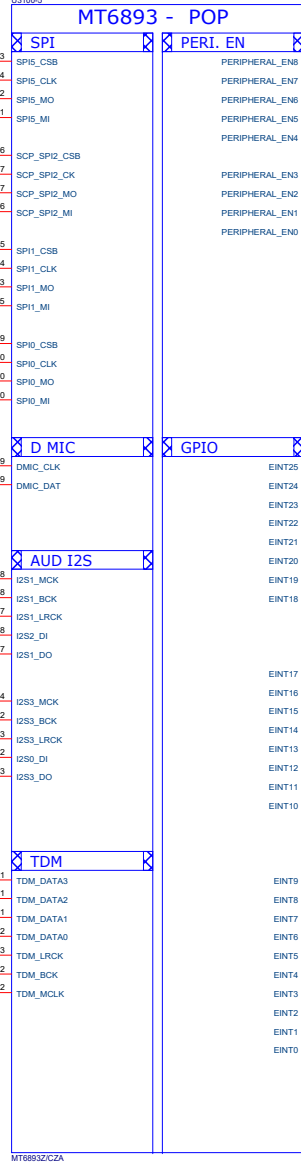
MI PHONE			
Title		MT6873_5GA	
Size	Document Number		Rev
A3	Reserved(for Analog FM)		P0
Date:	Friday, March 05, 2021	Sheet	31 of 73

JTAG TEST POINT

NFC

A+G

FOD



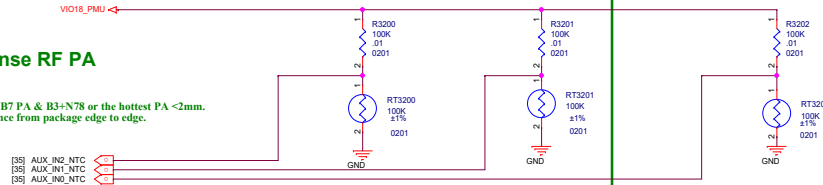
PERI. EN

GPIO



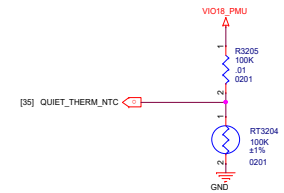
Thermistor to sense RF PA temperature

1. RT3200 & RT3201 must close to B7 PA & B3+N78 or the hottest PA <2mm.
2. The distance is the shortest distance from package edge to edge.



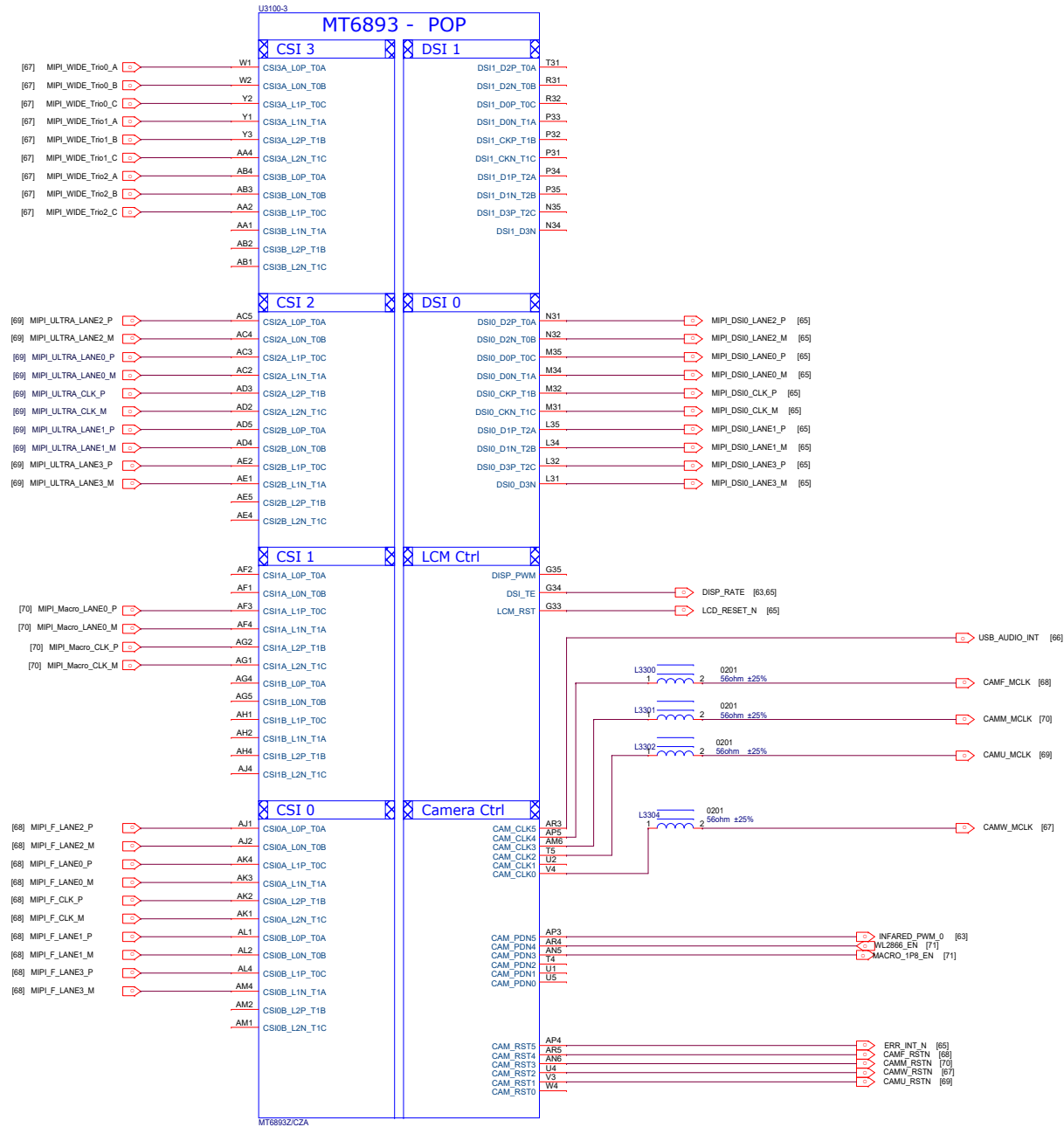
Thermistor to sense AP temperature

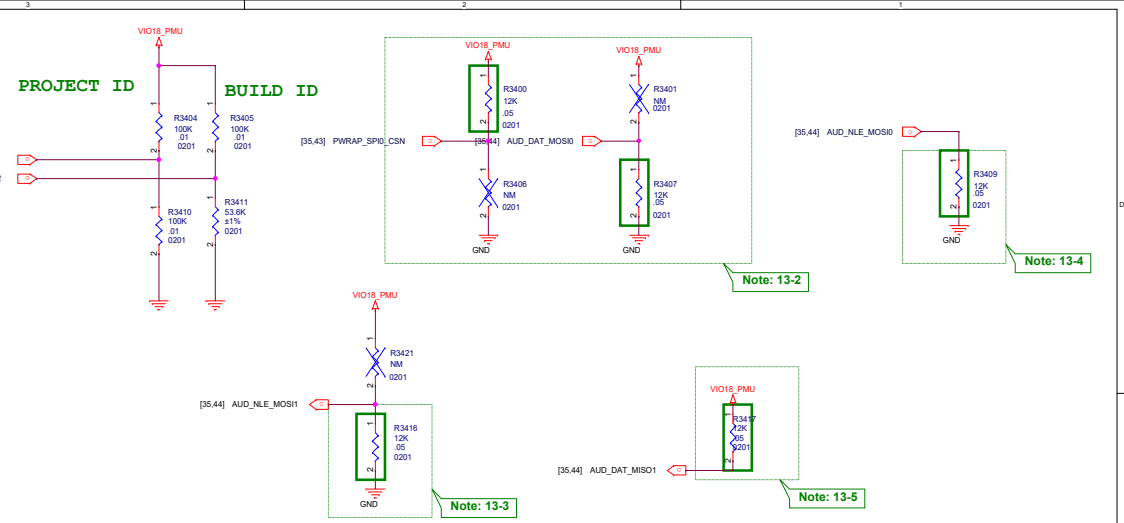
1. RT3202 must keep a distance about 6-8 mm away from AP and far from other heat sources 10 mm at least.
2. The distance is the shortest distance from package edge to edge.



Title		MT6873_5GA	
Size	Document Number	MT6893_GPIO_2	
Date		Sheet	of
Friday, March 05, 2021		33	73

2M depth, 2M Marco one csi phy



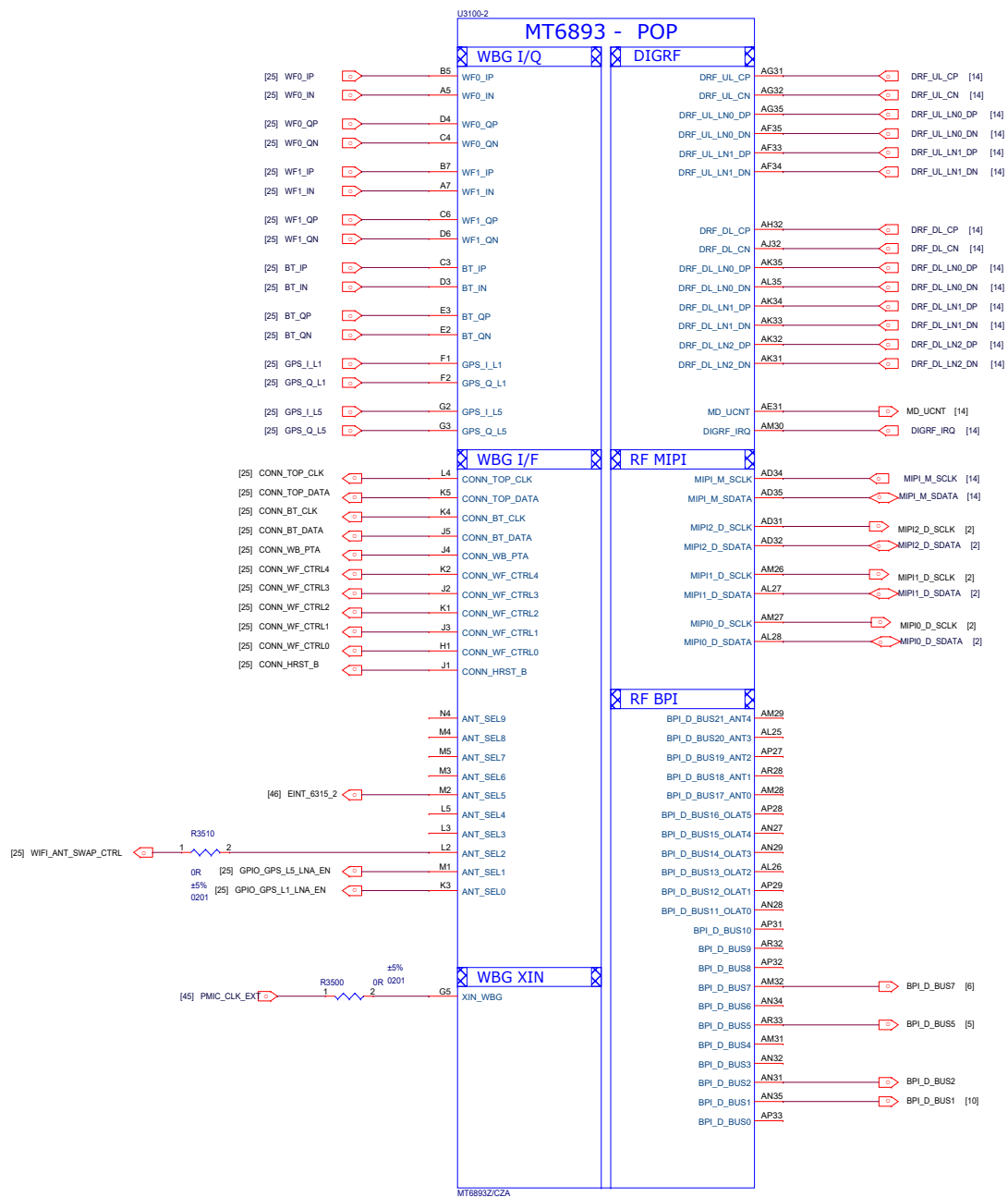


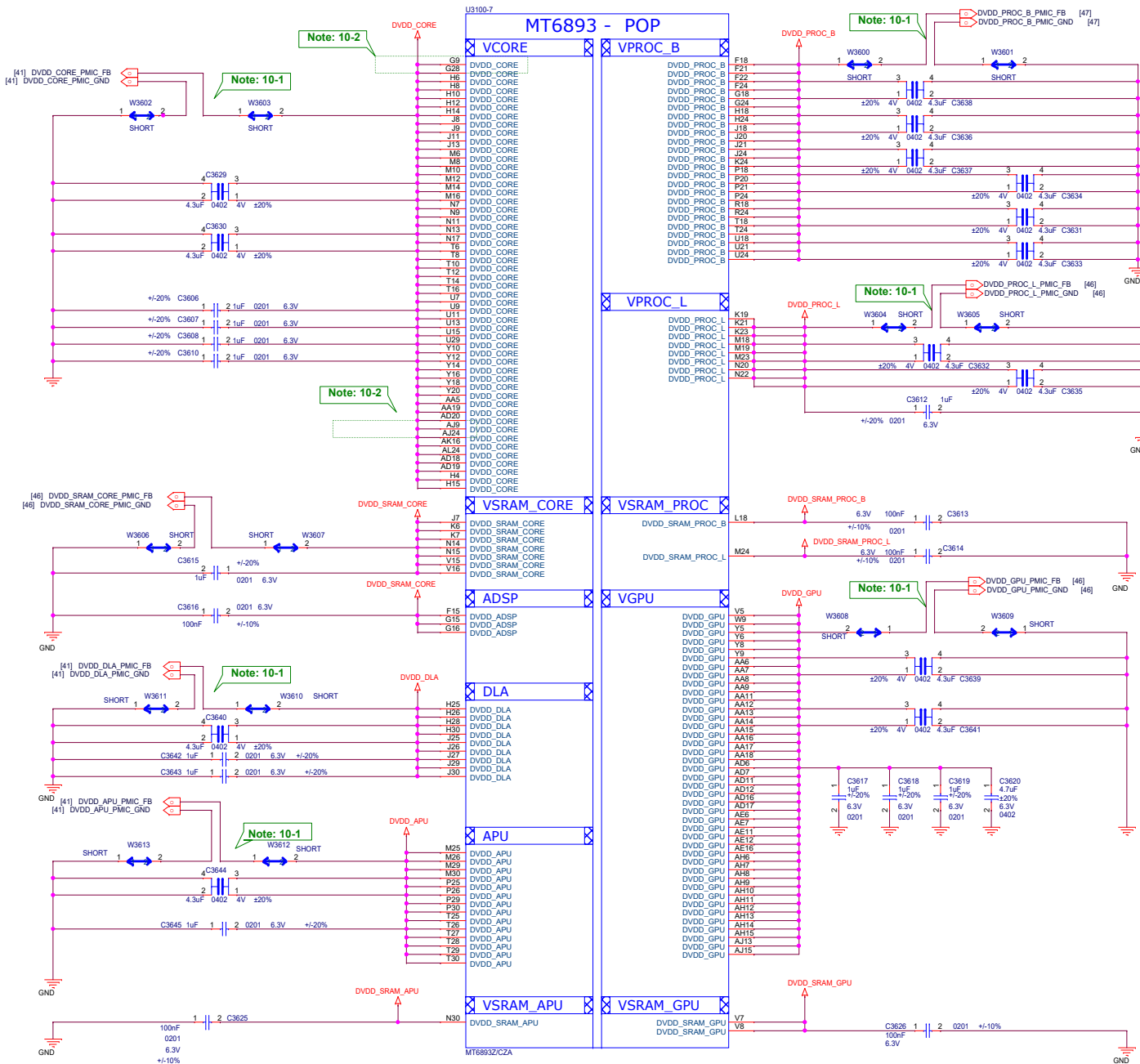
PWRAP_SPI0_CSN	AUD_DAT_MOSIO	AP JTAG	IO JTAG
H (Default)	L (Default)	N/A	N/A
H	H (by external PU)	SPI0_CSB, SPI0_CLK, SPI0_M0, SPI0_M1, DMIC_CLK	N/A
L (by external PD)	L	SPI0_CSB, SPI0_CLK, SPI0_M0, SPI0_M1, DMIC_CLK	TDM_LRCK/TDM_BCK/TDM_MCK/ TDM_DATA0/TDM_DATA1
L (by external PD)	H (by external PU)	MSDC1_CLK, MSDC1_CMD, MSDC1_DAT0, MSDC1_DAT1, MSDC1_DAT2	N/A

AUD_NLE_MOSI1	USB JTAG
L (Default)	USB 2.0
H (by external PU)	DP/DM output JTAG

AUD_NLE_MOSI0	Storage Booting
L (Default)	UFS only
H (by external PU)	eMMC only

AUD_DAT_MISO1	VEMC Voltage
L (Default)	VEMC=3.0V UFS2.1
H (by external PU)	VEMC=2.5V UFS3.0





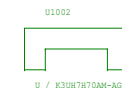
Schematic design notice of "10_BB_POWER_PDN" page.

Note 10-1: Differential pair of PMIC remote sense must be close to BB's ball.

Remote sense trace with GND shielding to PMIC (Differential)

Note 10-2: DVDD_CORE PDN cap must be under BB's ball G28/AJ24/AJ9/G9

The purpose of this symbol is used for including POP LPDDR4X in BOM.



4.3UF三端子电容可酌情替换为4.7UF和1UF的两端子电容

File	MT6873_5GA
Size	Document Number
C	MT6893_PWR1
Date	Friday, March 05, 2021
	Sheet

U3100-9		
MT6893 - POP		
GND		GND
G1 DVSS		Y31
G4 DVSS		Y34
G7 DVSS		AA3
G8 DVSS		AA32
G13 DVSS		AB6
G14 DVSS		AB7
G17 DVSS		AB11
G19 DVSS		AB12
G23 DVSS		AB16
G25 DVSS		AB17
G26 DVSS		AB18
G27 DVSS		AB20
G29 DVSS		AB22
H2 DVSS		AB24
H3 DVSS		AB26
H5 DVSS		AB28
H16 DVSS		AB29
H19 DVSS		AB30
H23 DVSS		AC1
J15 DVSS		AC6
J17 DVSS		AC7
J19 DVSS		AC11
J22 DVSS		AC12
J23 DVSS		AC16
K6 DVSS		AC17
K10 DVSS		AC18
K12 DVSS		AC19
K14 DVSS		AC21
K16 DVSS		AC23
K18 DVSS		AC25
K20 DVSS		AC27
K22 DVSS		AC32
K26 DVSS		AD1
K30 DVSS		AE3
K31 DVSS		AE17
L6 DVSS		AE32
L7 DVSS		AE33
L9 DVSS		AE34
L11 DVSS		AE35
L13 DVSS		AF5
L15 DVSS		AF6
L17 DVSS		AF7
L19 DVSS		AF11
L23 DVSS		AF12
L24 DVSS		AF16
L25 DVSS		AF17
L26 DVSS		AF18
L30 DVSS		AF19
L33 DVSS		AF20
M33 DVSS		AF22
N18 DVSS		AF24
N19 DVSS		AF26
N21 DVSS		AF28
N23 DVSS		AF31
N24 DVSS		AF32
N25 DVSS		AG3
N26 DVSS		AG6
N29 DVSS		AG7
N33 DVSS		AG8
P6 DVSS		AG9
P8 DVSS		AG10
P10 DVSS		AG11
P12 DVSS		AG12
P14 DVSS		AG13
P16 DVSS		AG14
P19 DVSS		AG15
P22 DVSS		AG16
P23 DVSS		AG17
R7 DVSS		AG18
R9 DVSS		AG20
R11 DVSS		AG21
R13 DVSS		AG24
R15 DVSS		AG25
R19 DVSS		AG27
R23 DVSS		AG28
R25 DVSS		AG30
R26 DVSS		AG33
R27 DVSS		AG34
R28 DVSS		AH3
R29 DVSS		AH6
R30 DVSS		AH11
R33 DVSS		AH33
T19 DVSS		AH34
T23 DVSS		AJ3
T32 DVSS		AJ5
T33 DVSS		AJ6
U19 DVSS		AJ7
U20 DVSS		AJ10
U22 DVSS		AJ11
U23 DVSS		AJ12
U25 DVSS		AJ14
U26 DVSS		AJ16
U27 DVSS		AJ23
U28 DVSS		AJ25
U30 DVSS		AJ27
U31 DVSS		AJ29
U33 DVSS		AJ31
V2 DVSS		AJ33
V6 DVSS		AJ34
V6 DVSS		AK12
V10 DVSS		AK14
V12 DVSS		AK15
V14 DVSS		AK17
V18 DVSS		AK18
V20 DVSS		AK19
W5 DVSS		AK20
W6 DVSS		AL3
W7 DVSS		AL6
W8 DVSS		AL31
W10 DVSS		AL32
W11 DVSS		AL33
W13 DVSS		AM3
W15 DVSS		AM16
W17 DVSS		AM33
W19 DVSS		AM35
W21 DVSS		ANA
W23 DVSS		AP10
W25 DVSS		AP20
W27 DVSS		AR10
W29 DVSS		AR14
W30 DVSS		AR16
Y11 DVSS		AR22
Y13 DVSS		AR25
Y15 DVSS		AR27
Y17 DVSS		AR31

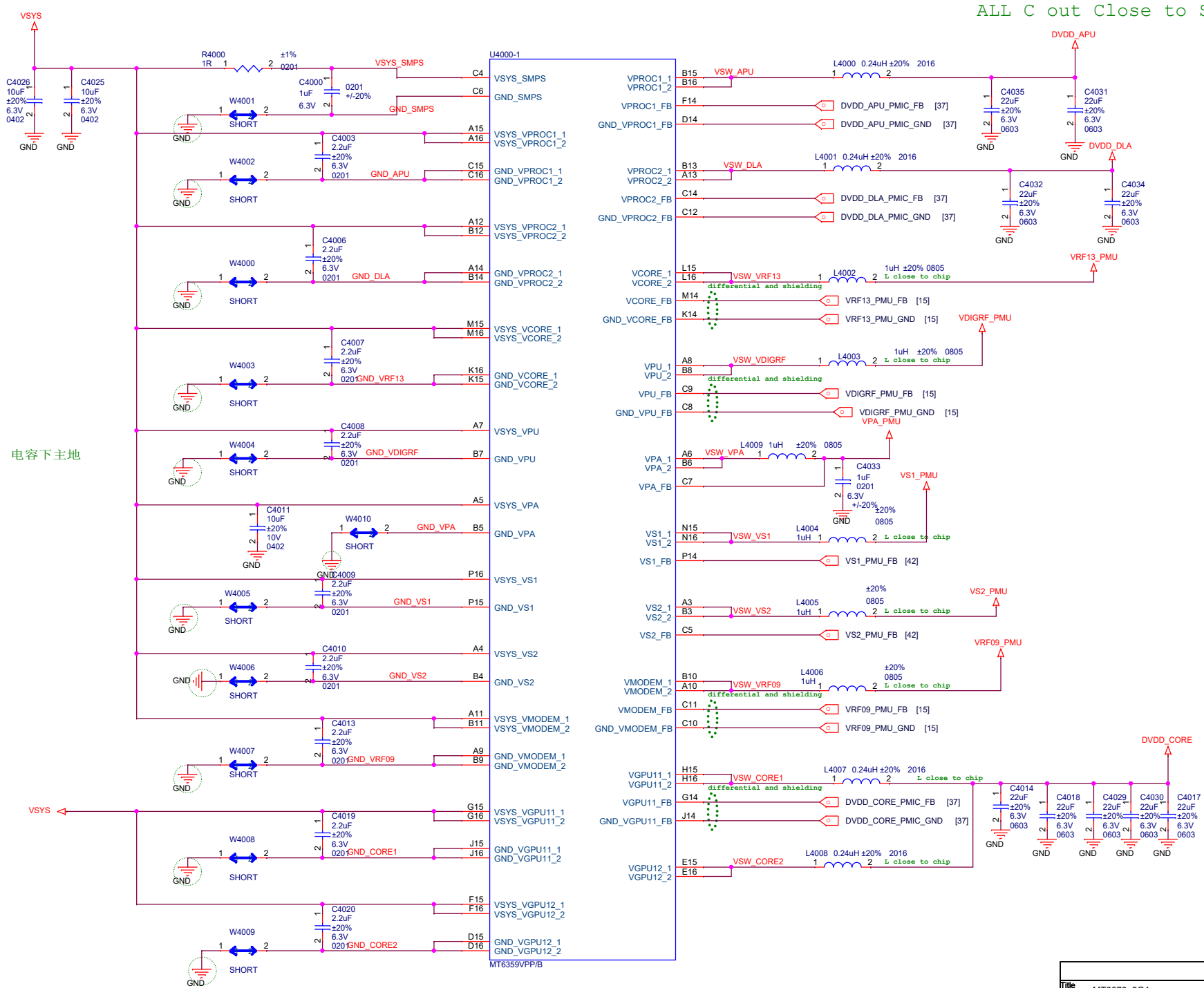
MT6893Z/CZA

GND

GND

File	MT6873_5GA		
Size	Document Number		
C	MT6893_GND		
Date	Friday, March 05, 2021		Sheet

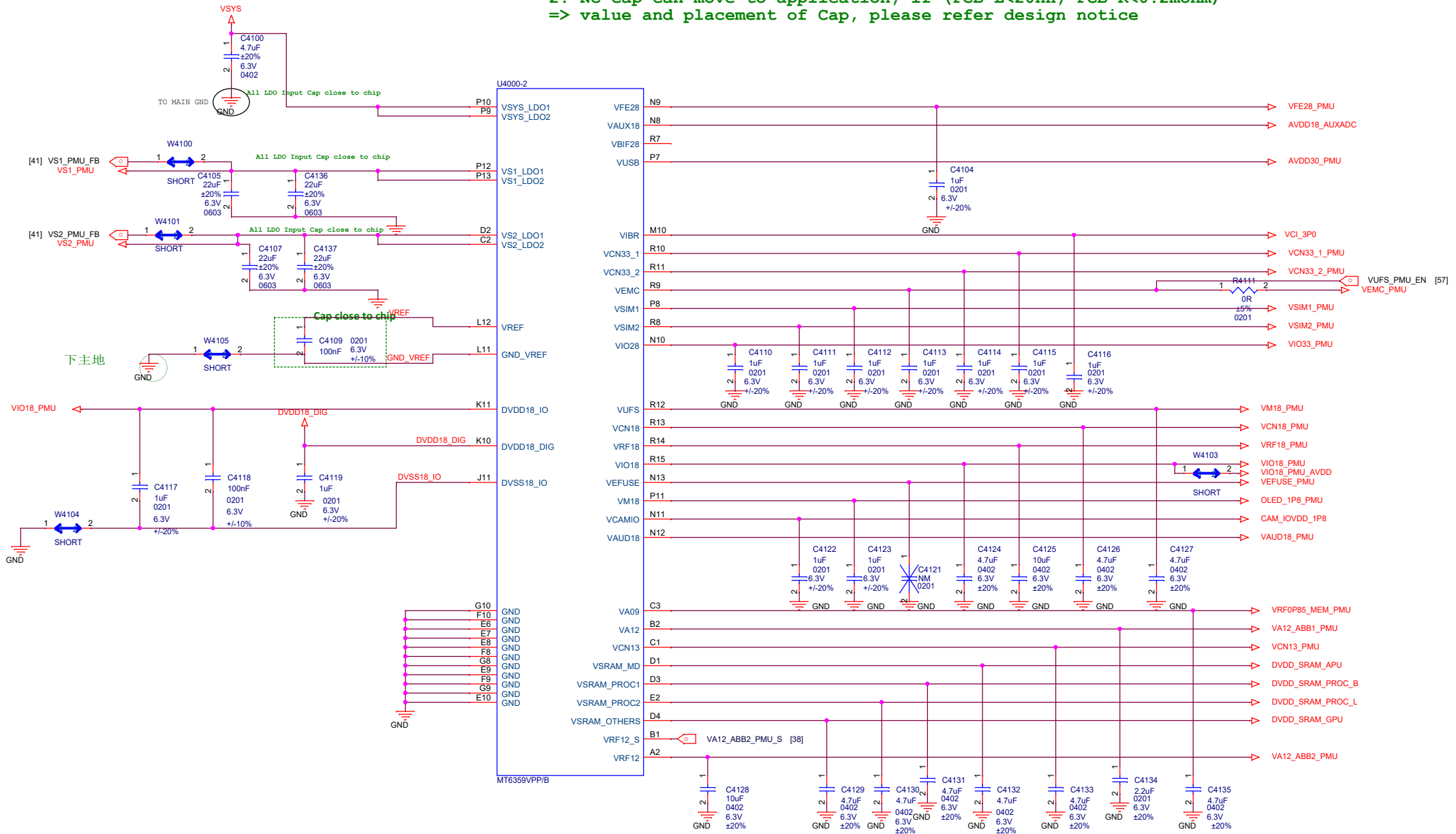
ALL C out Close to SOC



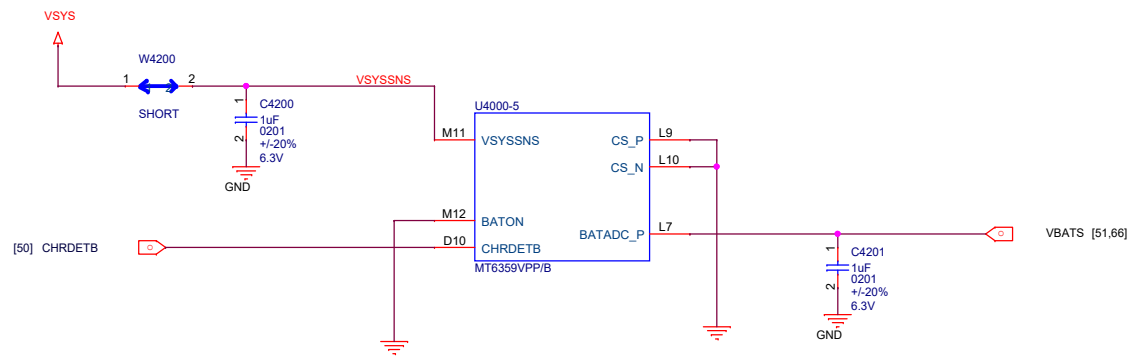
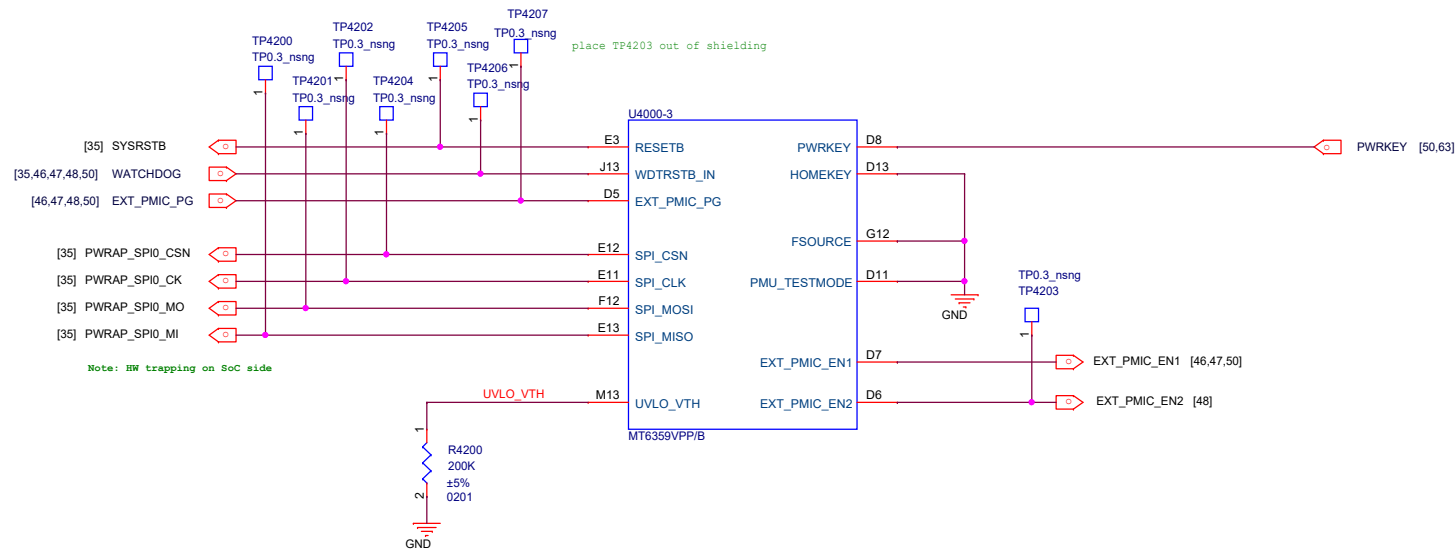
电容下主地

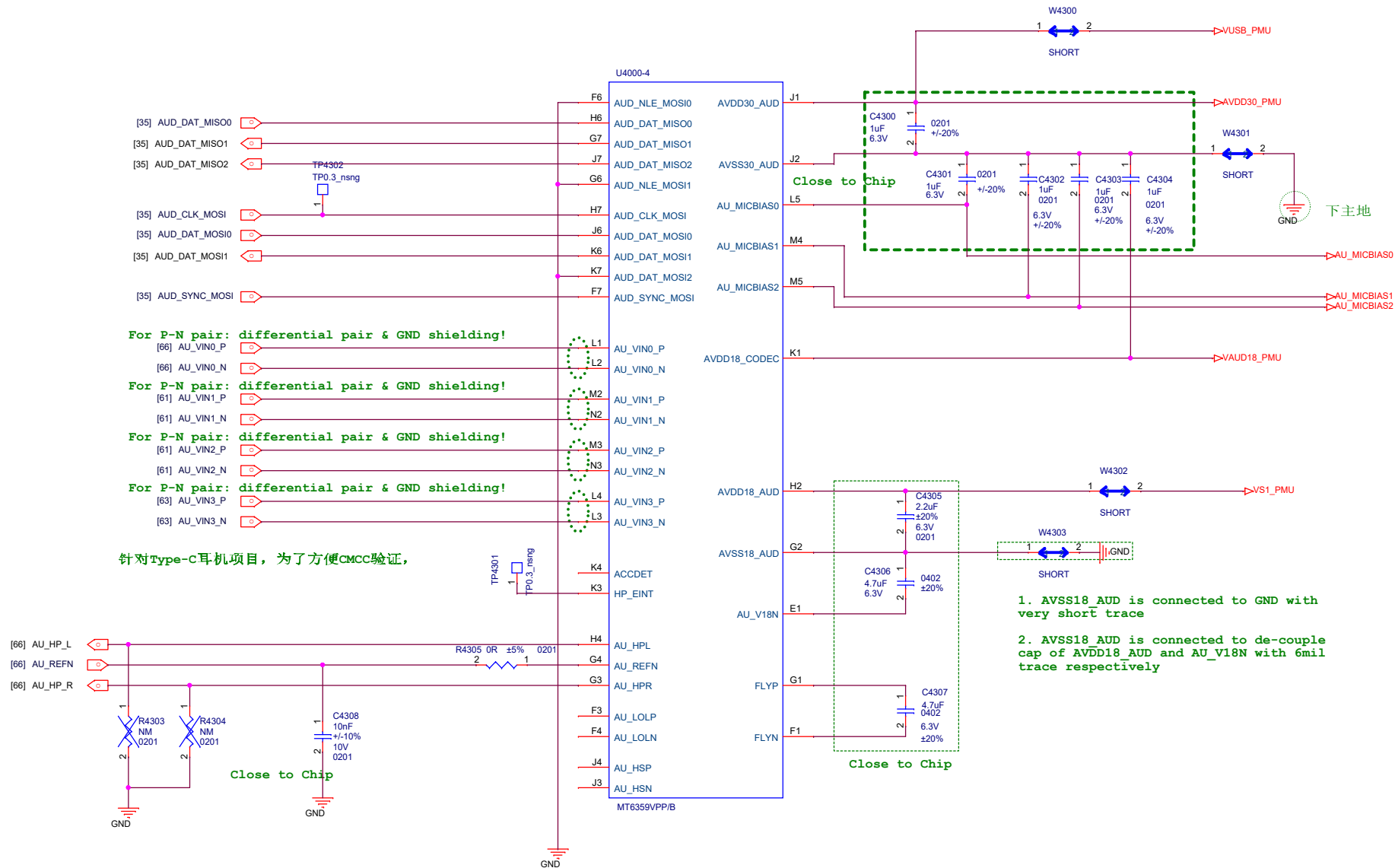
Title		MT6873_5GA
Size	Document Number	
A3	MT6359_Buck	
Date:		Friday, March 05, 2021

1. "Typical Cap" defined in design notice is the minimum cap. to LDO Cout.
 2. NC cap can move to application, if (PCB L<20nH, PCB R<0.2mohm)
- => value and placement of Cap, please refer design notice



Title		MT6873_5GA
Size		Document Number
A3		MT6359_LDO
Date:		Friday, March 05, 2021





-AU_HPL and AU_HPR should be routed as single end signal and be guarded by GND, up and down, left and right respectively

-The suggested layout pattern of AU_HPL/ AU_HPR/ AU_REFN is " GND AU_HPL AU_REFN AU_HPR GND"

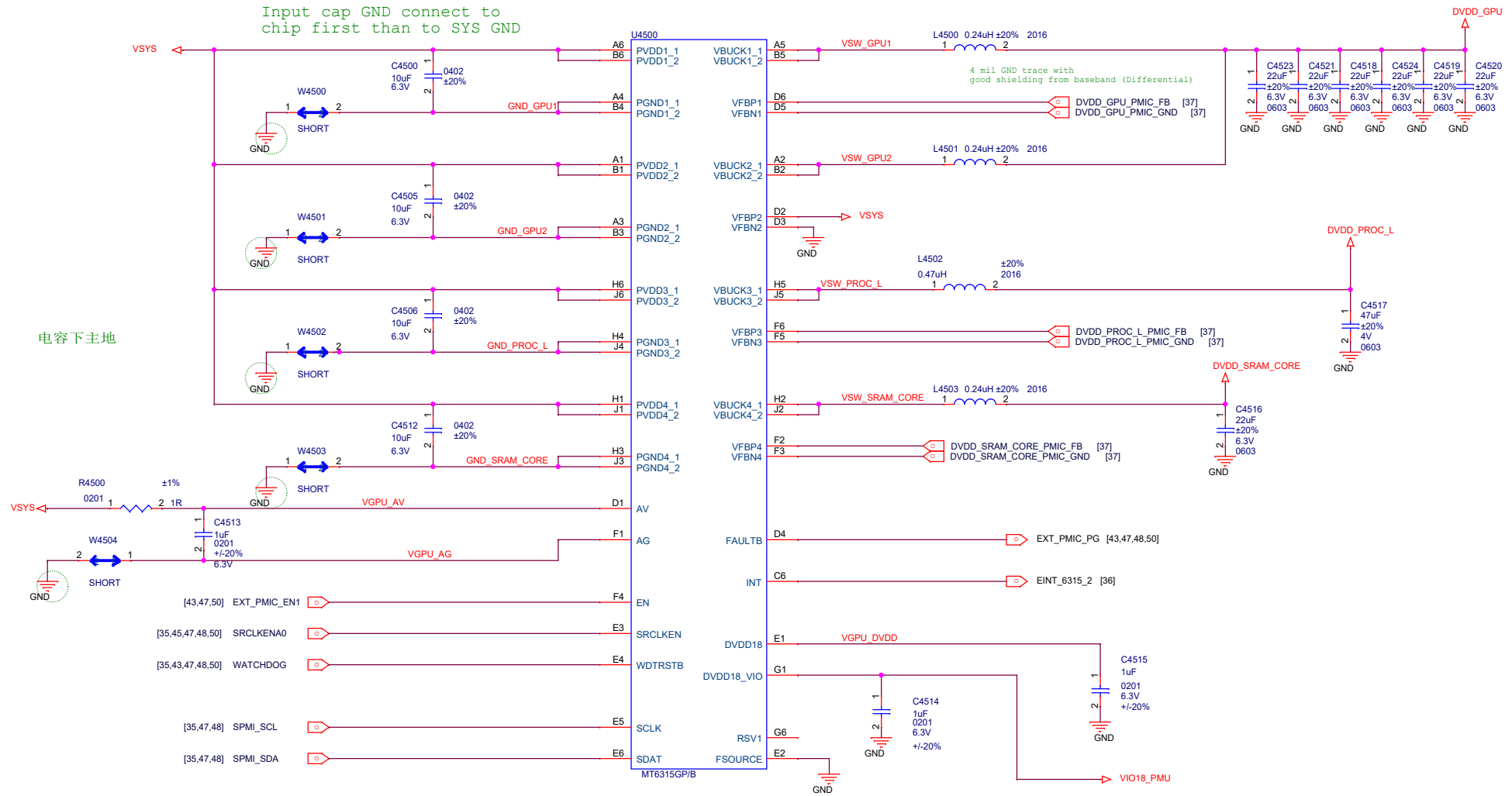
Title		MT6873_5GA
Size	Document Number	MT6359_AUDIO
A3	Date:	Friday, March 05, 2021

MT6315 4-Phase Buck (GPU)

ALL C out Close to SOC

Input cap GND connect to
chip first than to SYS GND

电容下主地



Title	MT6873_5GA
Size	Document Number
A3	MT6315_VGPU
Date:	Friday, March 05, 2021

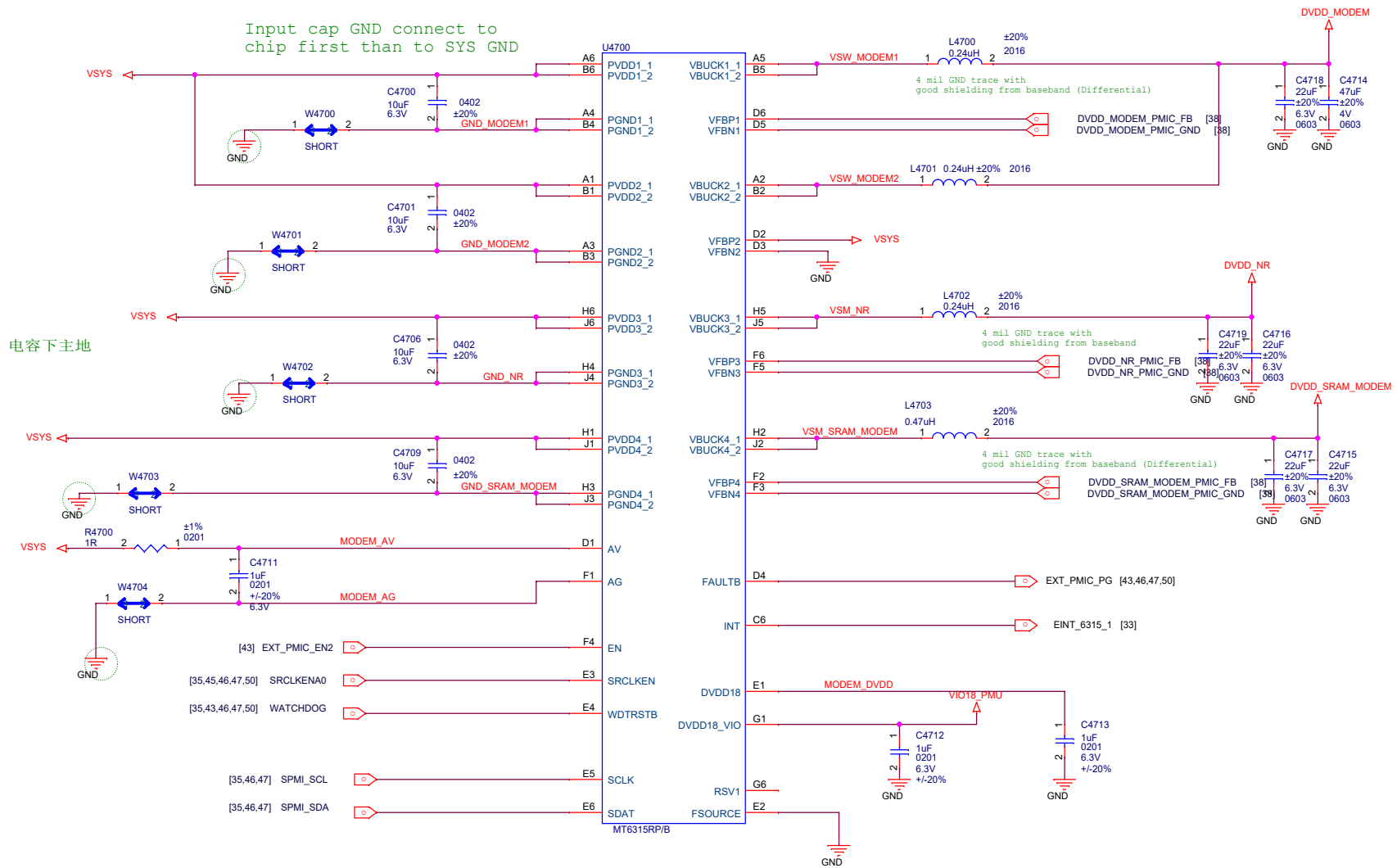
Input cap GND connect to
chip first than to SYS GND

DVDD_PROC_B



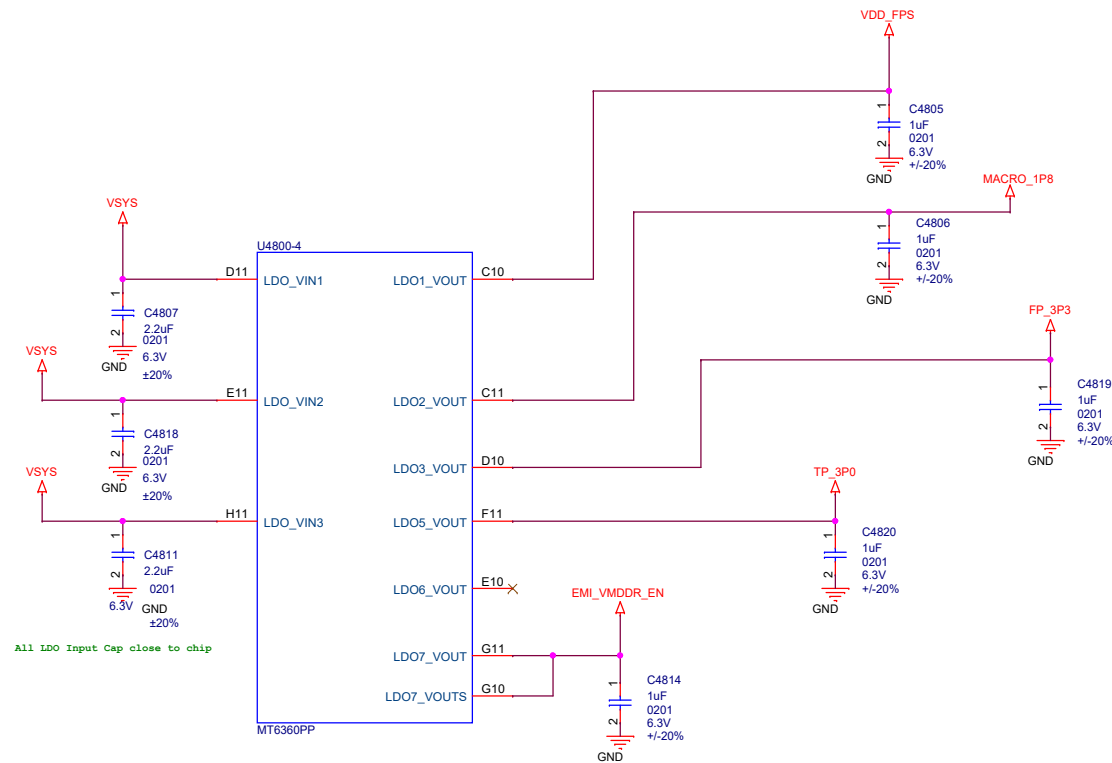
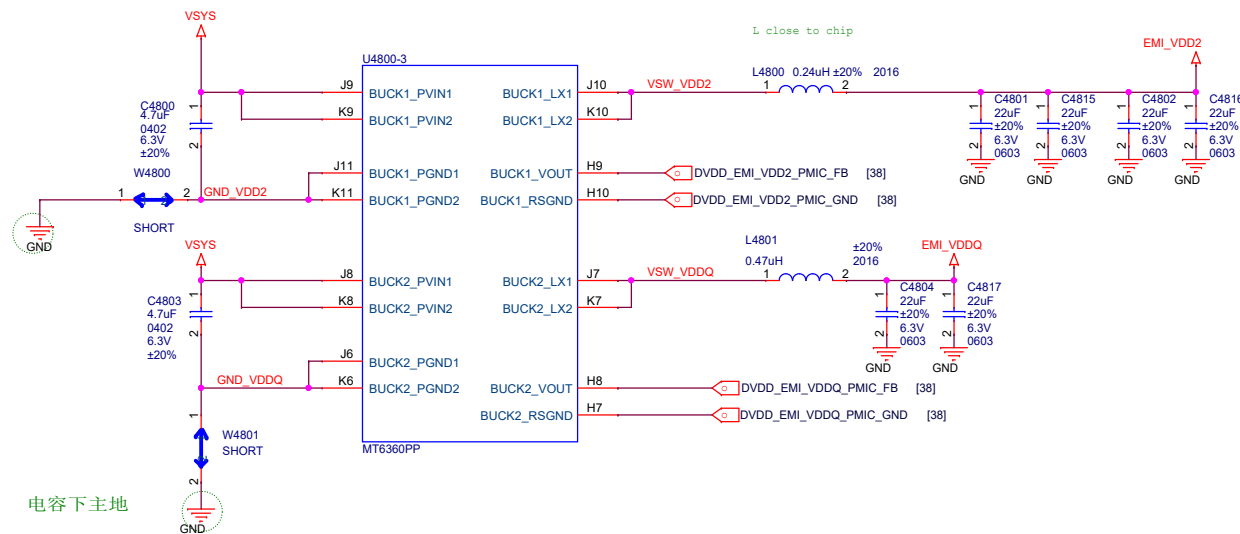
Title		MT6873_5GA	
Size	Document Number		
A3	MT6315_VCPU		
Date:	Friday, March 05, 2021		

ALL C out Close to SOC



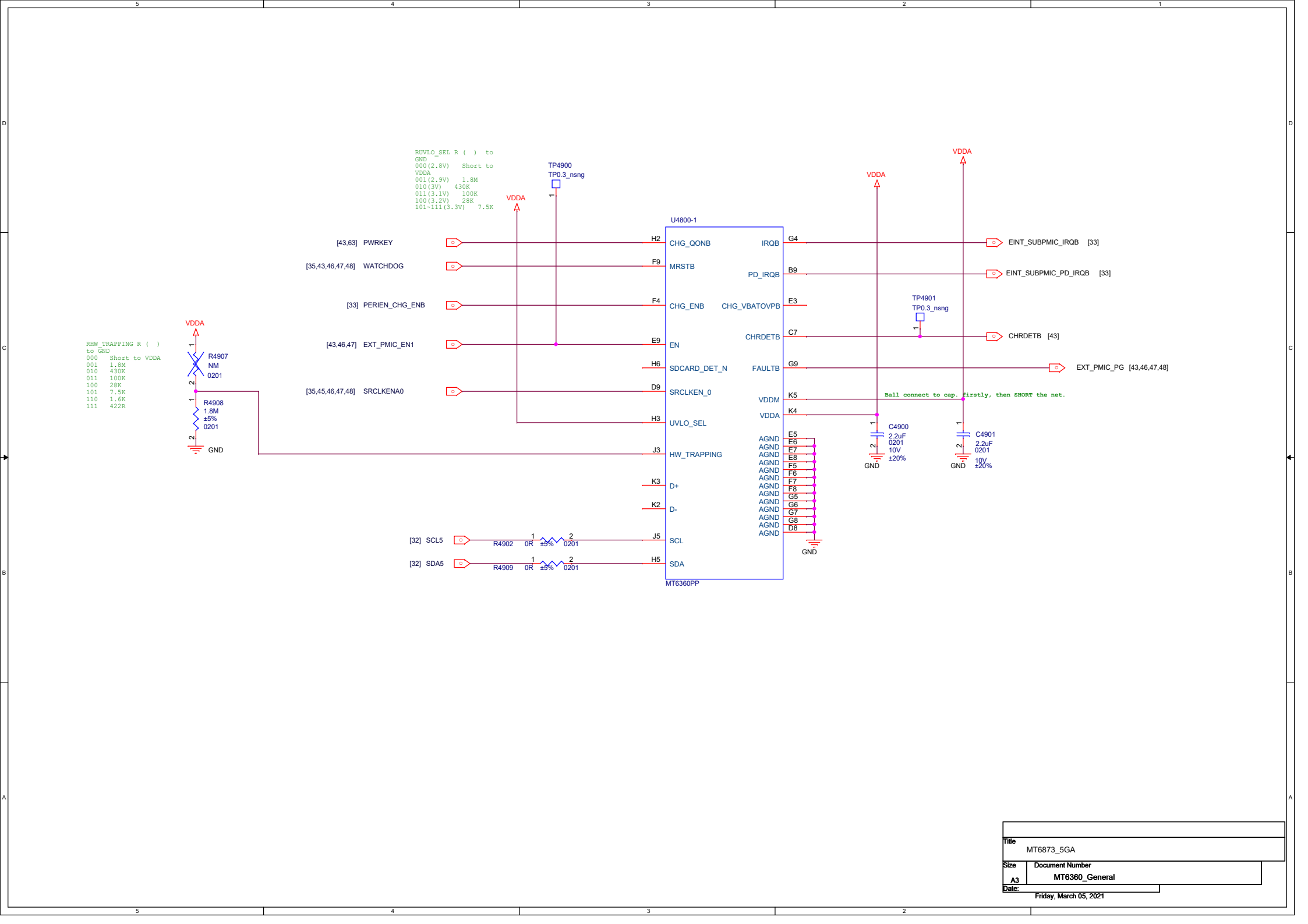
Title	
MT6873_5GA	
Size	Document Number
A3	MT6315_VMD
Date:	
Friday, March 05, 2021	

ALL C out Close to DDR

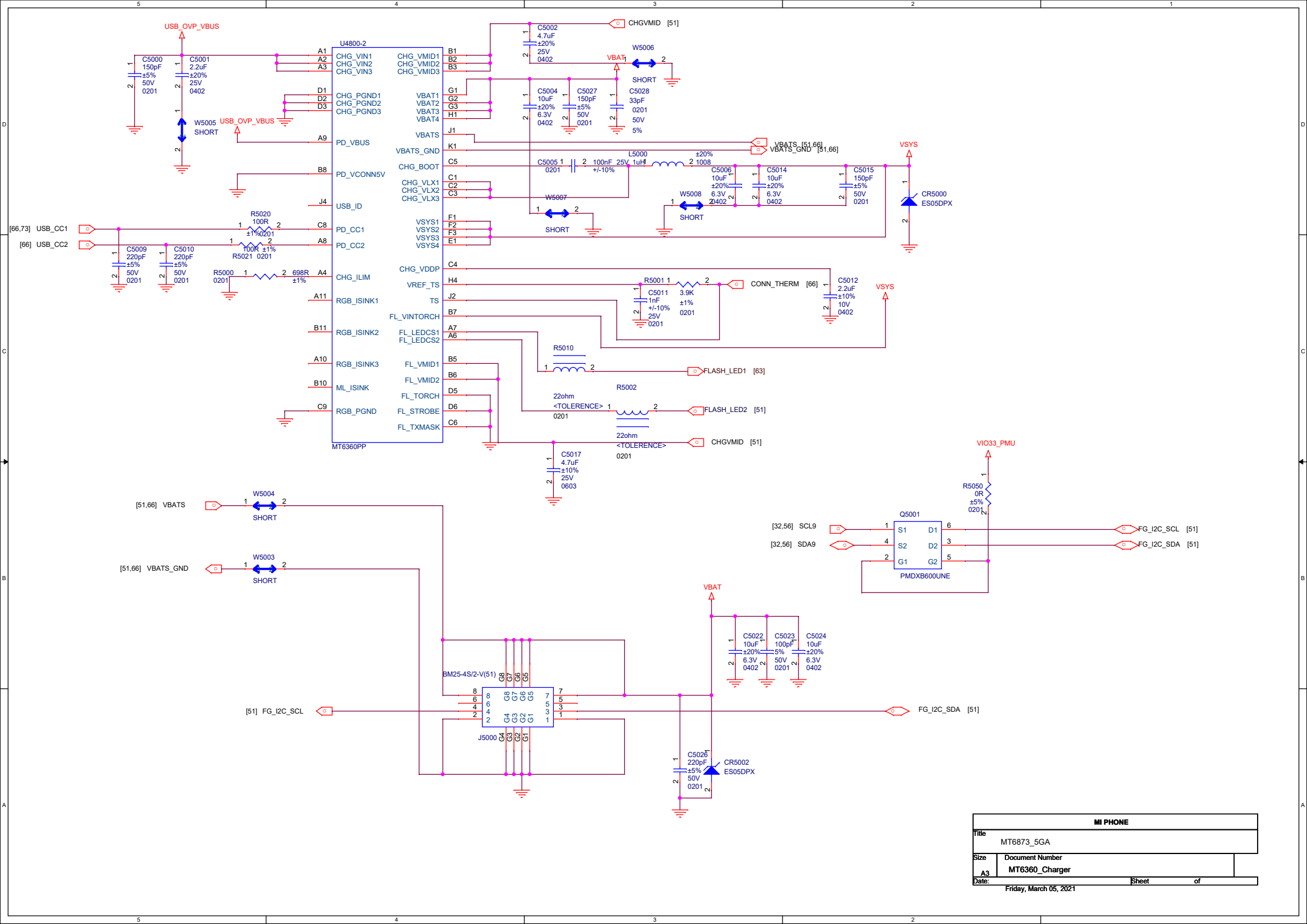


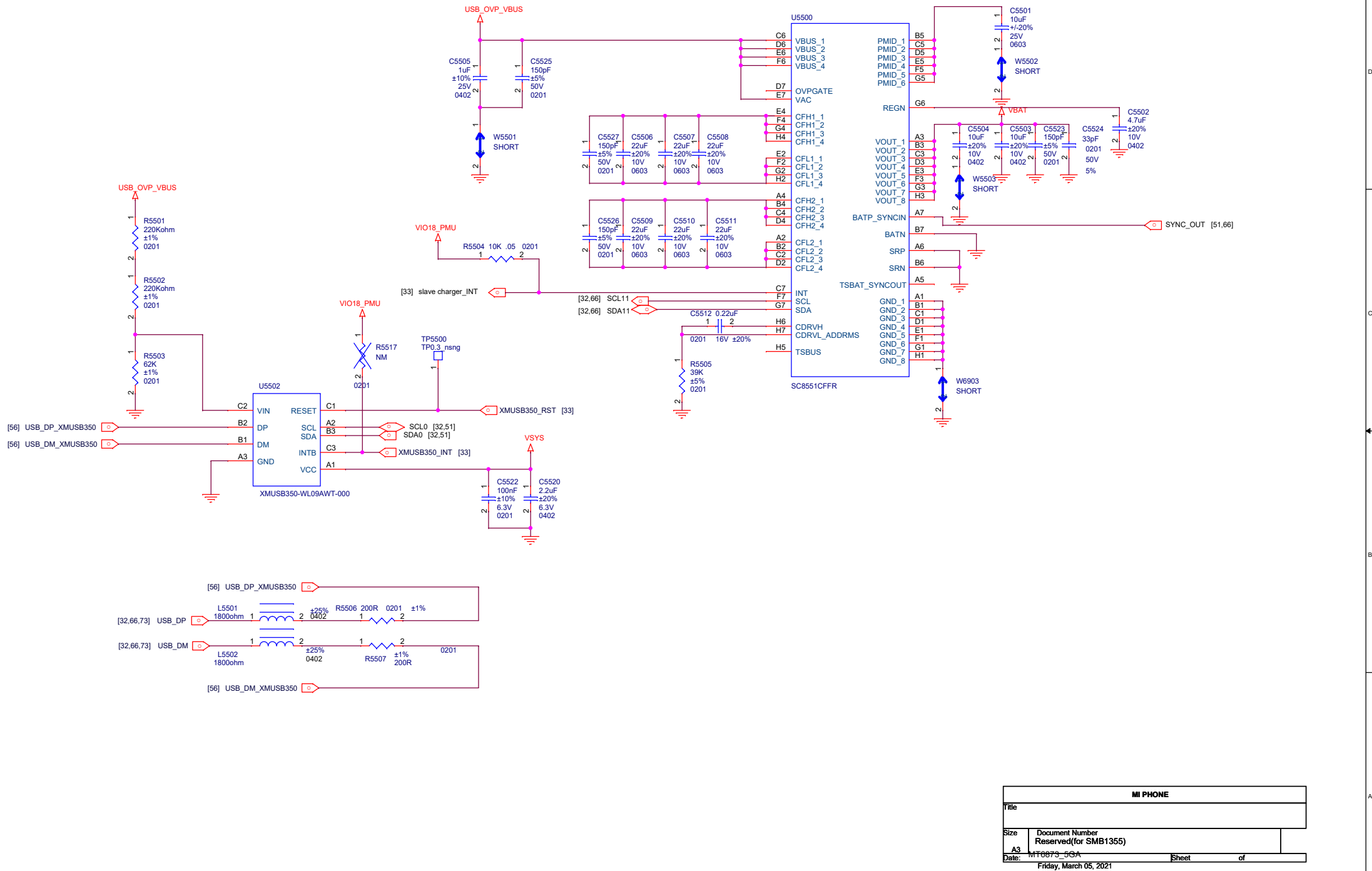
LDO1 VFP
LDO2 VTP
LDO3 VMC
LDO5 SD_CARD_2P95
LDO6
LDO7 VMDDR_EN

Title		MT6873_5GA
Size	Document Number	MT6360_BUCK_LDO
A3		
Date:	Friday, March 05, 2021	

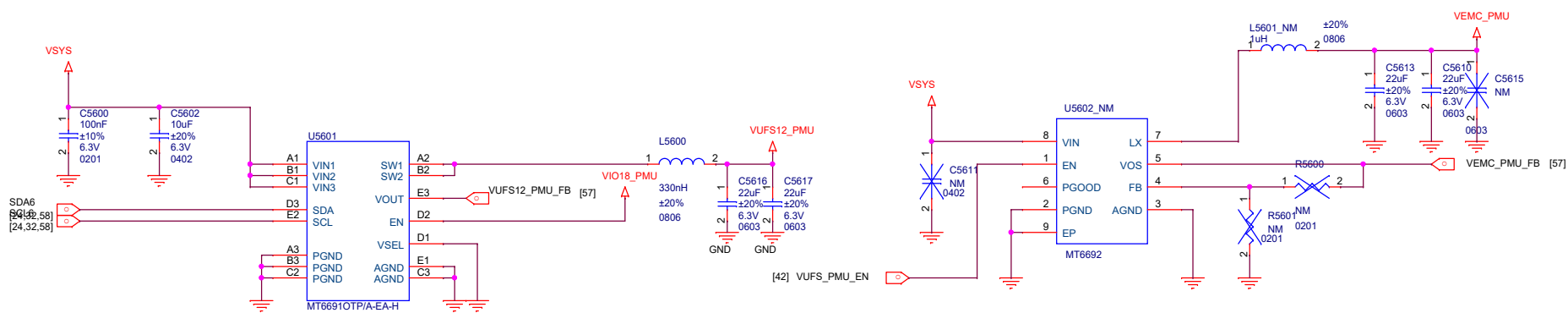
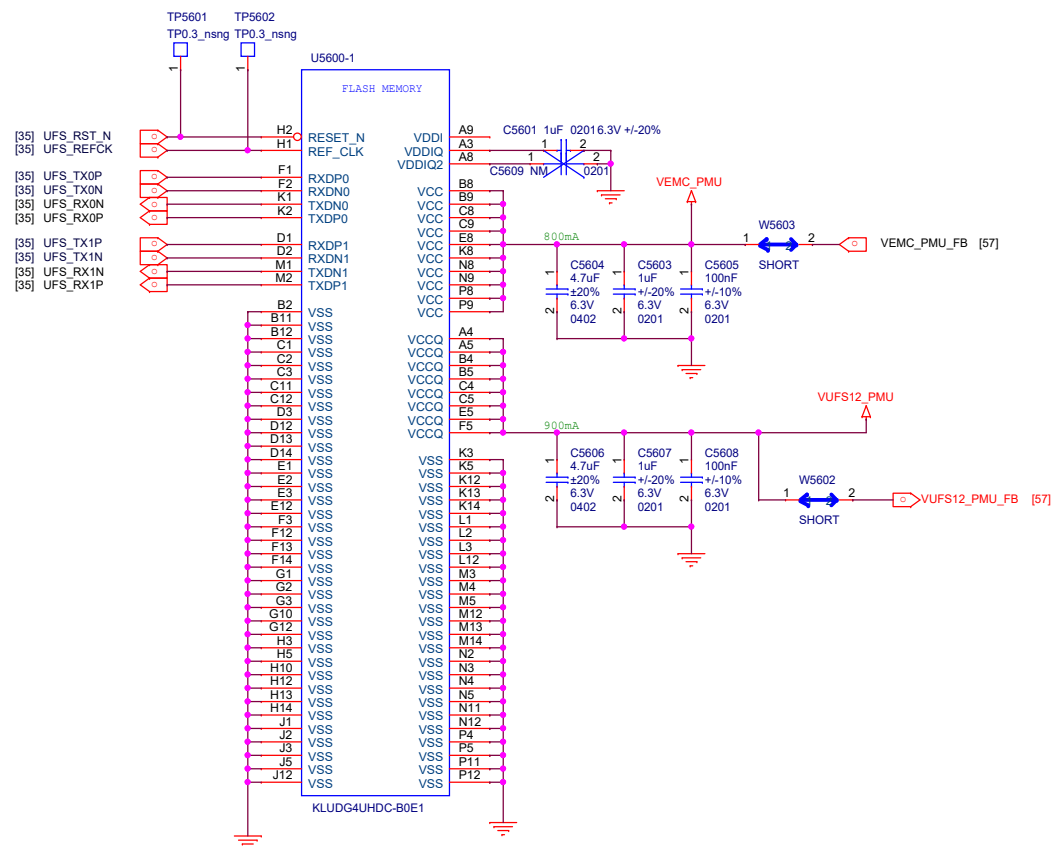
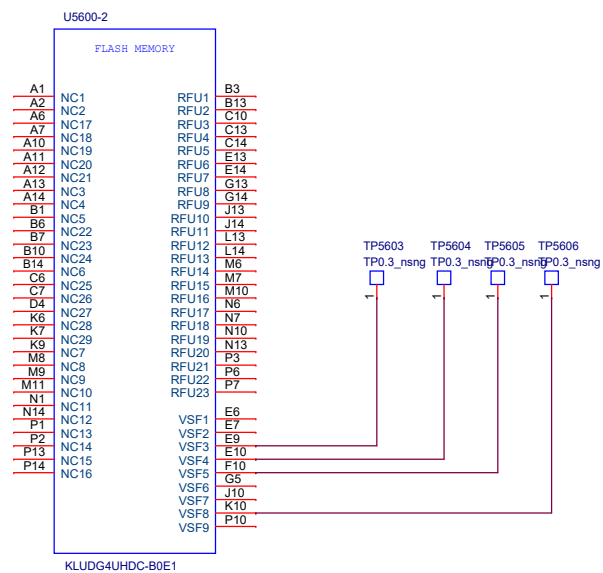


Title	
MT6873_5GA	
Size	Document Number
A3	MT6360_General
Date:	
Friday, March 05, 2021	



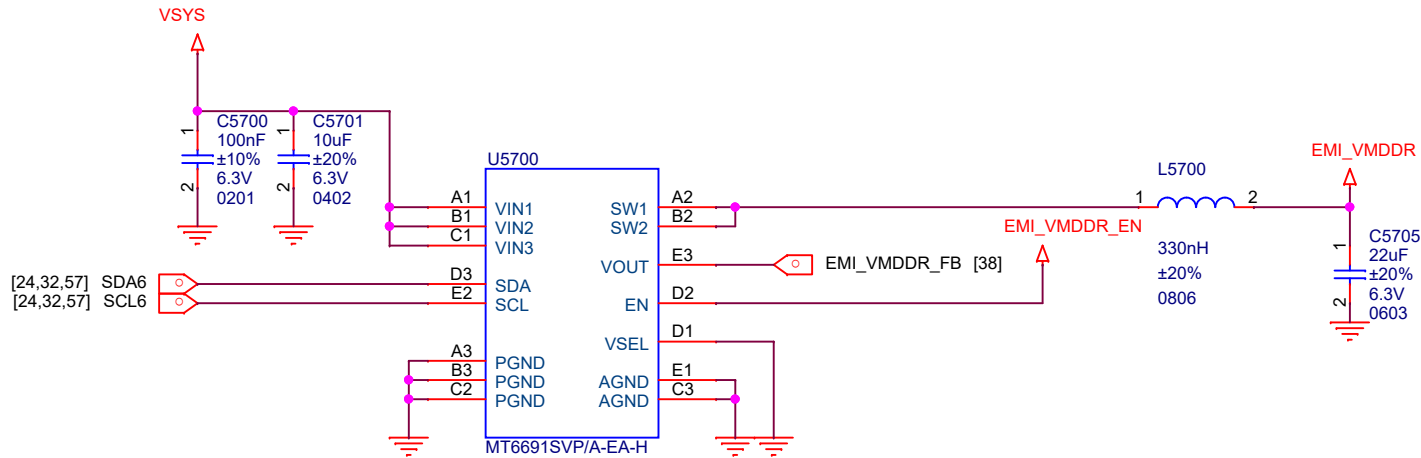


UFS 3.1

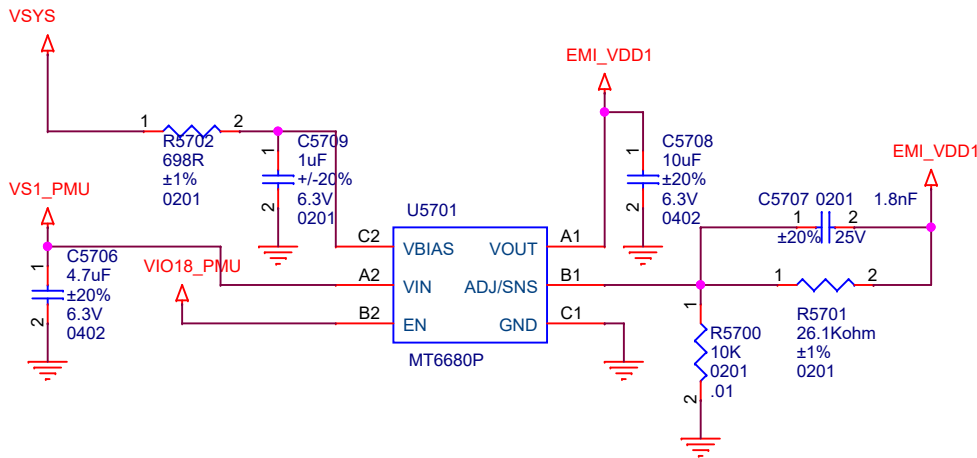


Title	
Memory_1	
Size	Document Number
A3	MEMORY
Date:	Friday, March 05, 2021

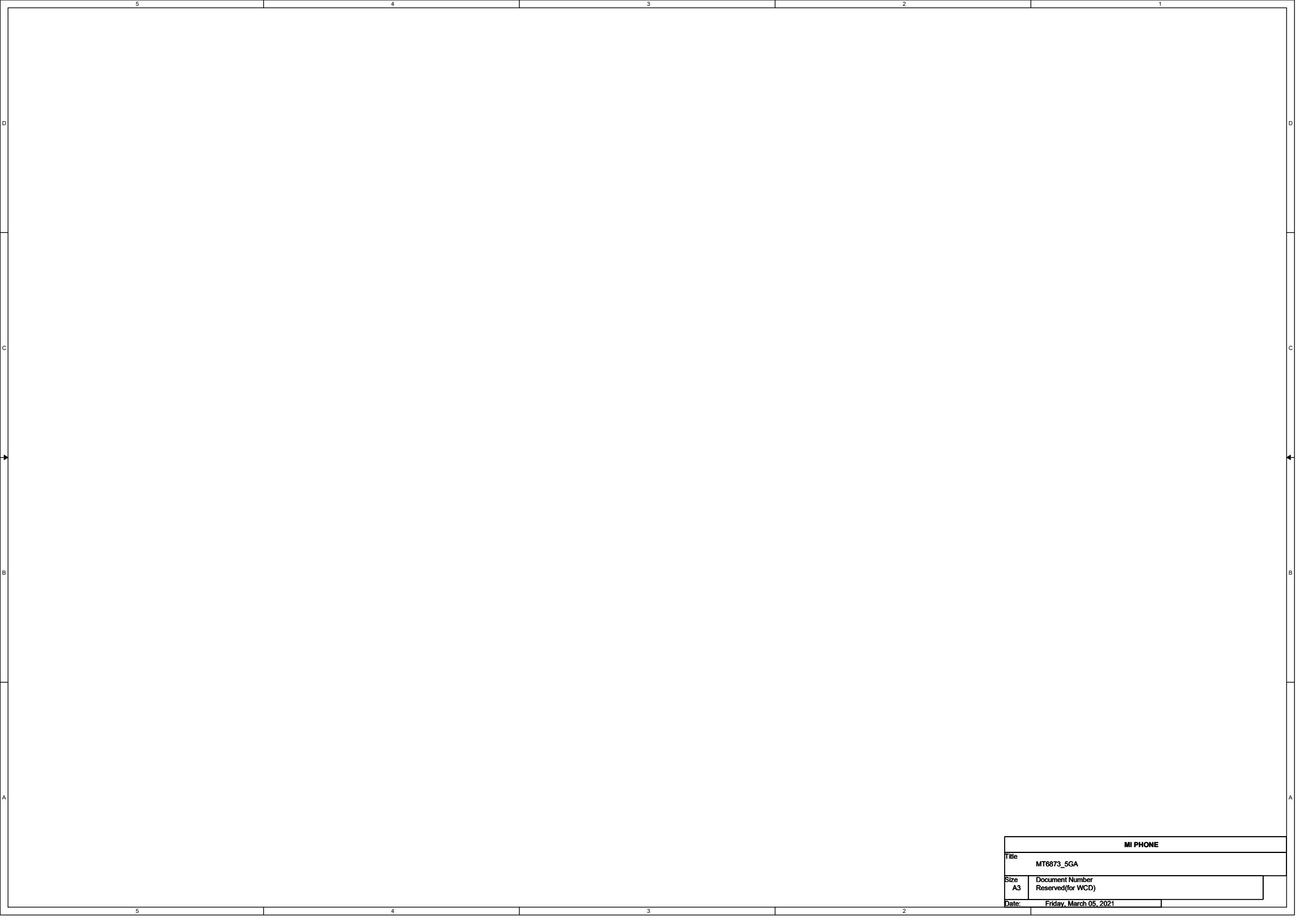
LP4X DRAM: VMDDR



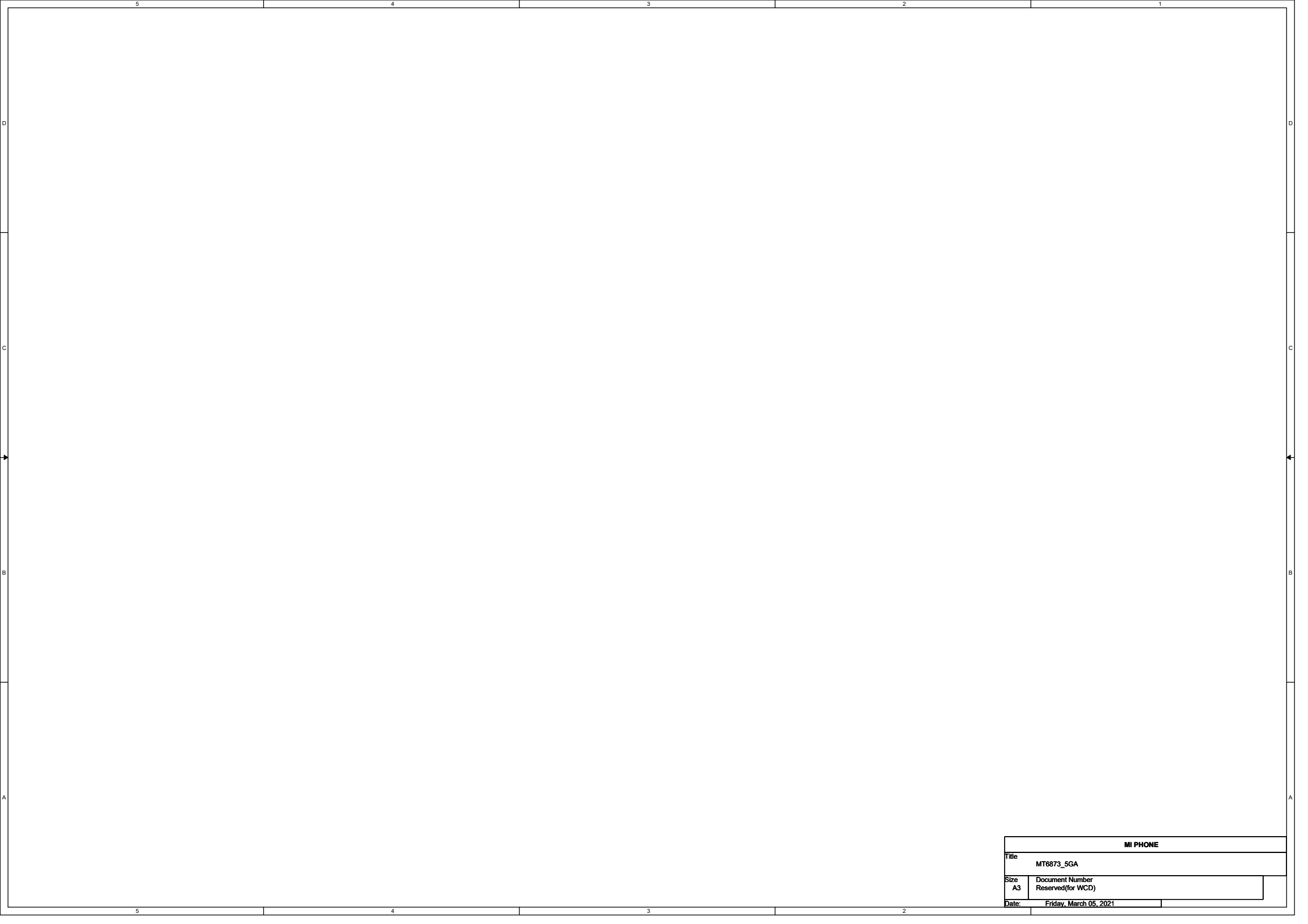
LP4X DRAM: VDD1



Title		
Size	Document Number	
A4	Memory_2	
Date:	Friday, March 05, 2021	

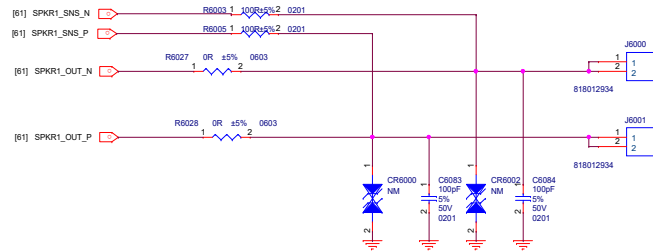


MI PHONE	
Title	MT6873_5GA
Size A3	Document Number Reserved(for WCD)
Date:	Friday, March 05, 2021



MI PHONE	
Title	MT6873_5GA
Size A3	Document Number Reserved(for WCD)
Date:	Friday, March 05, 2021

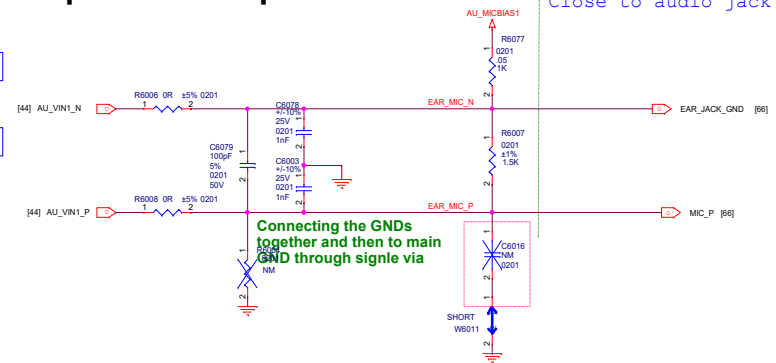
SPK



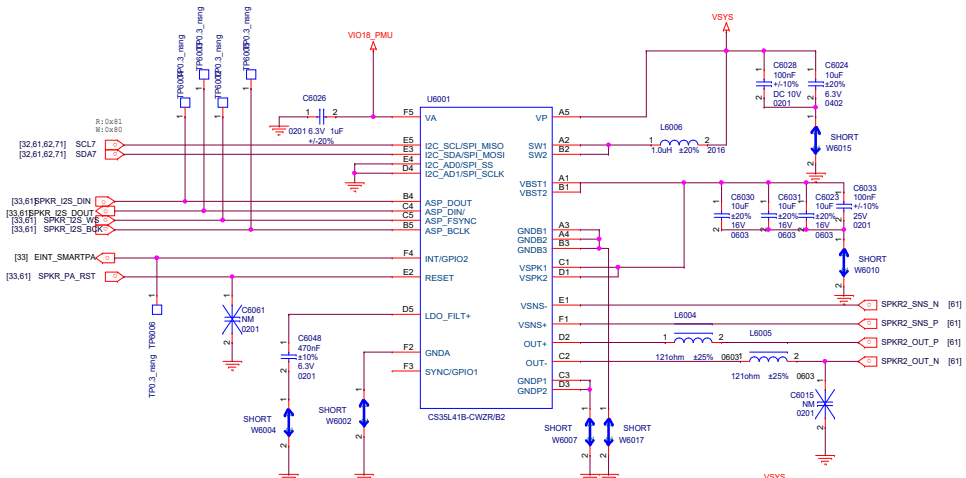
Earphone Microphone

Close to PMIC

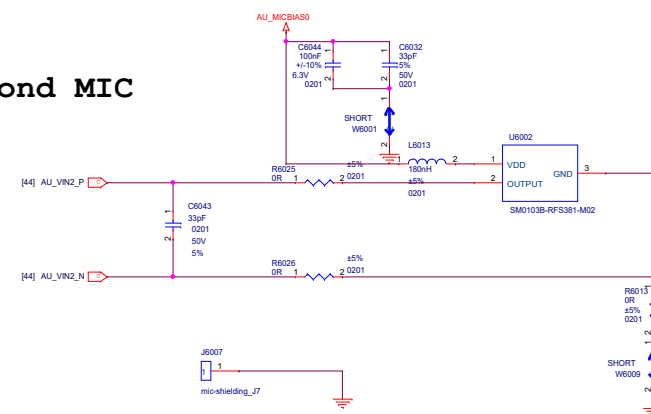
Close to audio jack



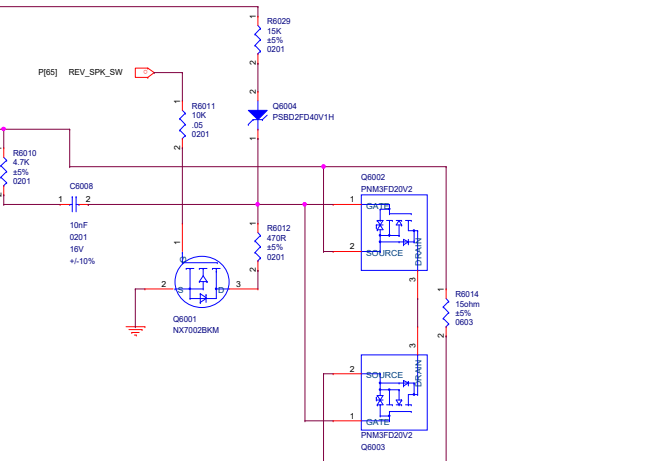
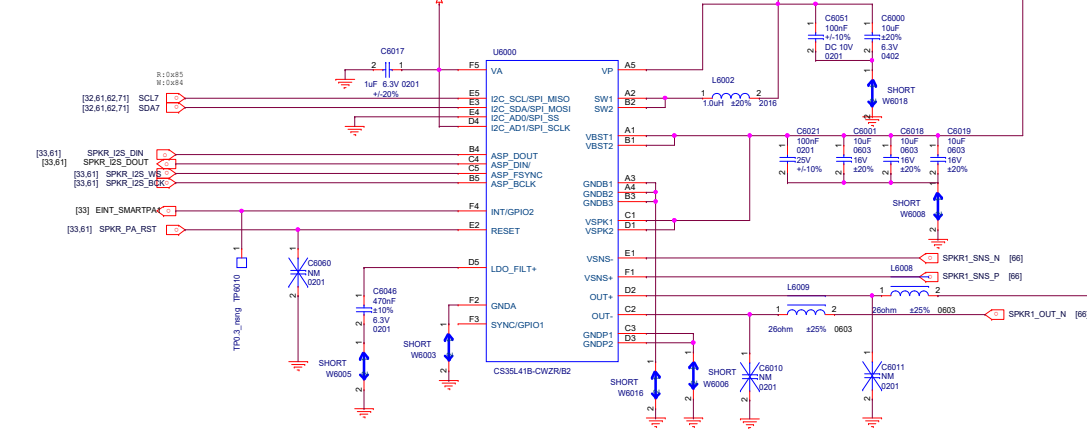
SPKR PA



Second MIC

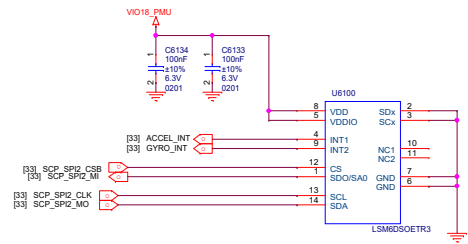


SPK2 PA

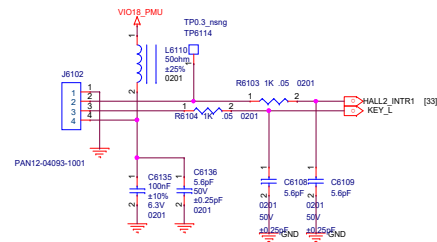
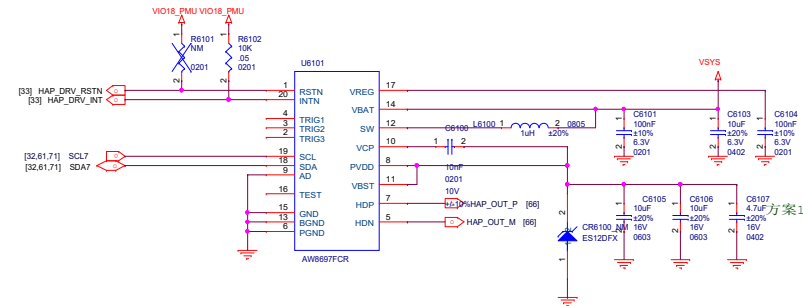


File	MT8873_5GA
Size	Document Number
A2	AUDIO
Date	Friday, March 05, 2021

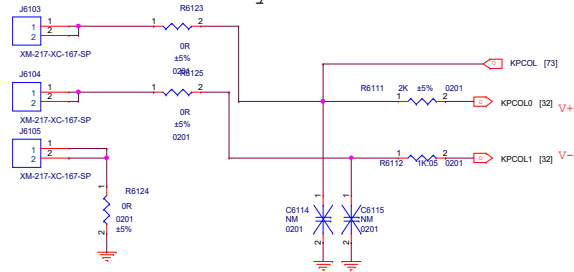
A+G1



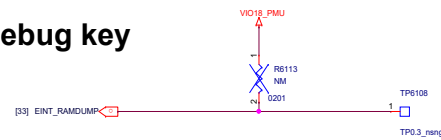
HAPTICS



Sidekey

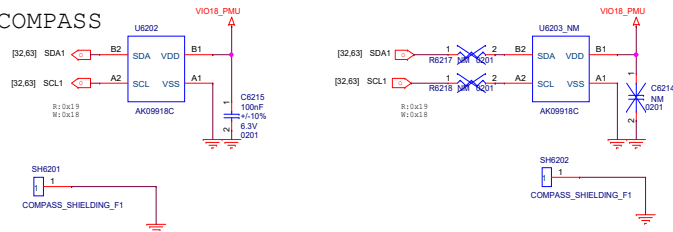


RAMDUMP debug key

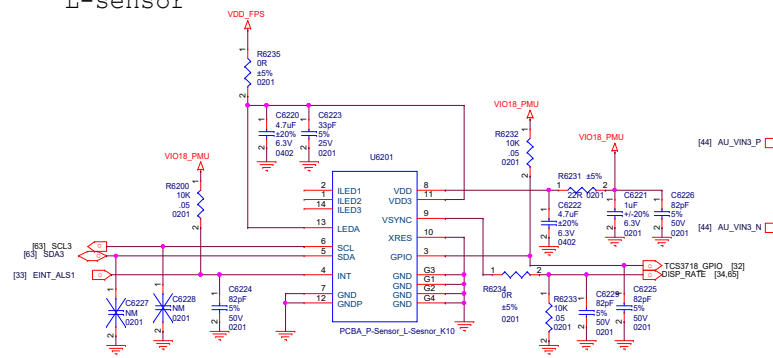


Title		J11 MAIN
Size	Document Number	A2
Size	Document Number	A2
Date	Friday, March 05, 2021	

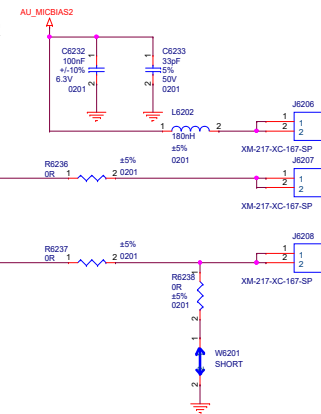
COMPASS



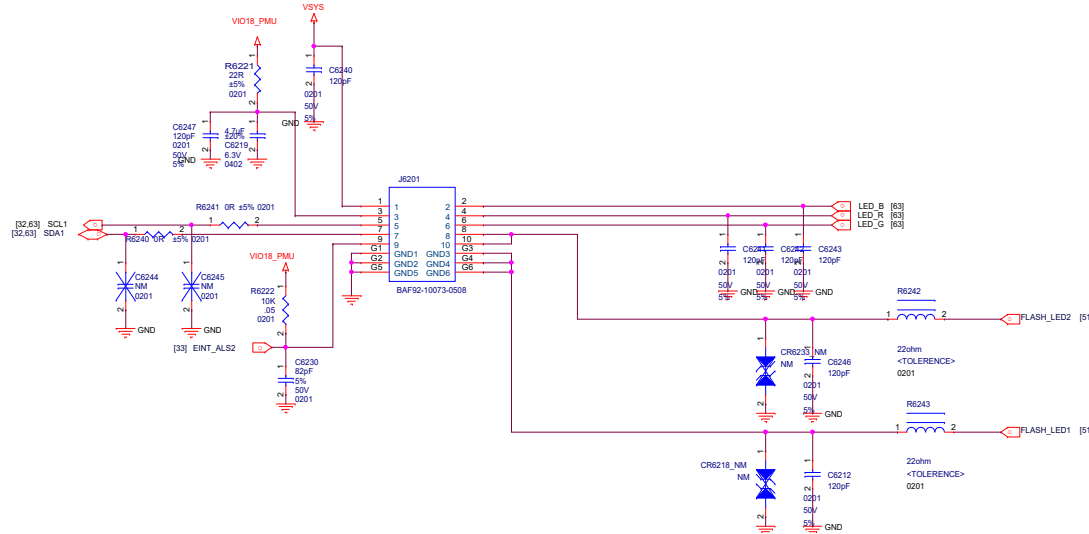
L-sensor



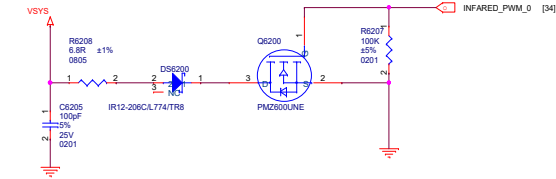
BACK MIC



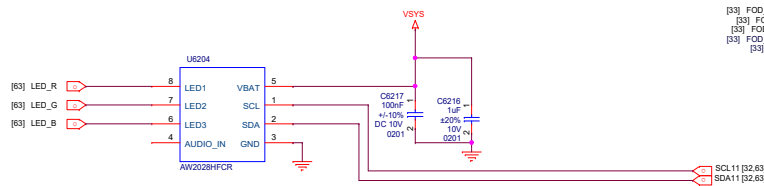
RGB+ALS2+FLASH



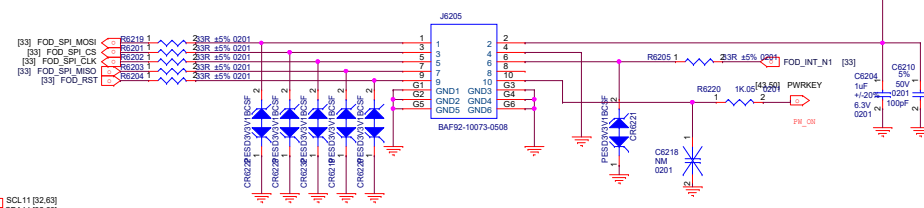
INFRARED



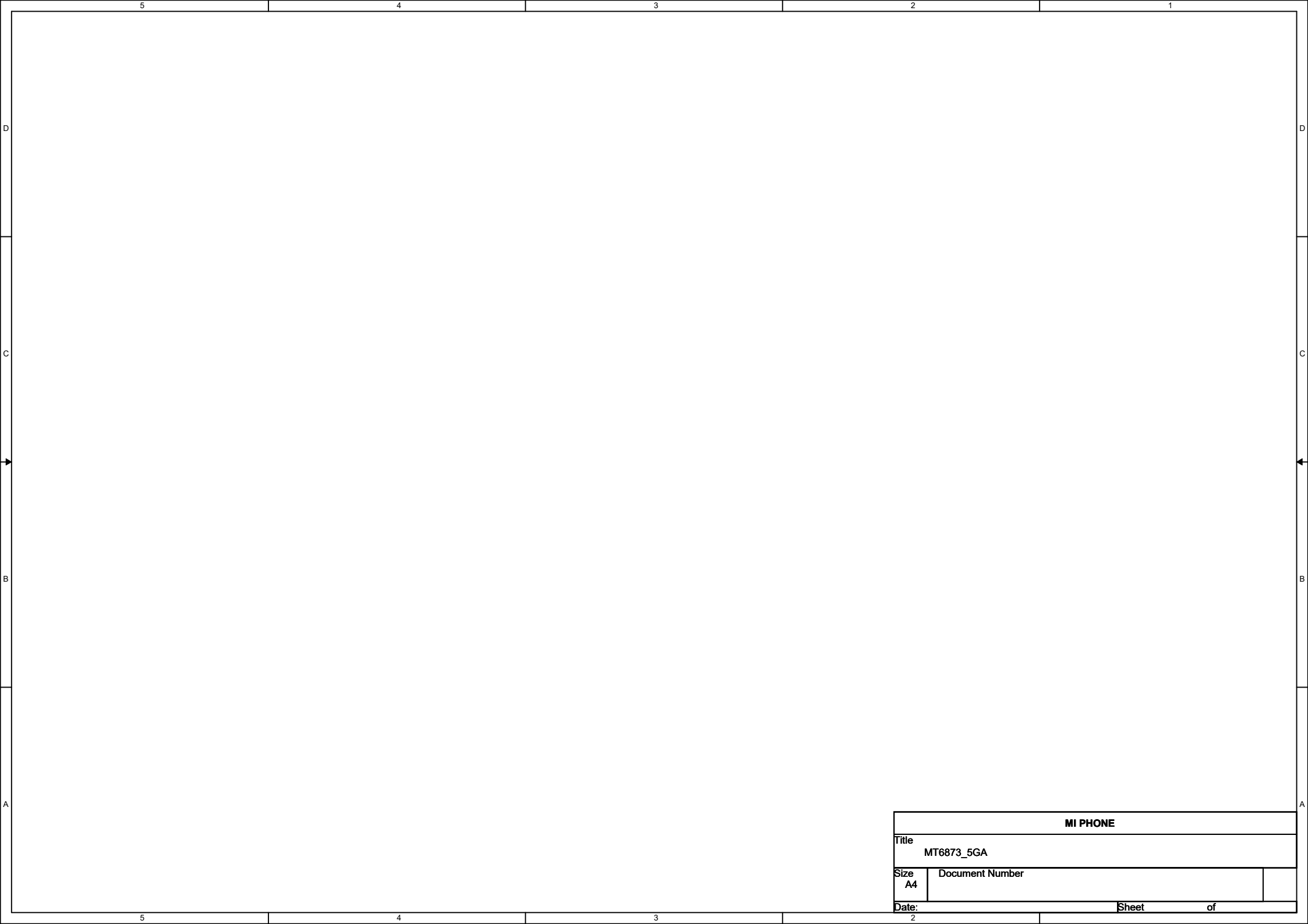
RGB DRIVER

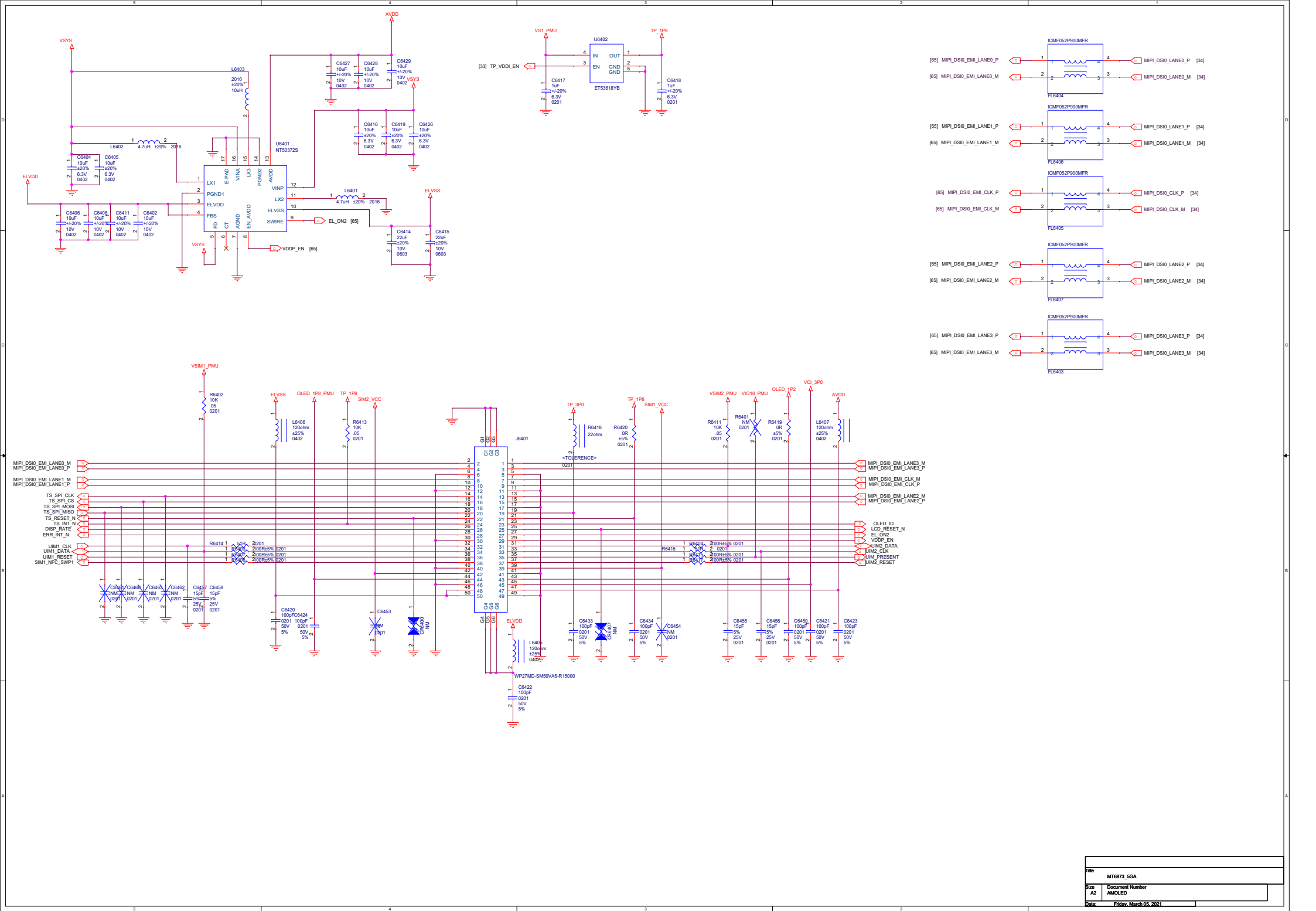


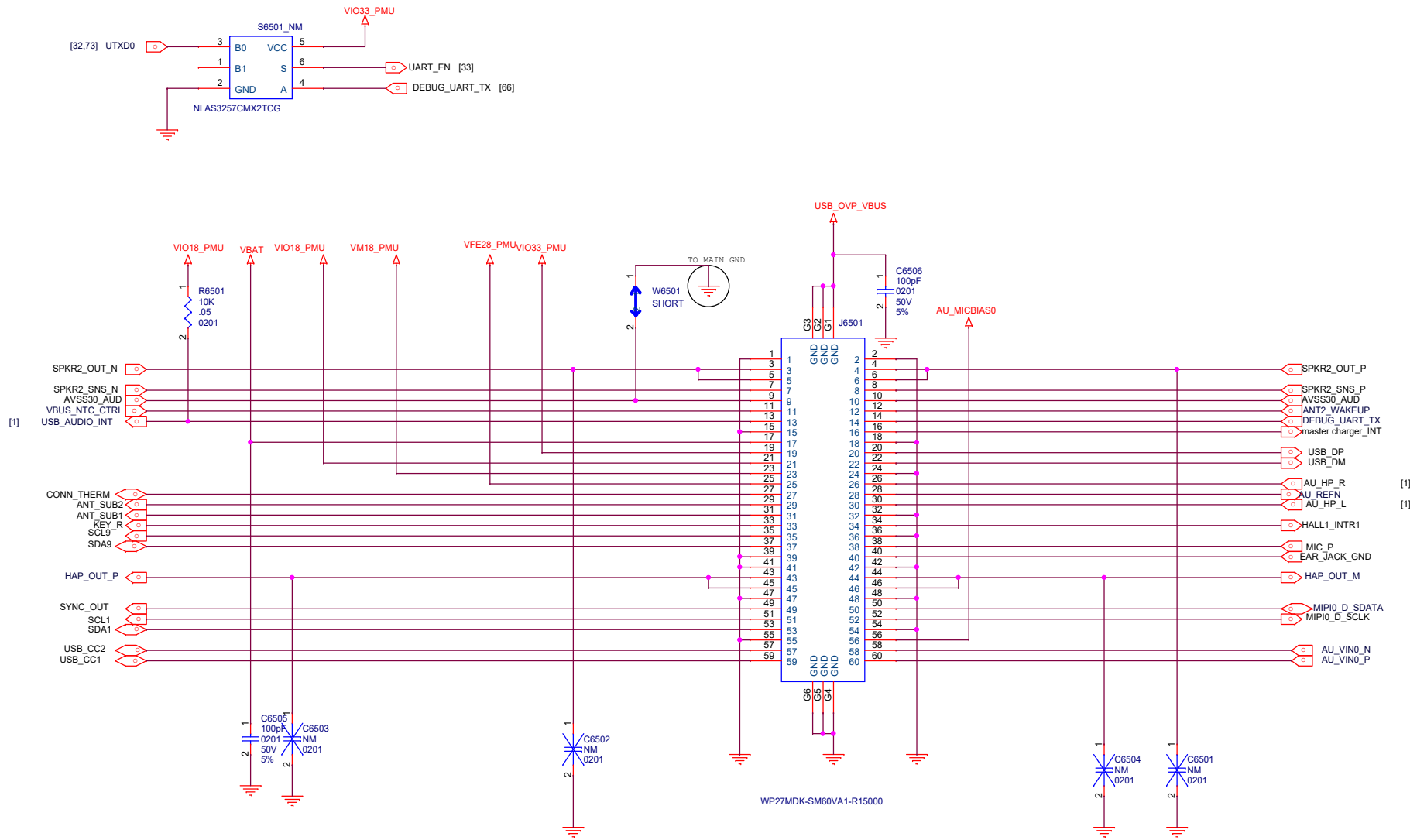
FP



Title		MT0873_5GA
Size		Document Number
A2		SENSOR-2
Date		Friday, March 05, 2021

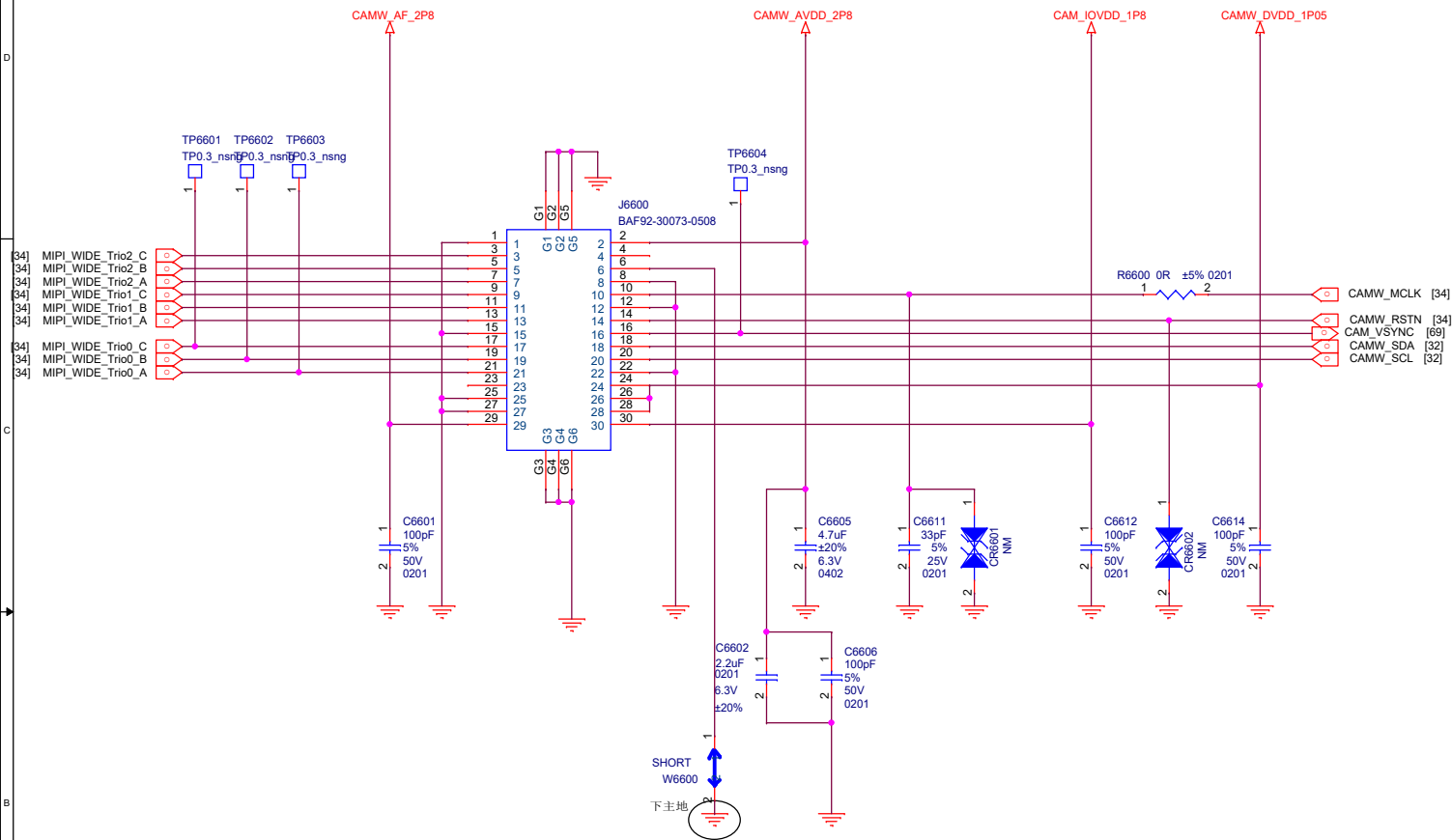






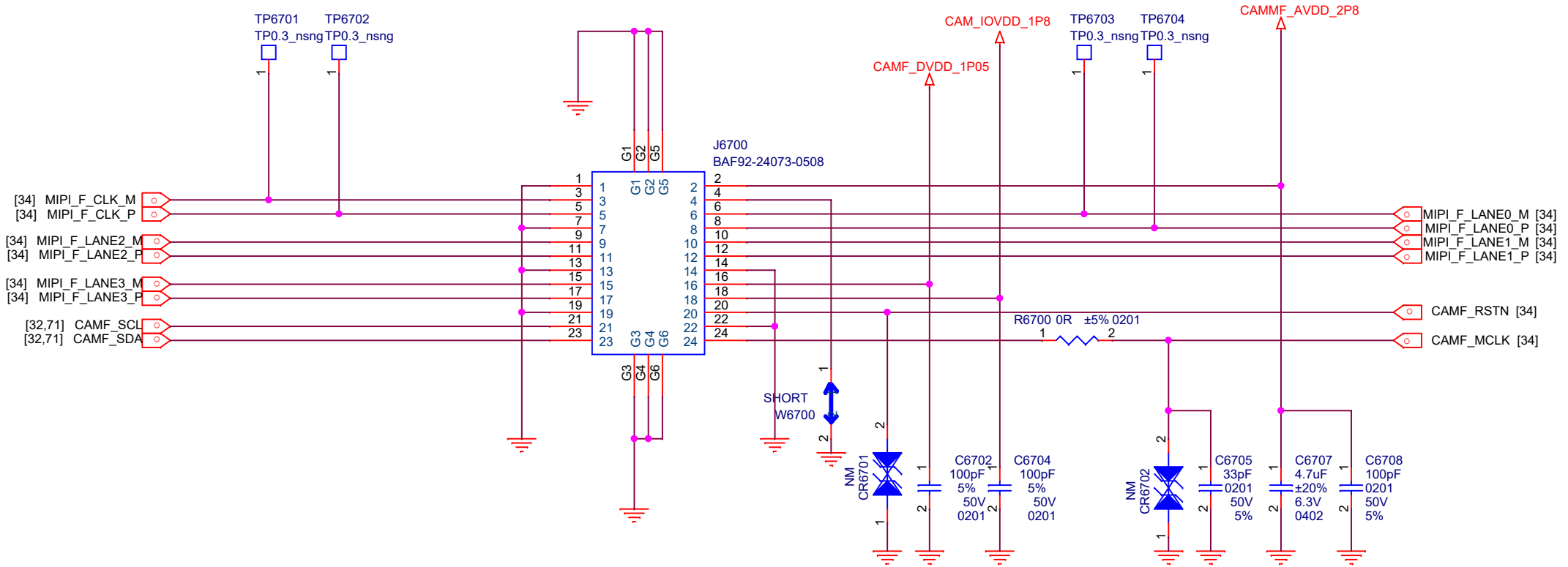
Title	
J11 MAIN	
Size	Document Number
A3	BTB
Date: Friday, March 05, 2021	

WIDE 64M IMX686



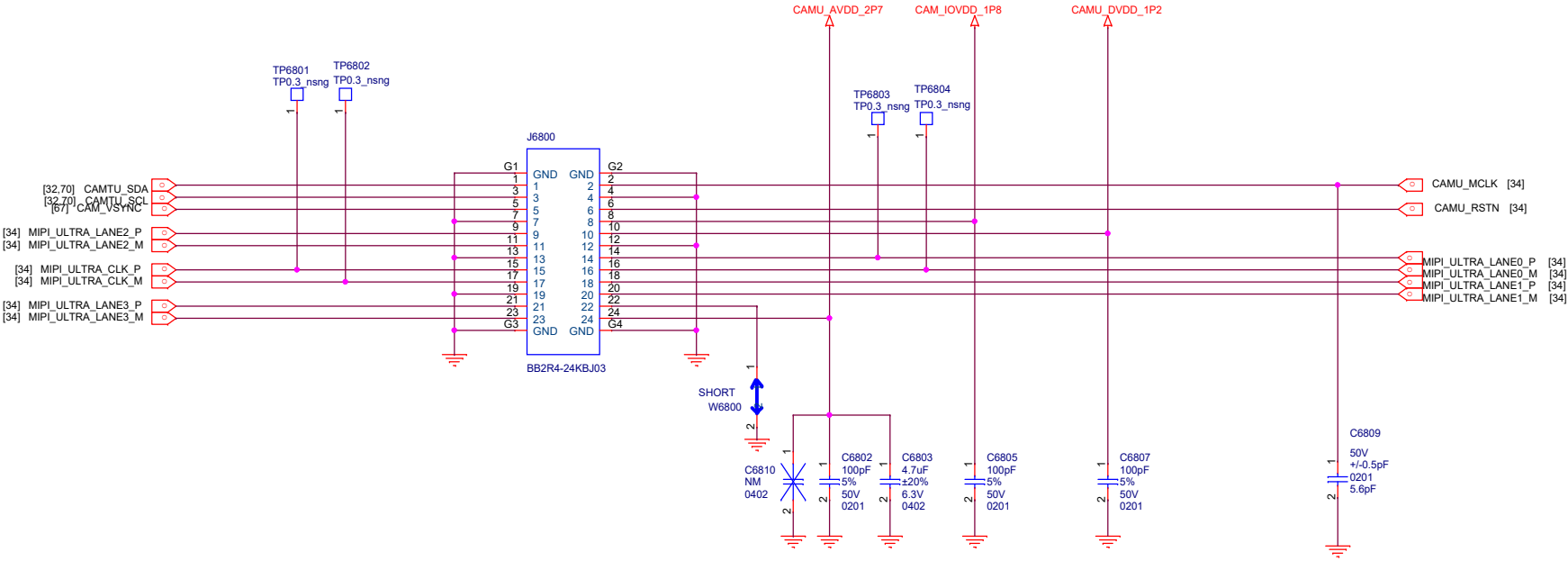
Title	
J11 MAIN	
Size	Document Number
A3	WIDE&TELE CAMERA
Date:	Friday, March 05, 2021

front 16M S5K3T2



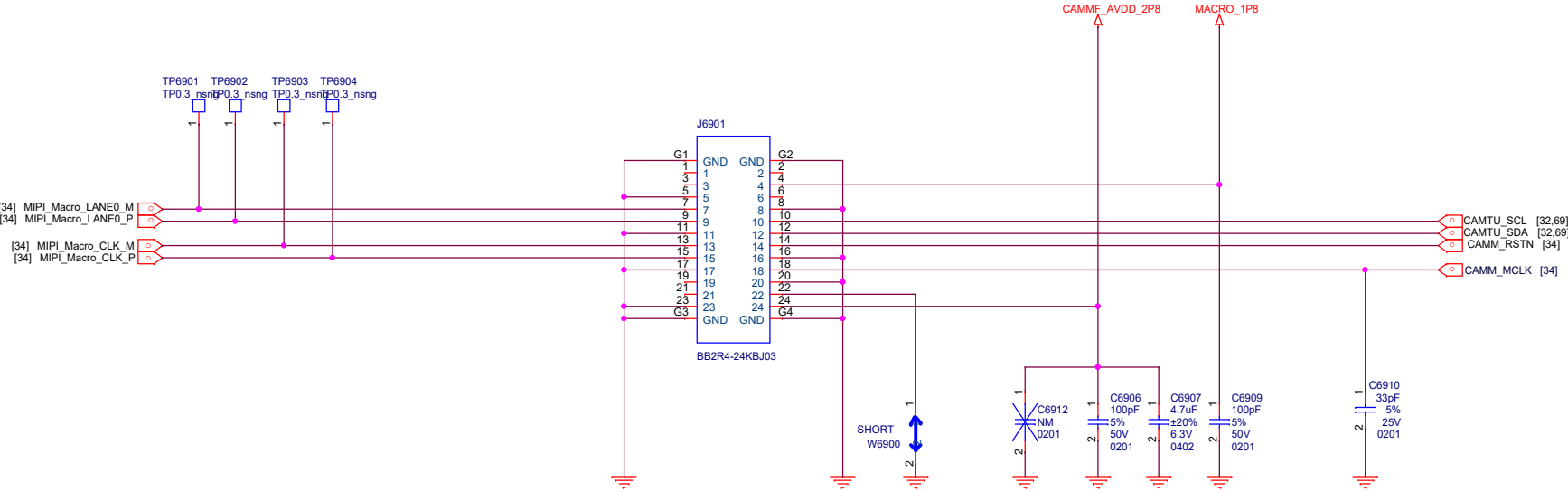
Title		
J11 MAIN		
Size	Document Number	
A4	Reserved(for Analog FM)	
Date:	Friday, March 05, 2021	Sheet

ULTRA 8M IMX355

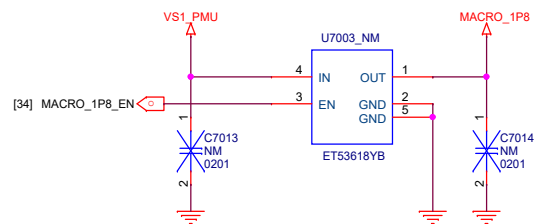
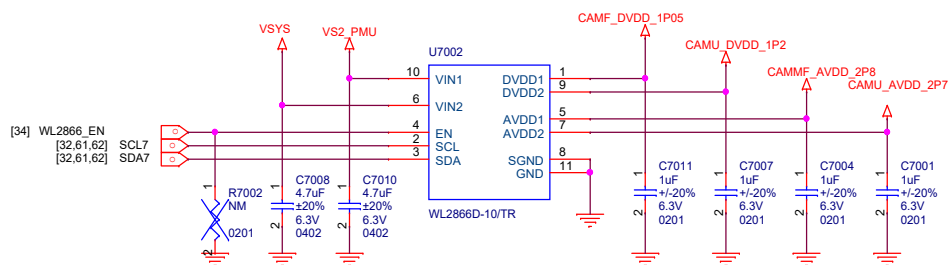
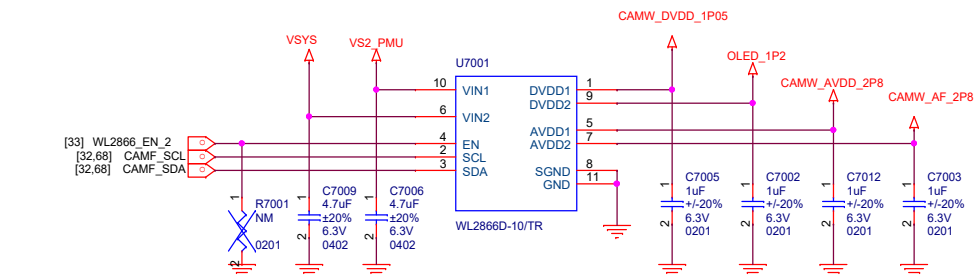


Title	
J11 MAIN	
Size	Document Number
A3	ULTRA CAMERA
Date:	Friday, March 05, 2021

MACRO 2M



Title	
J11 MAIN	
Size	Document Number
A3	ULTRA CAMERA
Date:	Friday, March 05, 2021



Title	
<Title>	
Size	Document Number
A3	<Doc>
Date:	Friday, March 05, 2021
Sheet	

Schematic design notice of "90_DEBUG_IO" page.

Note 90-1: UART Default support : UART0,
JTAG Default support : AP JTAG/MD_IO_JTAG

